



Collapse and Ejection in the N-body problem and the Formation of Rubble Pile Asteroids

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Motivating Questions

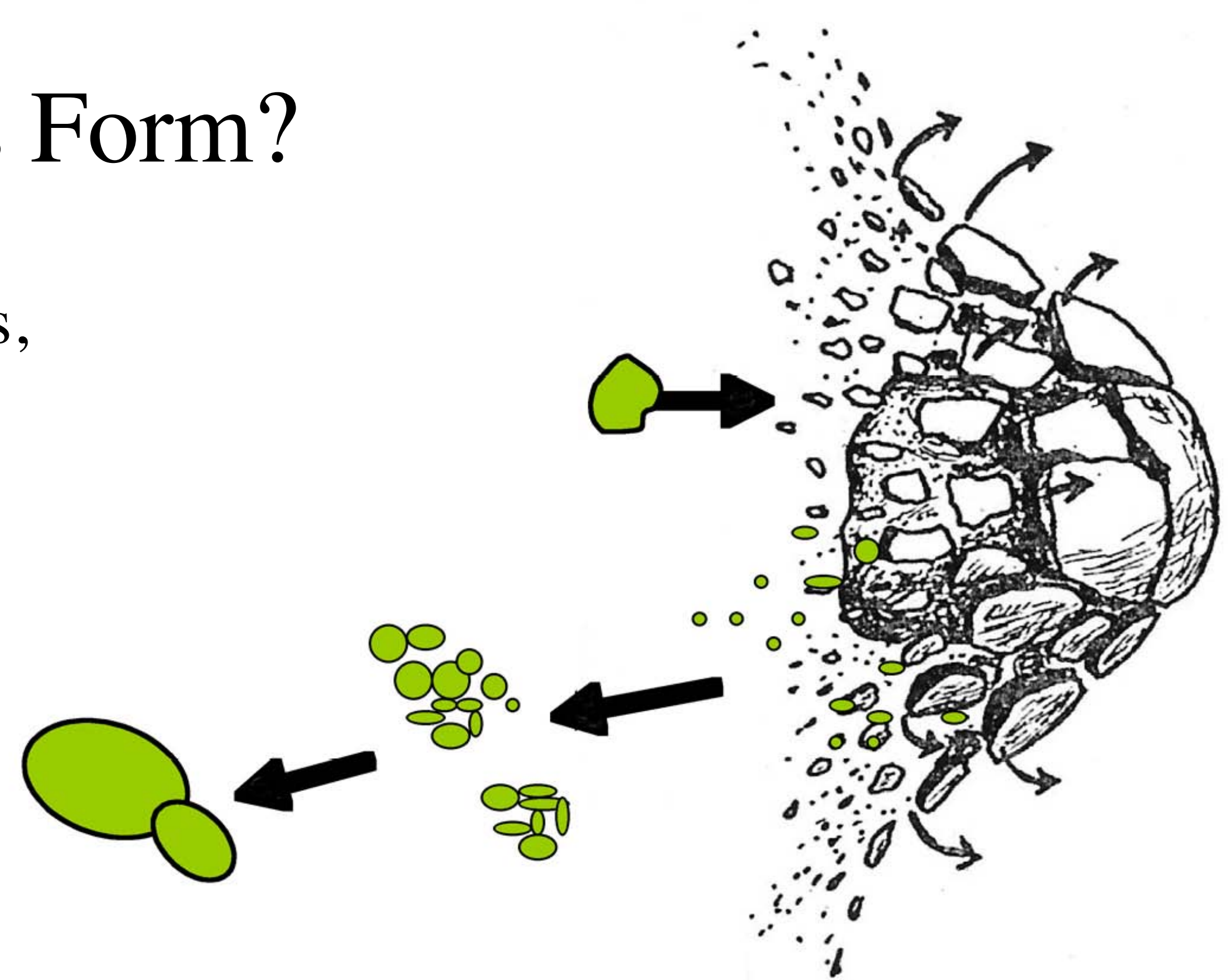


How do Rubble Piles Form?

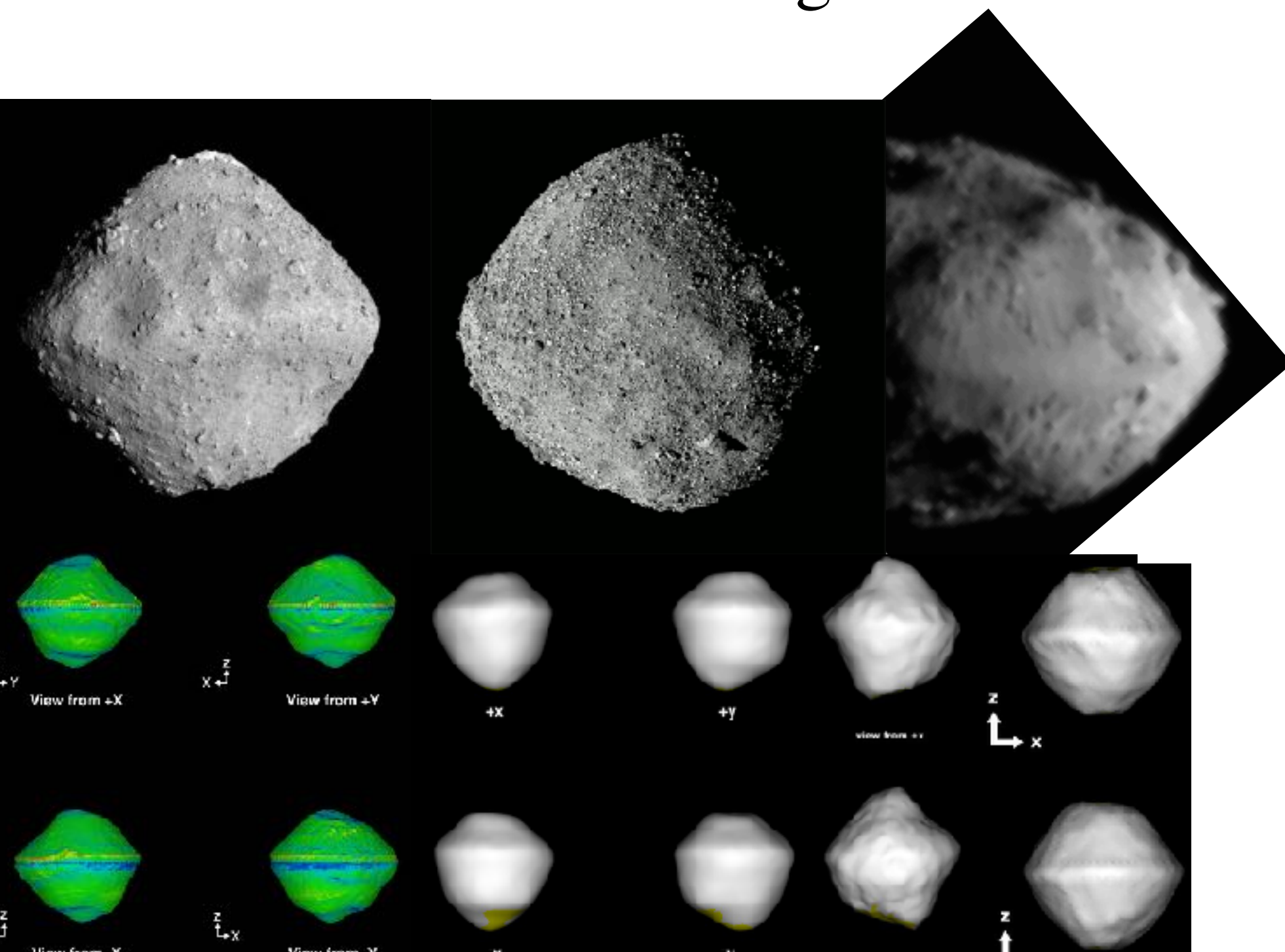
Components can be treated like rigid bodies, due to small pressures and impact speeds.

How efficient is the condensation process?

How does the initial state influence its final configurations and shapes?

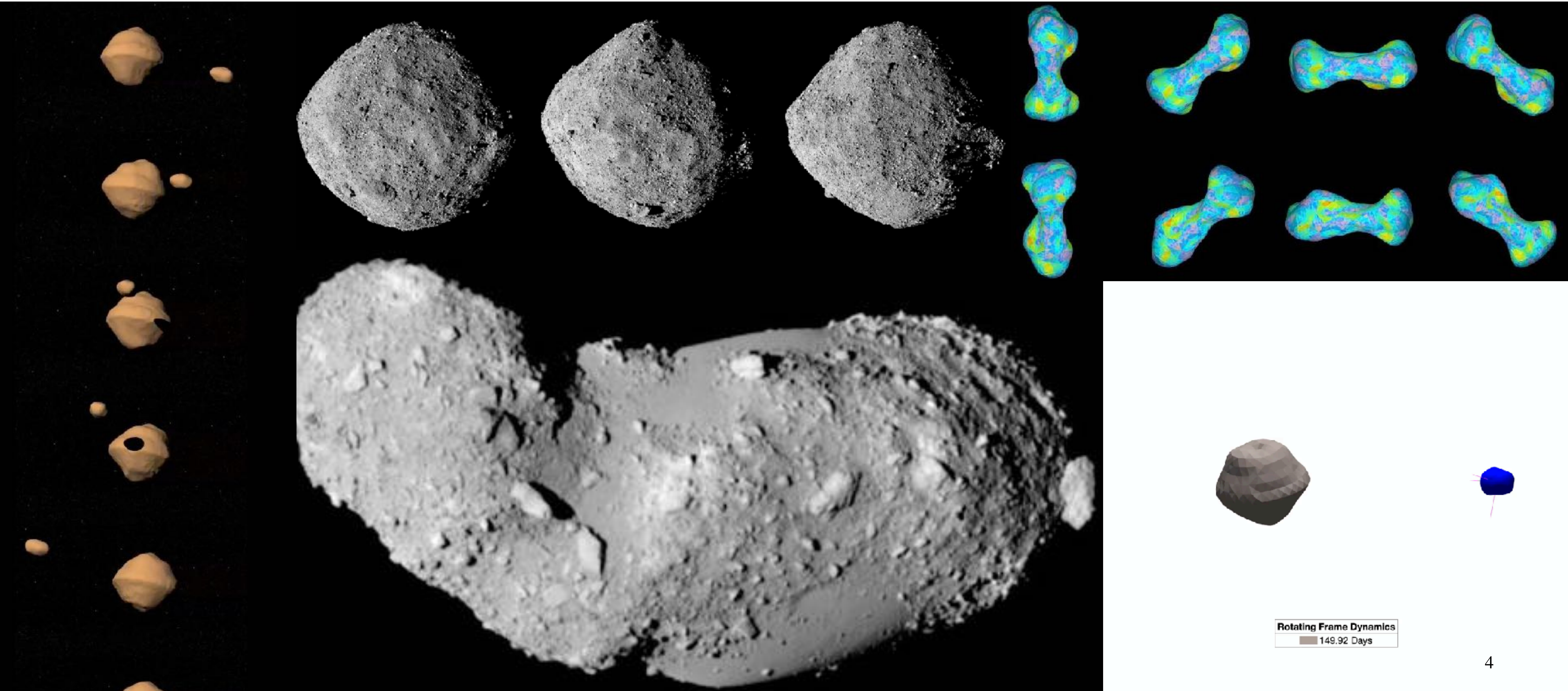


Formation model for Itokawa proposed in Fujiwara et al., Science 2006.



Convergent evolution independent of asteroid mineralogy indicates a fundamentally mechanical process for the formation of rubble pile asteroids.

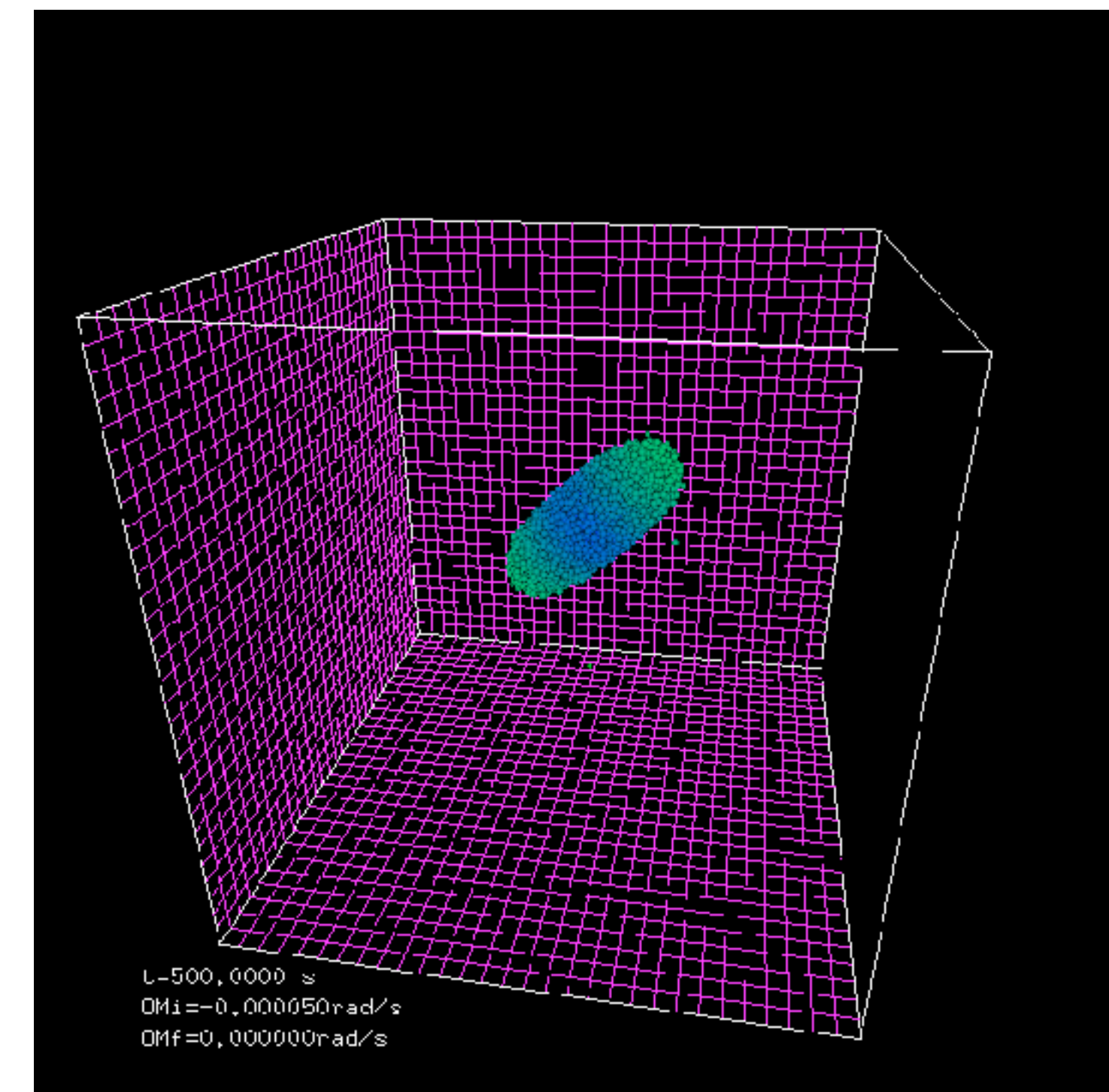
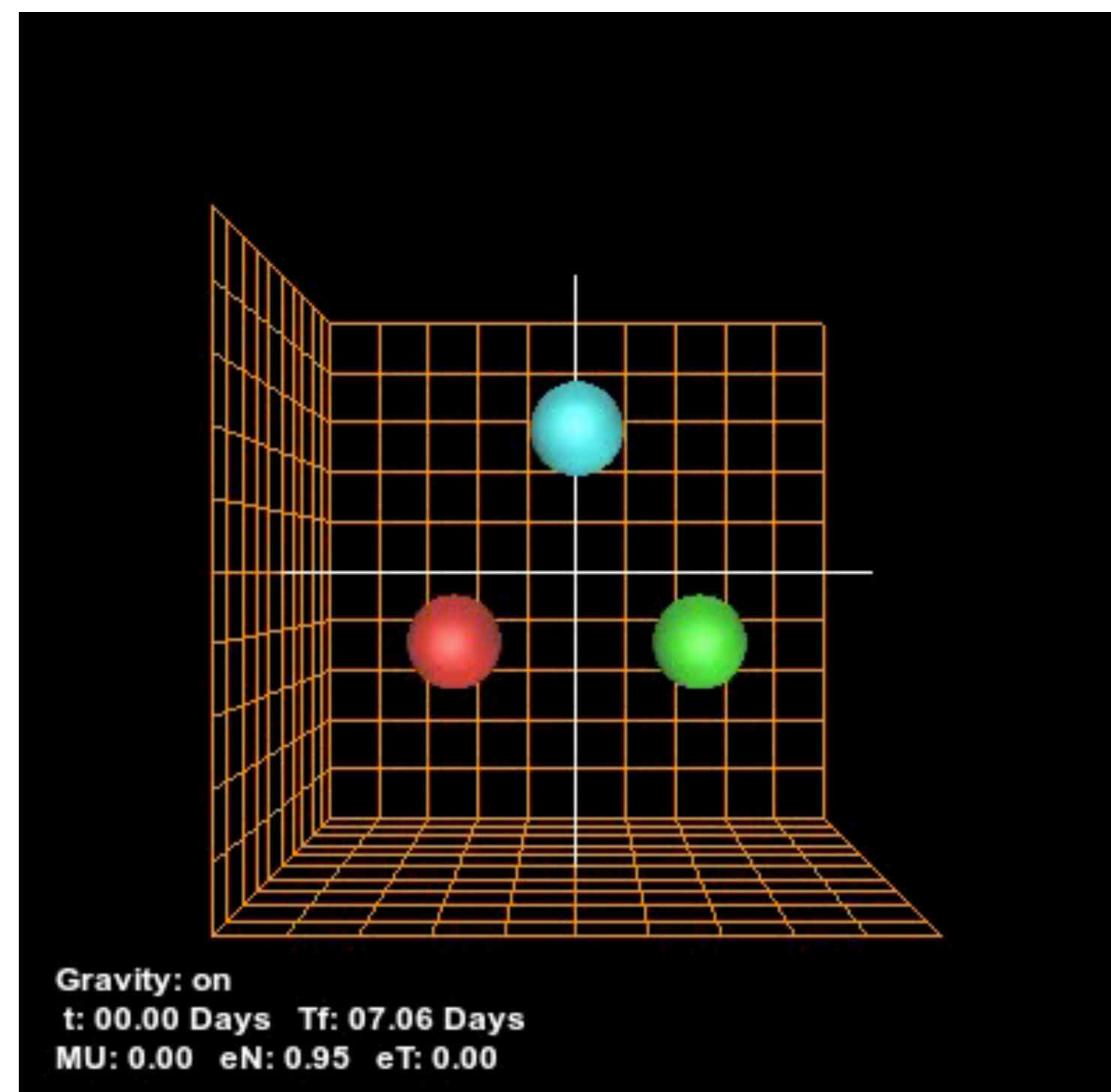
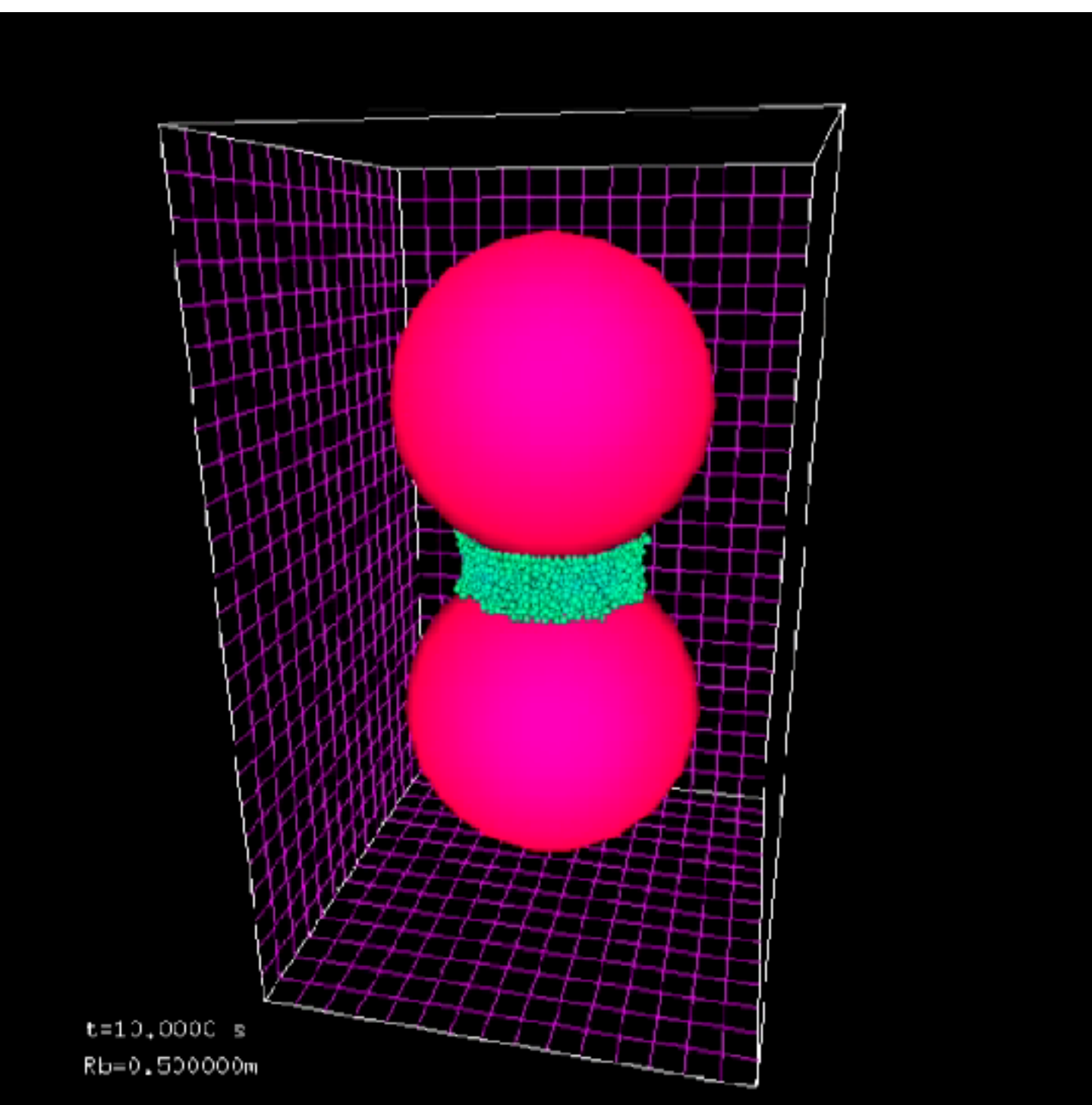
What Celestial Mechanics constraints exist for a collection of self-gravitating grains?



Finite Density Celestial Mechanics



- To analyze this question we can numerically study the relationship between the energy, angular momentum and configuration of a system of finite density components interacting via gravity and surface forces
 - For a closed system, Angular Momentum will be conserved but energy will not be
 - We wish to delineate how these relationships are interrelated and how they constrain the evolution of rubble pile asteroids and similar systems





Our approach utilizes the Amended Potential

- The Amended Potential $\mathcal{E}$ (or Minimum Energy Function) defines a fundamental relationship
 - It naturally combines a system's total angular momentum and energy into one relationship
 - It naturally couples both rotational and translational motion in a system

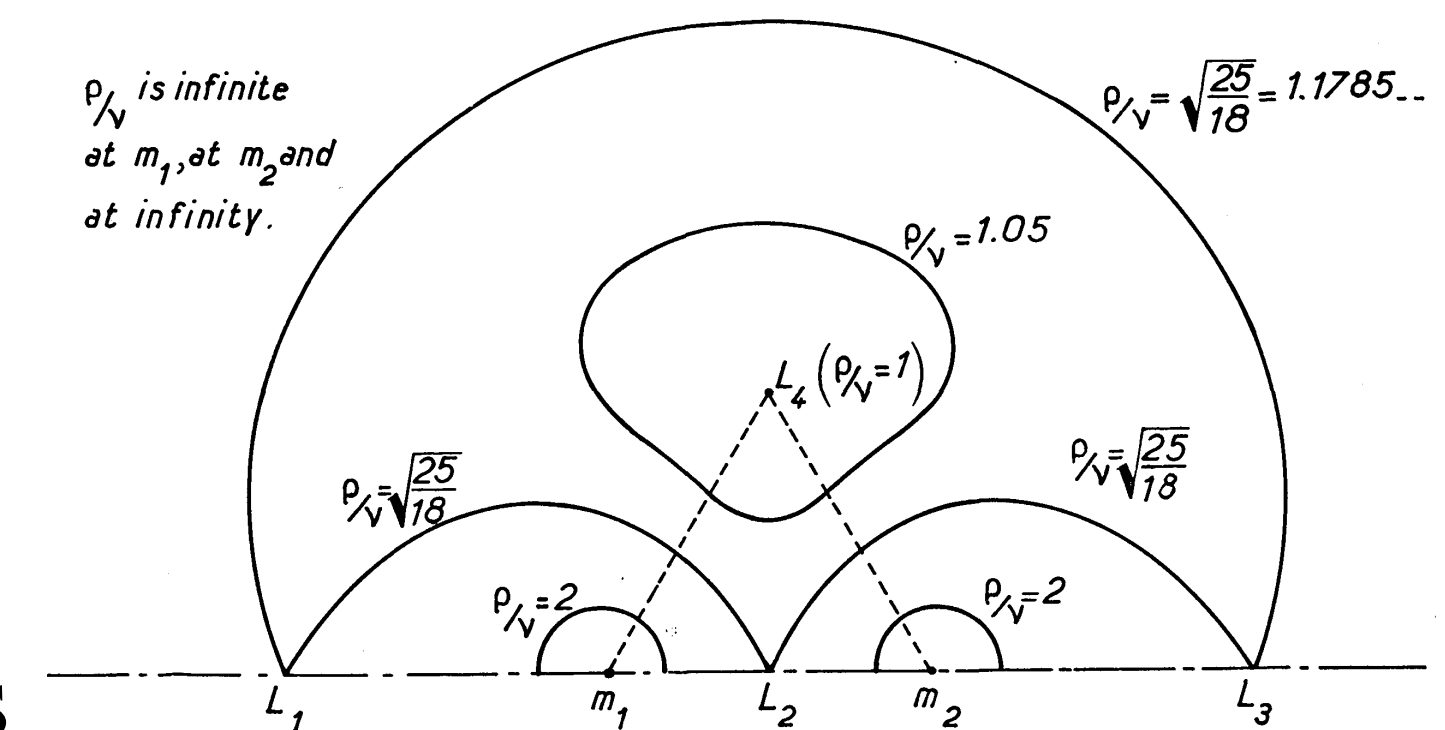
$$\mathcal{E} = \frac{H^2}{2I_H(\mathbf{Q})} + \mathcal{U}(\mathbf{Q}) \leq E$$

- Constants:
 - H is the total system angular momentum
 - E is the total system energy, which can be lost through dissipative interactions
- System configuration / Degrees of Freedom:
 - $\mathbf{Q} = \{\text{Minimal coordinates needed to specify the system configuration}\}$
- Functions of the system configuration (coordinates & orientations) that change dynamically
 - I_H is the system instantaneous total moment of inertia about the angular momentum direction
 - $\mathcal{U}$ is the instantaneous total gravitational potential energy, and can include mutual and self potentials
- Analysis of the Minimum Energy Function can determine a system's dynamics, equilibrium states and stability using a unified treatment

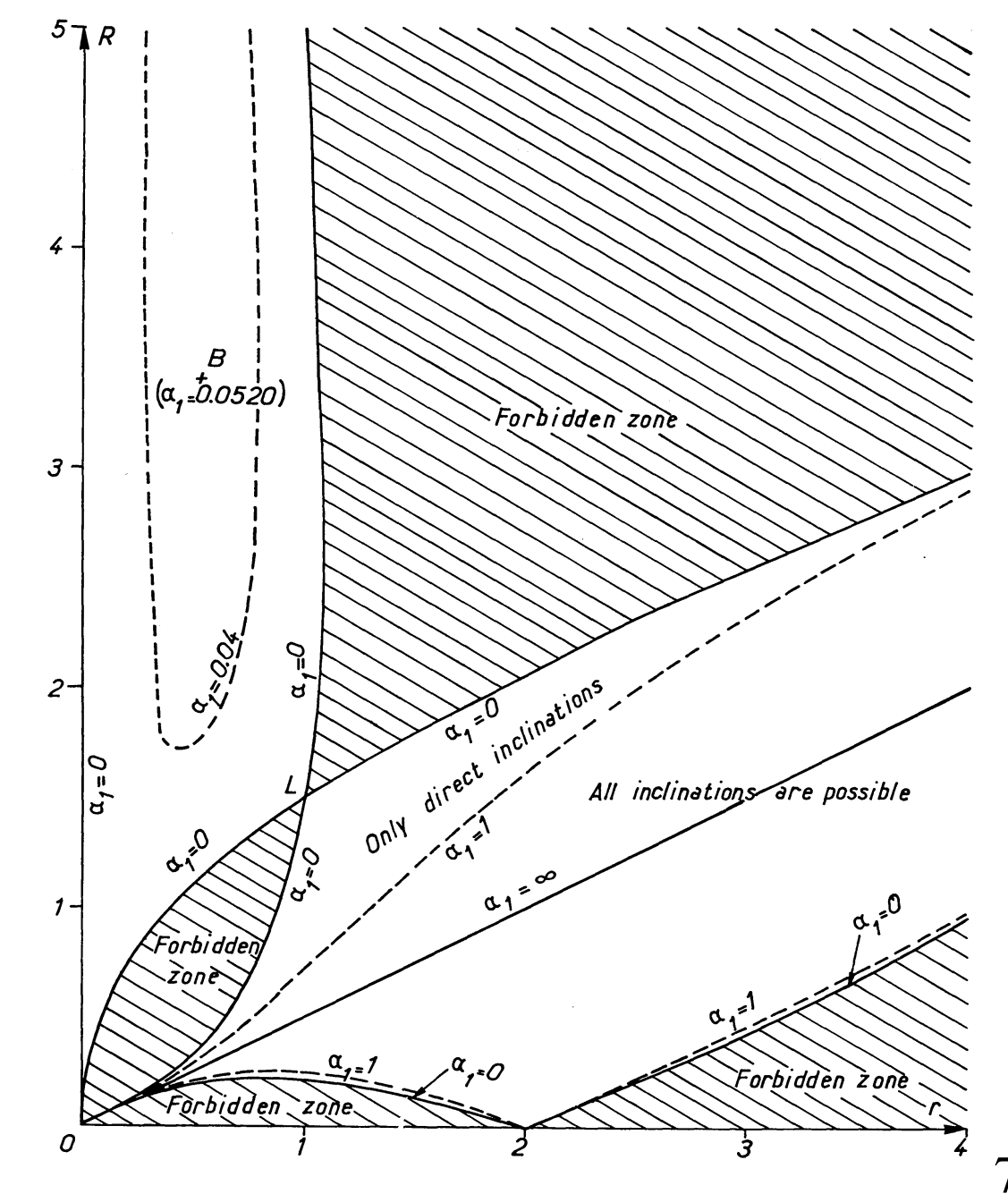
Relationship to Classical Studies of Escape in the N body problem



- In the classical, point-mass N body problem the necessary and sufficient conditions for escape have been long studied, and include some general results for the N body problem and detailed results for the 3 body problem
 - Relevant works by Chazy, Khimli, Marchal, Saari, Laskar and many others
- The “Full N Body Problem” is distinctive from these classical studies in that the presence of a rigid mass distribution opens up many more equilibrium conditions, pathways for energy dissipation, possible minimum energy configurations for $N \geq 3$ and much clearer thresholds for necessary conditions for escape
 - Aspects of these earlier studies are still relevant, but are not being specifically utilized in our current research
 - Can motivate future areas of work



Marchal & Saari 1975



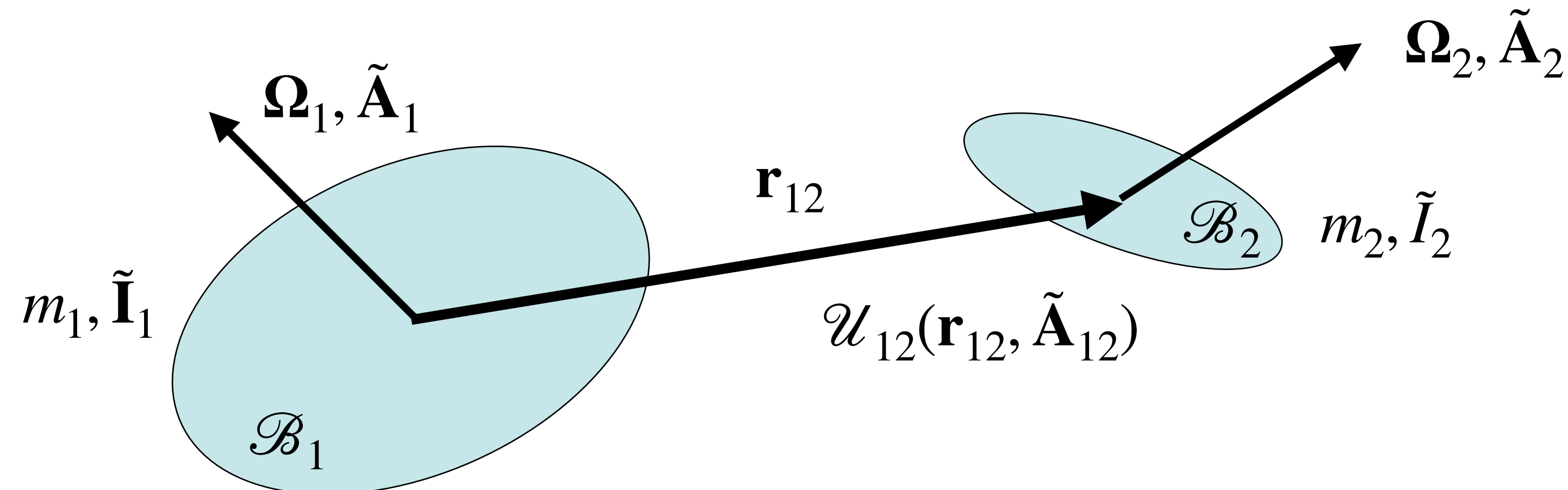
Generalized Integral Formulation



$$\mathbf{H} = \sum_{i=1}^N m_i [\mathbf{r}_i \times \cdot \mathbf{r}_i + \tilde{\mathbf{I}}_i \cdot \boldsymbol{\Omega}_i] = H \hat{\mathbf{H}} \quad \text{Total Angular Momentum}$$

$$I_H = \sum_{i=1}^N \left[m_i \left(r_i^2 - (\hat{\mathbf{H}} \cdot \mathbf{r}_i)^2 \right) + \hat{\mathbf{H}} \cdot \tilde{\mathbf{I}}_i \cdot \hat{\mathbf{H}} \right] \quad \text{Total Moment of Inertia about } \hat{\mathbf{H}} \text{ (point mass and inertia tensor)}$$

$$\mathcal{U} = -\mathcal{G} \sum_{1 \leq i < j \leq N} \int_{\mathcal{B}_i} \int_{\mathcal{B}_j} \frac{dm_i dm_j}{\rho_{ij}} - \frac{\mathcal{G}}{2} \sum_{i=1}^N \int_{\mathcal{B}_i} \int_{\mathcal{B}'_i} \frac{dm_i dm_{i'}}{\rho_{ii'}} \quad \text{Total mutual and self potential energy}$$



Sphere-Restricted Full N-Body Problem



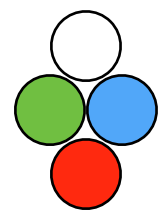
$$\mathbf{H} = \sum_{i=1}^N m_i [\mathbf{r}_i \times \dot{\mathbf{r}}_i + I_S \boldsymbol{\Omega}_i] = H \hat{\mathbf{H}} \quad \text{Total Angular Momentum}$$

$$I_H = \sum_{i=1}^N \left[m_i \left(r_i^2 - (\hat{\mathbf{H}} \cdot \mathbf{r}_i)^2 \right) + I_S \right] \quad \text{Total Moment of Inertia about } \hat{\mathbf{H}}$$

$$\mathcal{U} = -\mathcal{G} \sum_{1 \leq i < j \leq N} \frac{m_i m_j}{r_{ij}} - \frac{6\mathcal{G}}{5} \sum_{i=1}^N \frac{m_i^2}{d_i} \quad \text{Total potential energy, mutual and self}$$

Switching to constant-sized spheres creates a more tractable problem for analysis and presentation of results.

Our general relationships and results do not rely on this assumption, however.



Particle Mass m

Diameter d

$$\text{Moment of Inertia } I_S = \frac{1}{10} m d^2$$



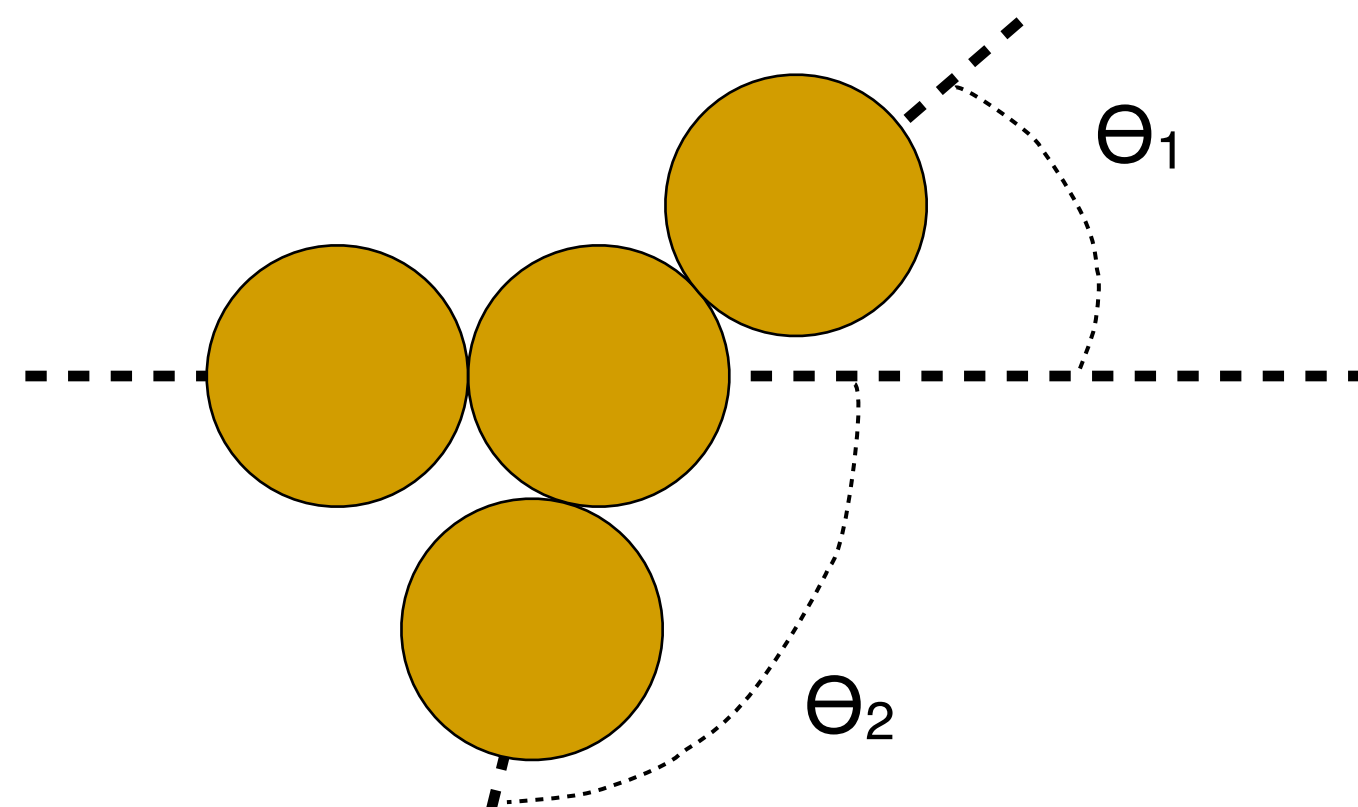
Relative Equilibria and Minimum Energy Configurations

- Relative equilibria and stability for N body systems are a function of angular momentum and configuration and can be determined using $\mathcal{E}$:

$$\mathbf{Q} = \left\{ \mathbf{r}_{ij} \mid r_{ij} \geq d \right\}$$

Relative Equilibrium

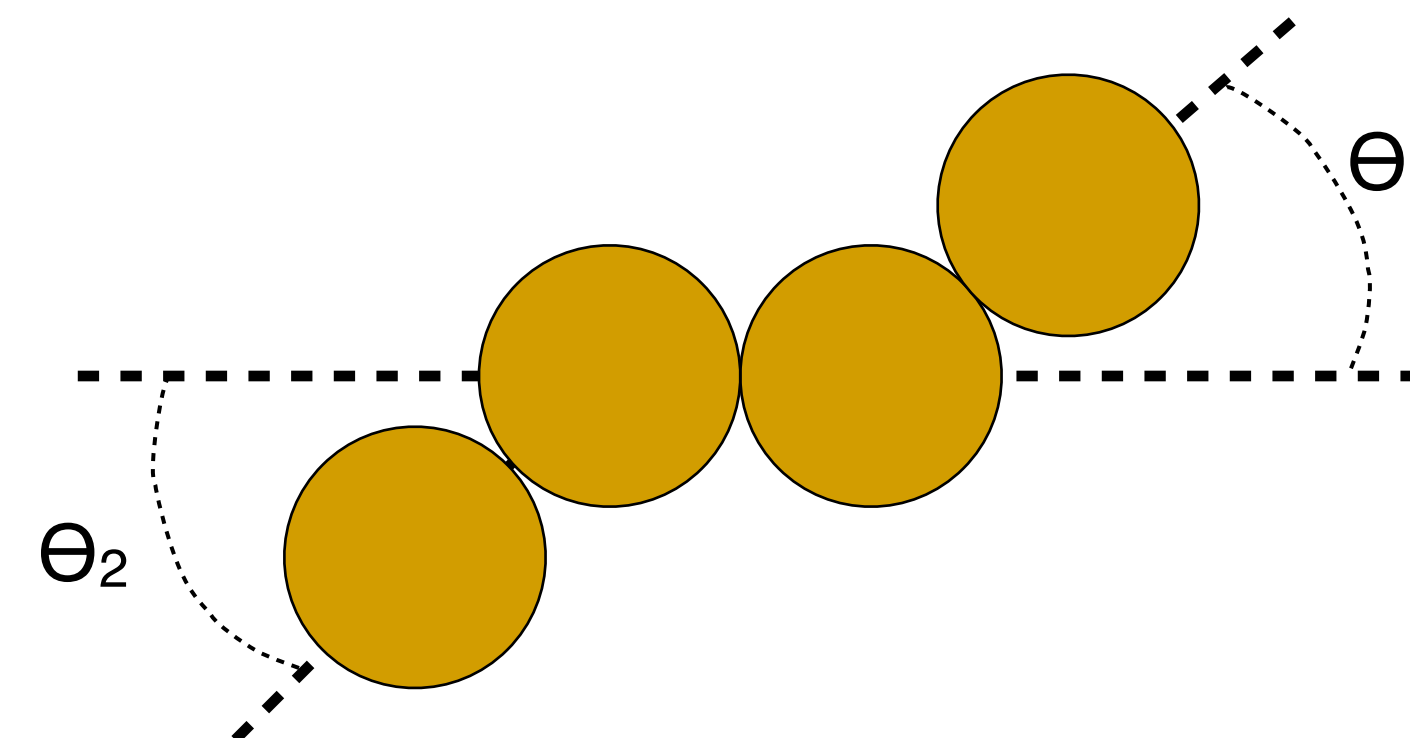
$$\delta \mathcal{E} = \frac{\partial \mathcal{E}}{\partial \mathbf{Q}} \cdot \delta \mathbf{Q} \geq 0$$



$$\mathcal{E} = \frac{H^2}{2 I_H(\mathbf{Q})} + \mathcal{U}(\mathbf{Q})$$

Stability

$$\delta^2 \mathcal{E} = \delta \mathbf{Q} \cdot \frac{\partial^2 \mathcal{E}}{\partial \mathbf{Q}^2} \cdot \delta \mathbf{Q} \geq 0$$

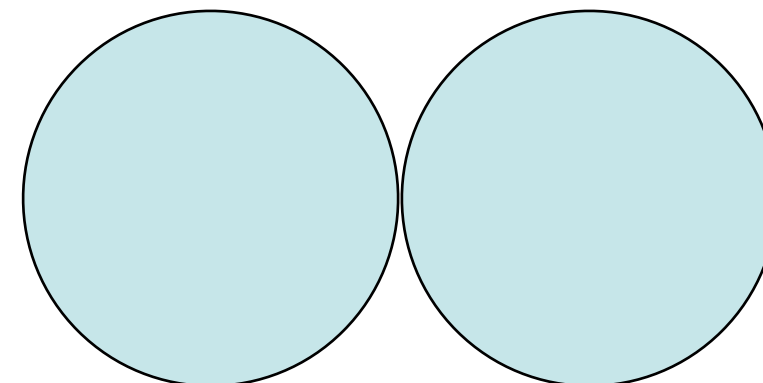




Relative Equilibria

- For $N = 2, 3, 4$ all relative equilibria for the Spherical Restricted Full NBP can be delineated and their stability regimes rigorously computed
 - Restricted to systems rotating about their maximum moment of inertia, to ensure energetic stability

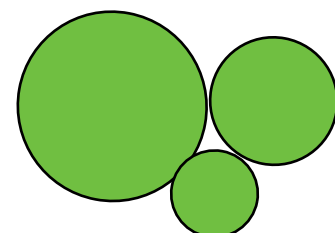
$$N = 2$$



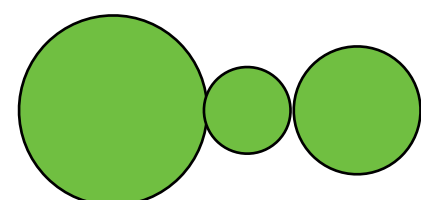
$$N = 3$$

Note: For $N \geq 3$ the “classical” central configurations are always energetically unstable (Moeckel)

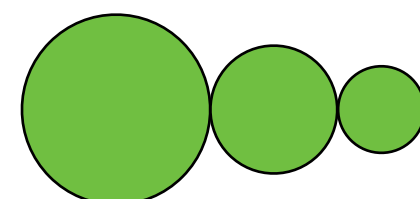
Lagrange Resting



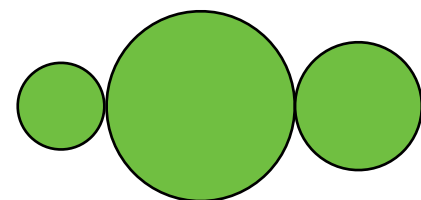
Euler Resting



ER132

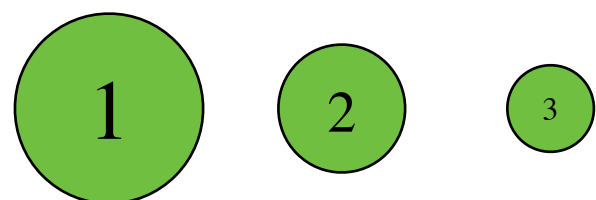


ER123

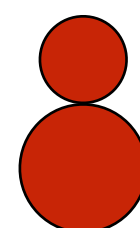


ER312

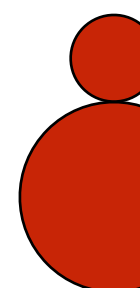
$$m_1 \geq m_2 \geq m_3$$



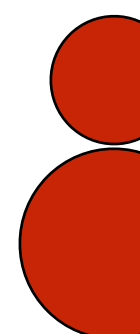
Isosceles



IS23-1

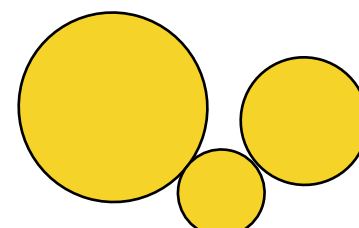


IS13-2

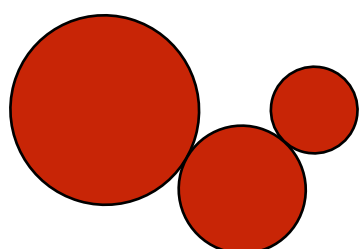


IS12-3

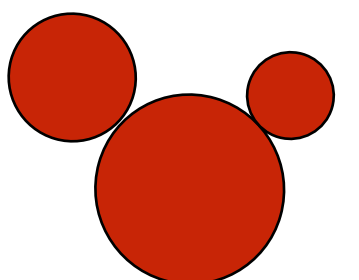
Transitional Resting



TR132

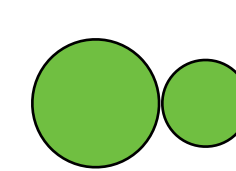


TR123

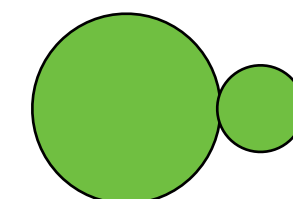


TR213

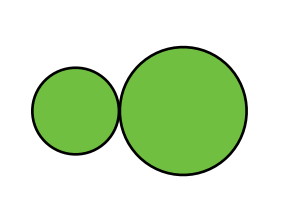
Euler Aligned



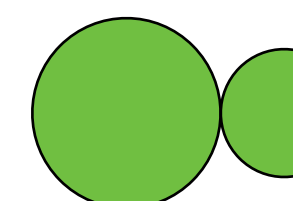
EA23-1



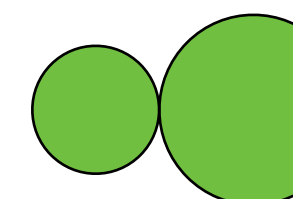
EA13-2



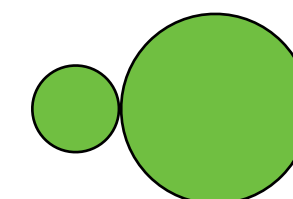
EA32-1



EA12-3

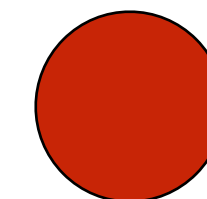
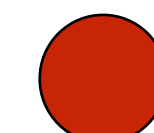


EA21-3

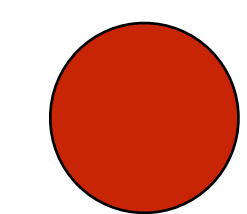


EA31-2

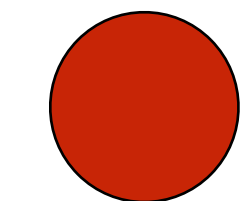
Lagrange Orbital



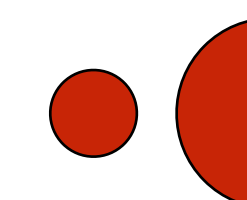
Euler Orbital



EO132



EO123



EO312

 stable for some values of masses and H

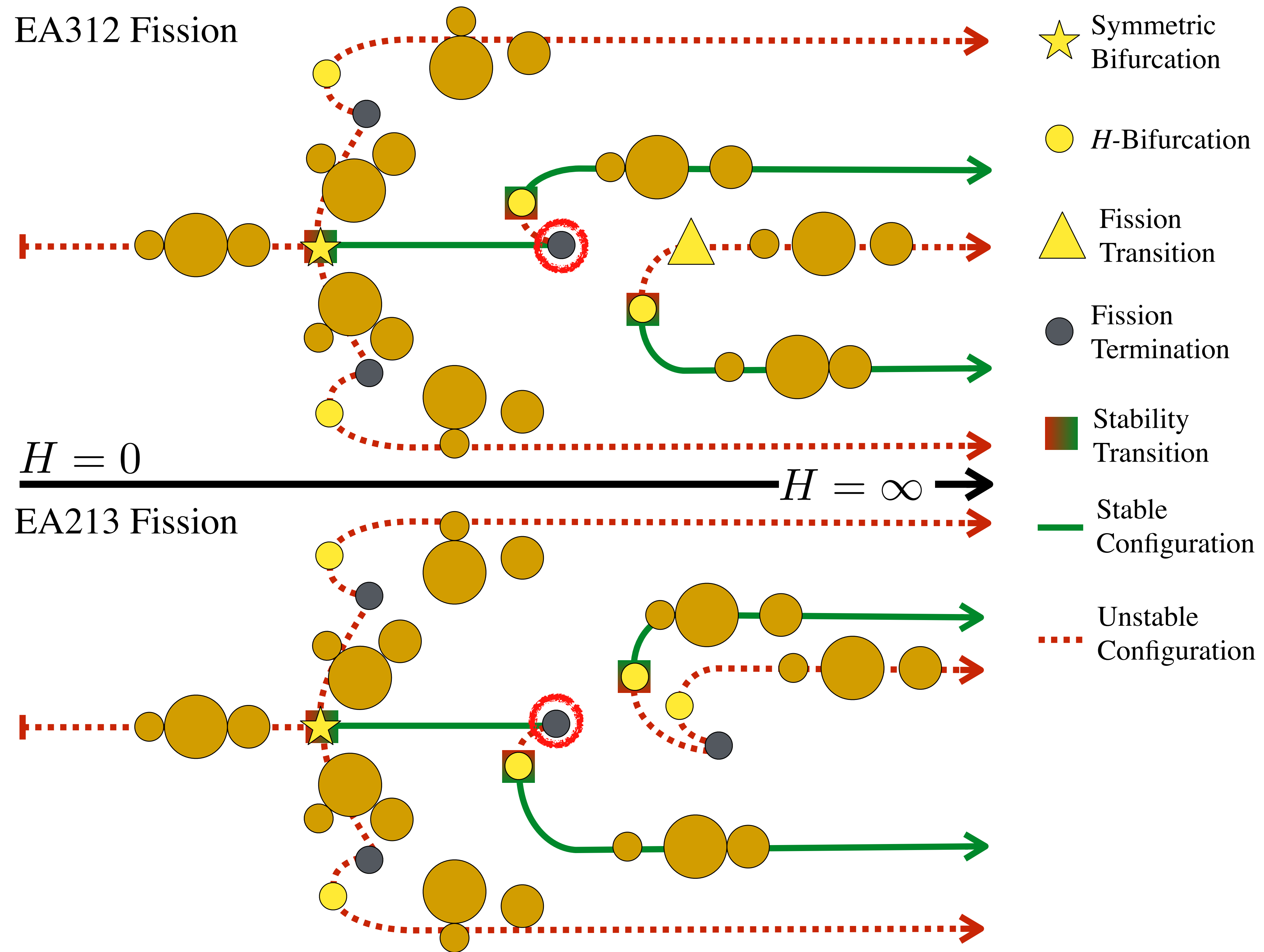
 stable for some values of H

 always unstable

 has dual orderings

Whether a configuration is stable depends on the precise range of angular momentum.

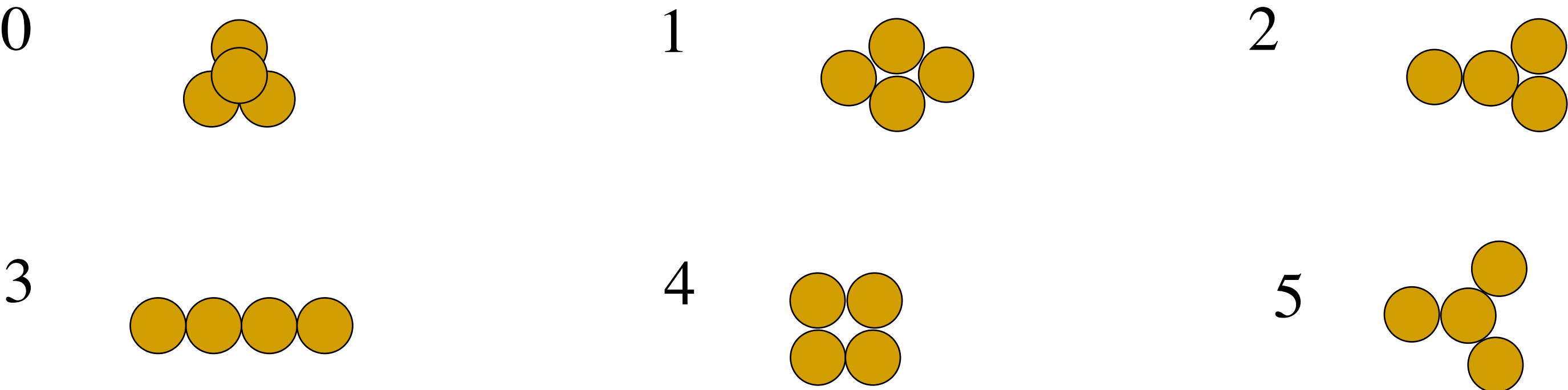
Multiple configurations are related through bifurcations.



Finite Density 4 Body Problem

$$N = 4$$

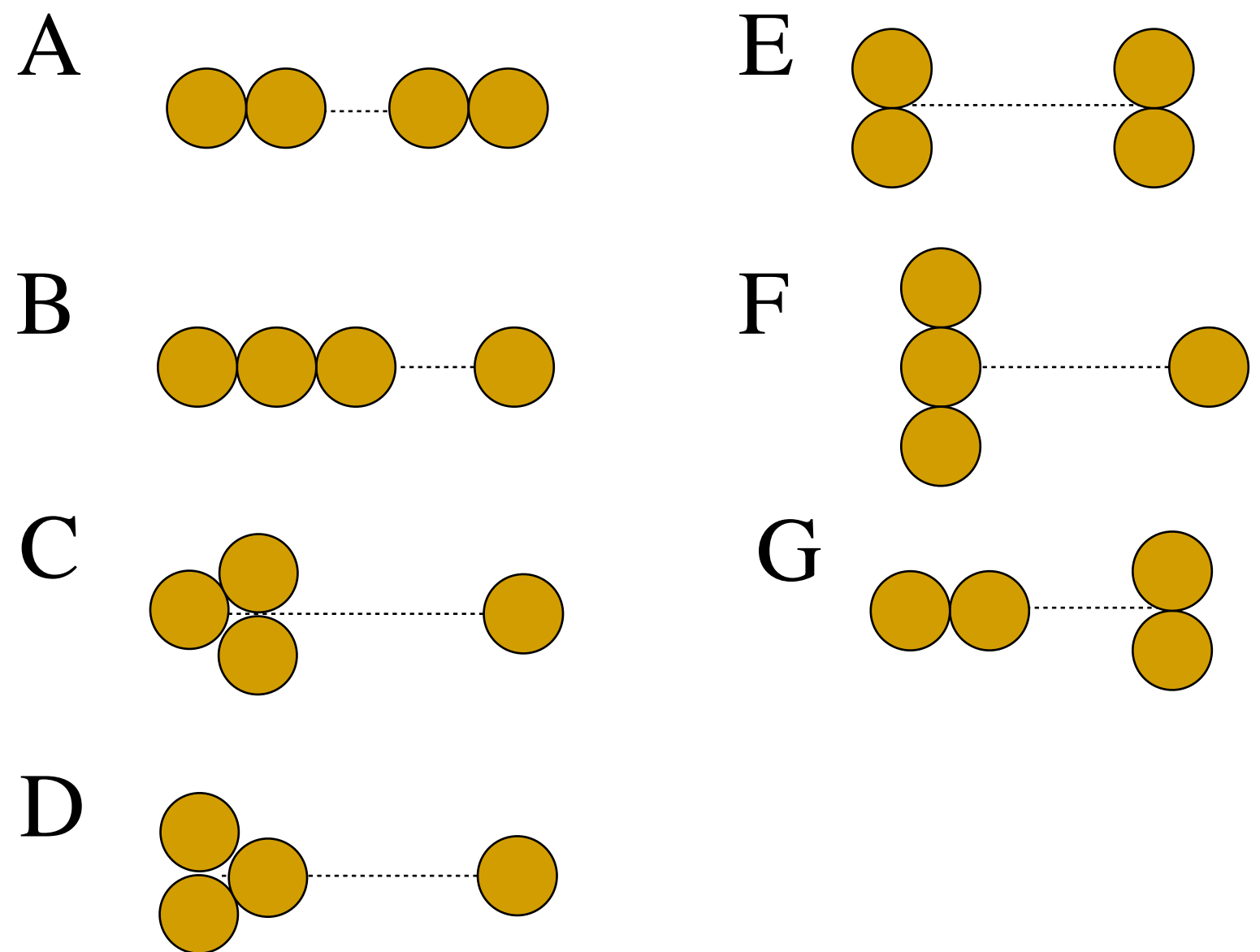
Static Resting Equilibrium Configurations



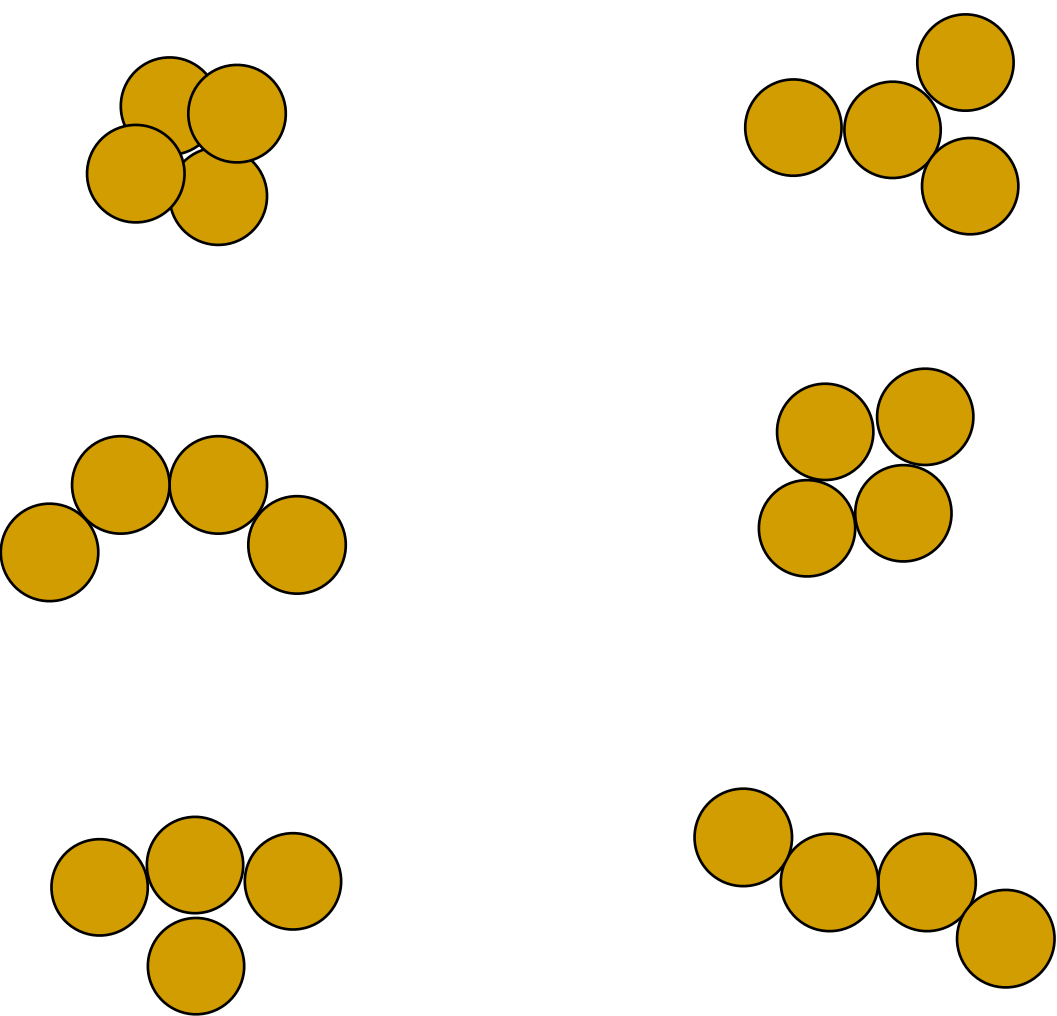
Non-contact equilibria are not shown.

Hypothesis that these represent all resting equilibrium solutions for the 4 equal size case.

Mixed Equilibrium Configurations



Variable Resting Equilibrium Configurations



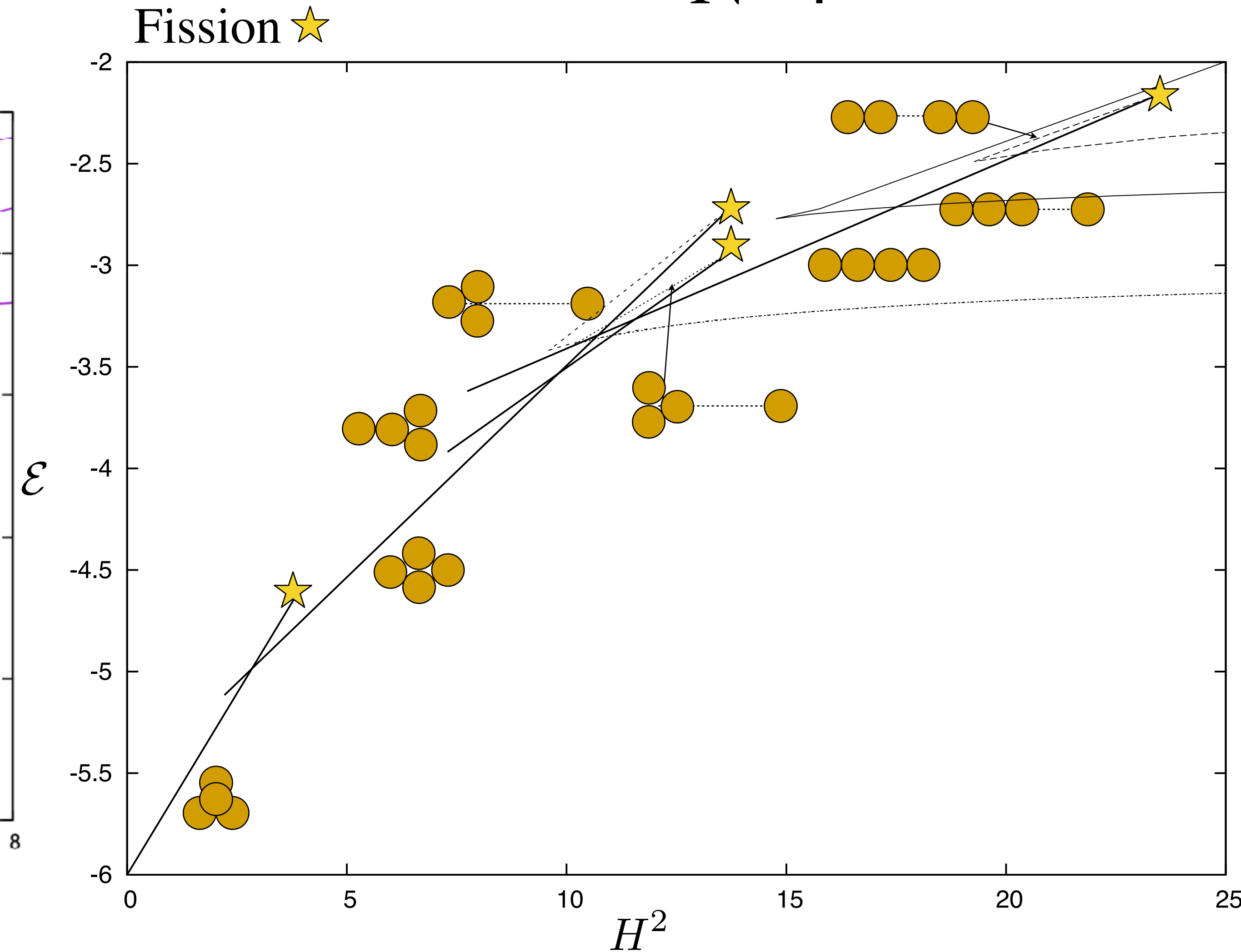
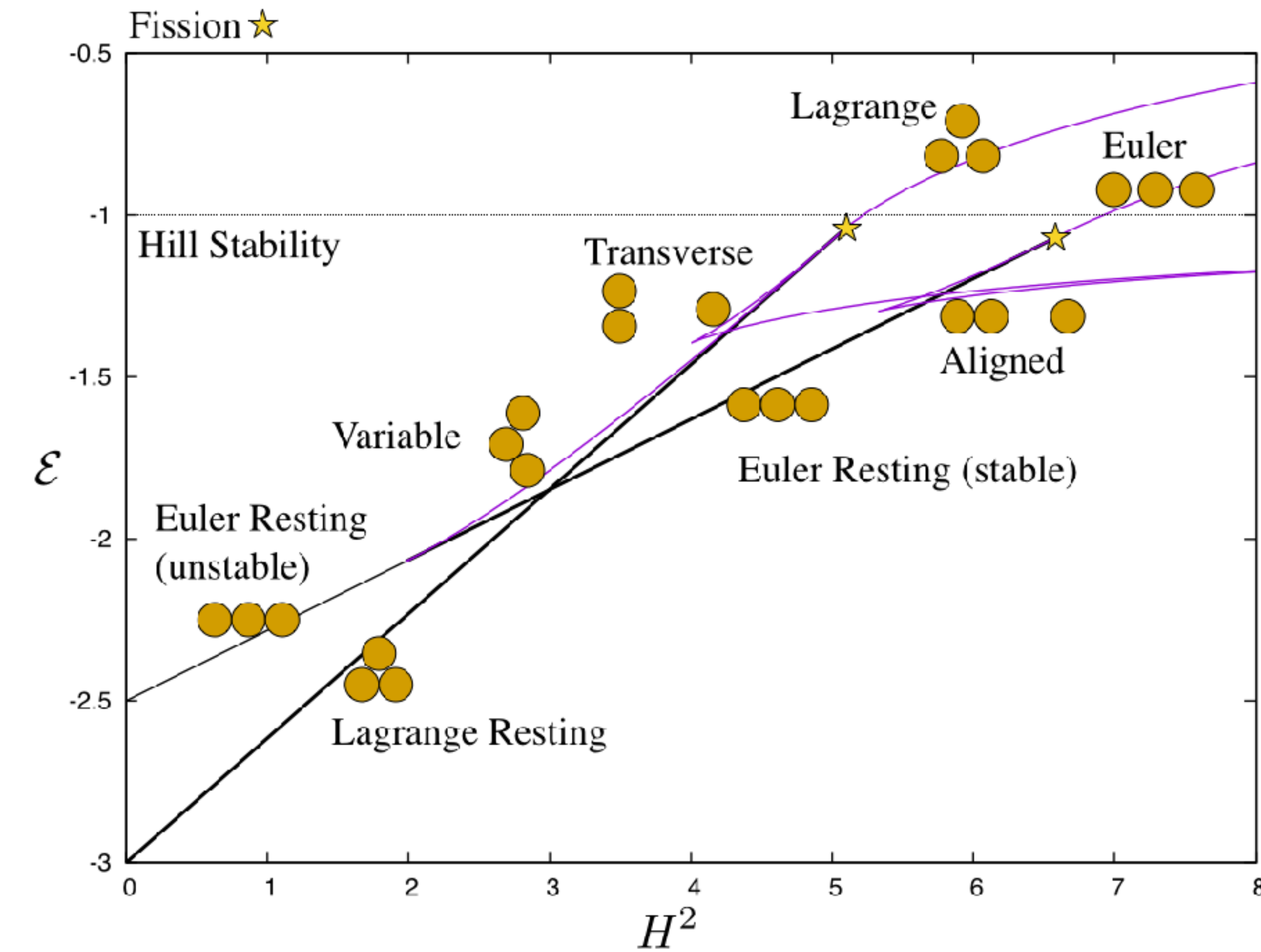


N=3

$$\mathcal{E} = \frac{H^2}{2I_H} + \mathcal{U} \leq E$$



N=4



Useful to chart these as functions of angular momentum and energy.



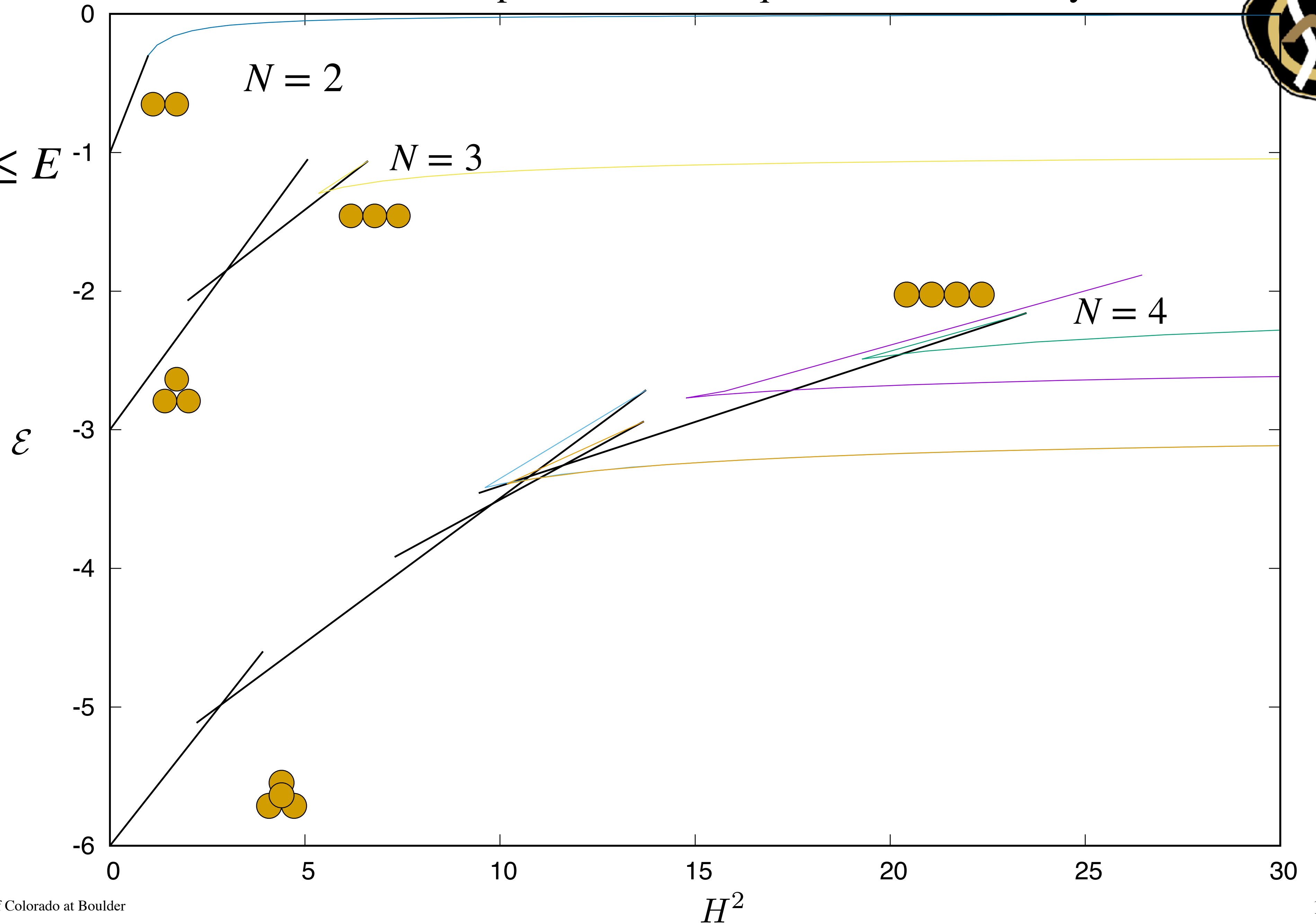
Combined Graph of Relative Equilibria and Stability

$$\mathcal{E} = \frac{H^2}{2I_H} + \mathcal{U} \leq E$$

Can normalize to directly compare energies, opens up the analysis of inclusion and ejection of bodies.

How do we increase N ?

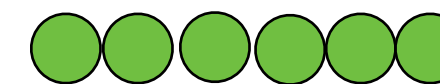
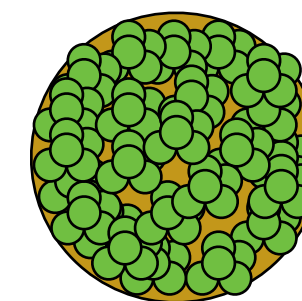
How do we account for ejection?

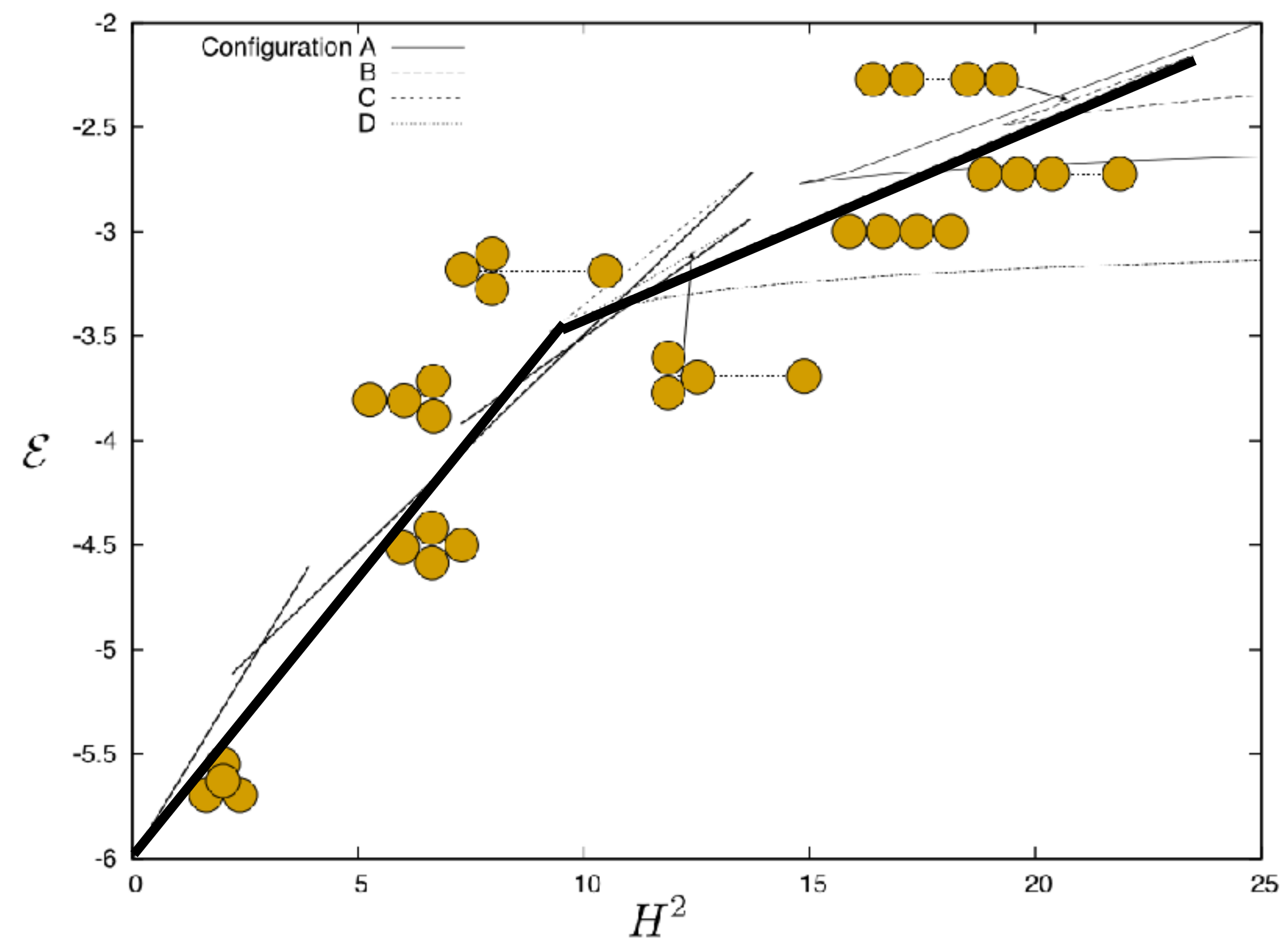
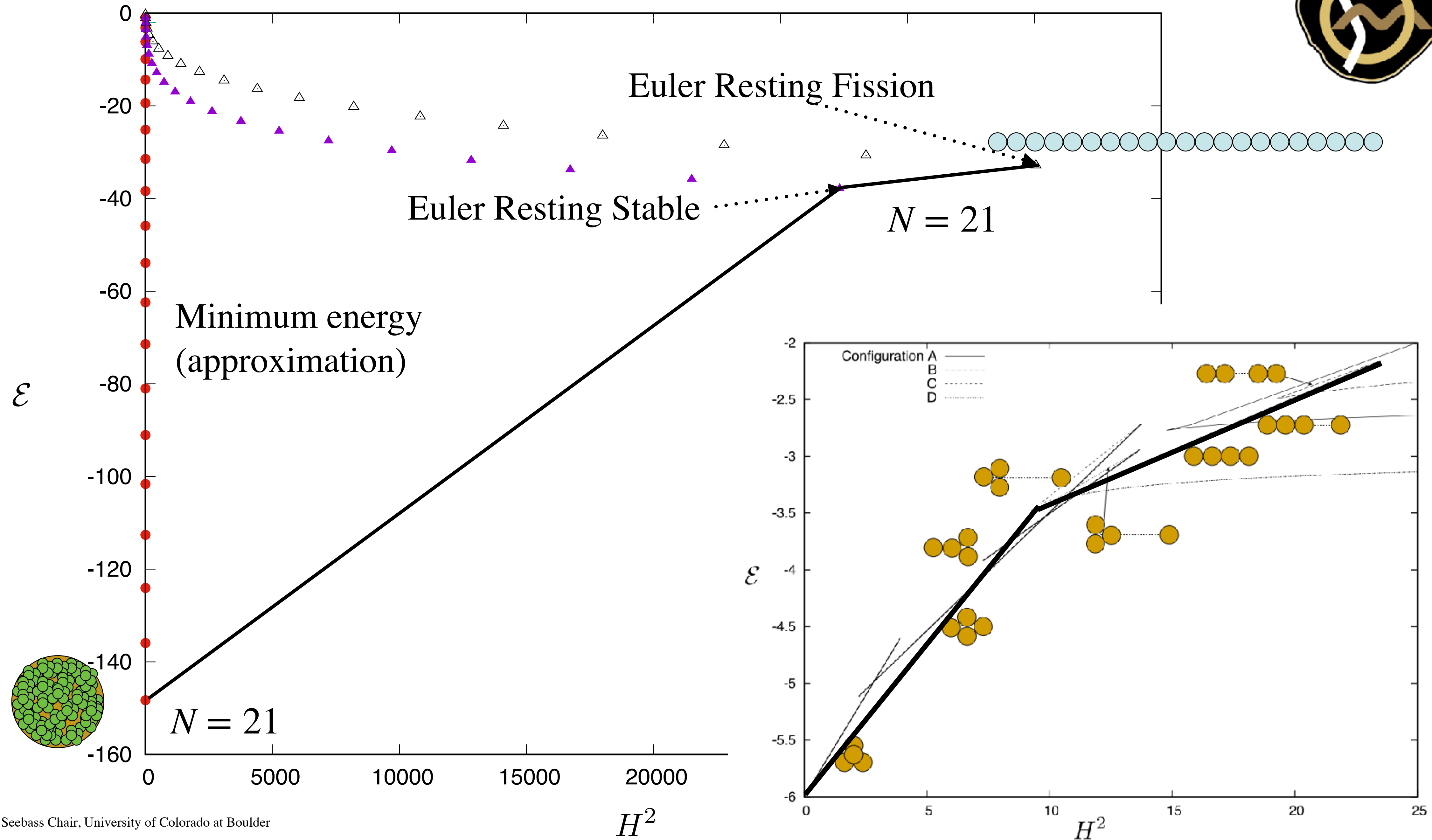




Extending the Analysis beyond $N = 4$

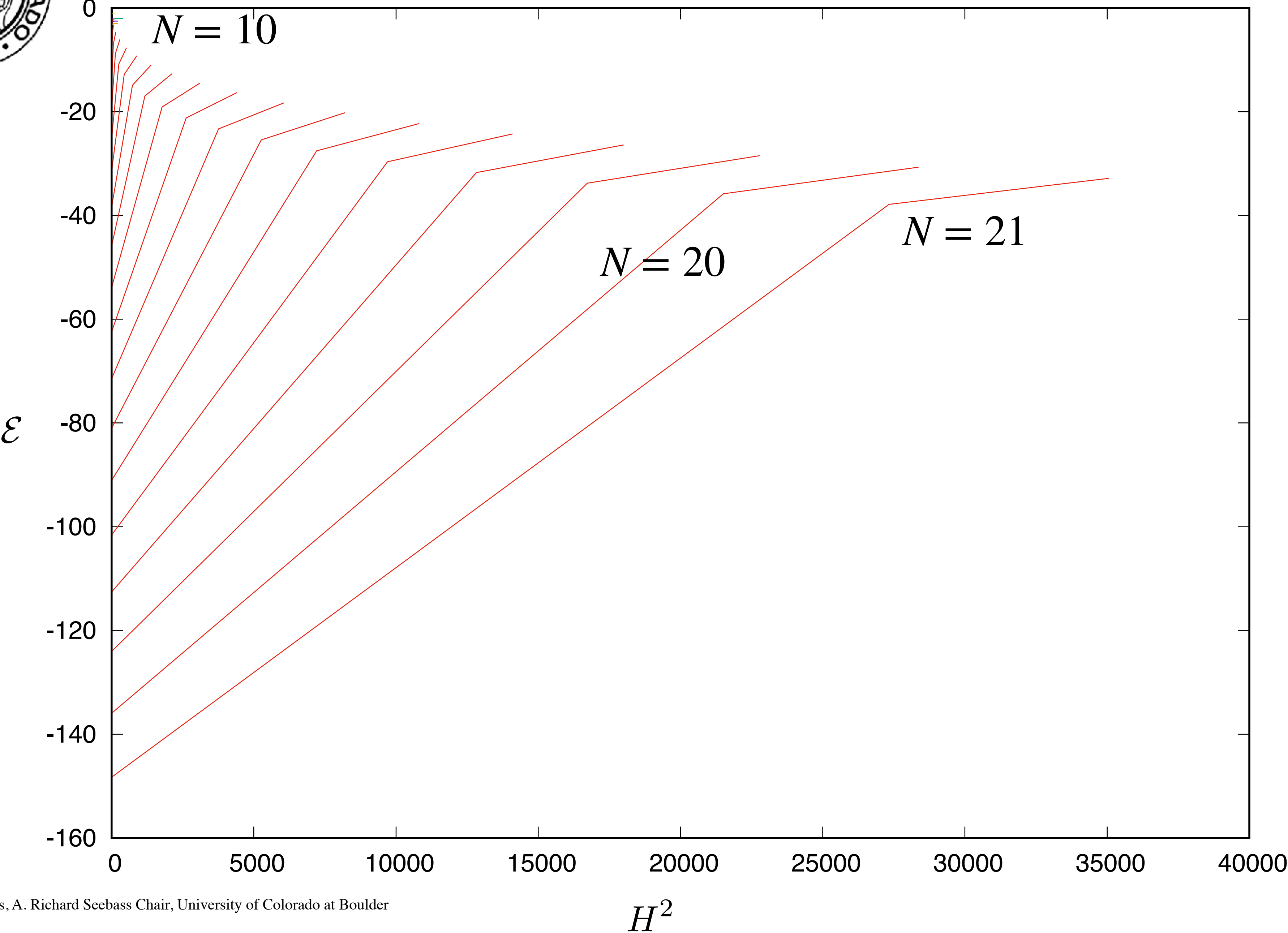
- There are several computable points in the evolution diagram when $N \gg 1$
- Minimum energy configuration when $H^2 = 0$
 - For $N \gg 1$ is defined as a quasi-sphere
 - Can be approximated as a continuum sphere with an appropriate packing fraction
- Angular momentum limits for the highest energy resting equilibrium structures:
 - Values of H^2 at which the Euler Aligned configuration becomes stable and fissions
 - Only valid for $N \leq 21$, for $N > 21$ will use Jacobi Ellipsoid limits (future work)
- Limiting energy bounds on binary orbit configurations
 - Lower limit corresponds to the resting energy of the $N - 1$ configuration:
 - Upper limit corresponds to twice the potential of the $(N/2)$ Euler Resting configuration:





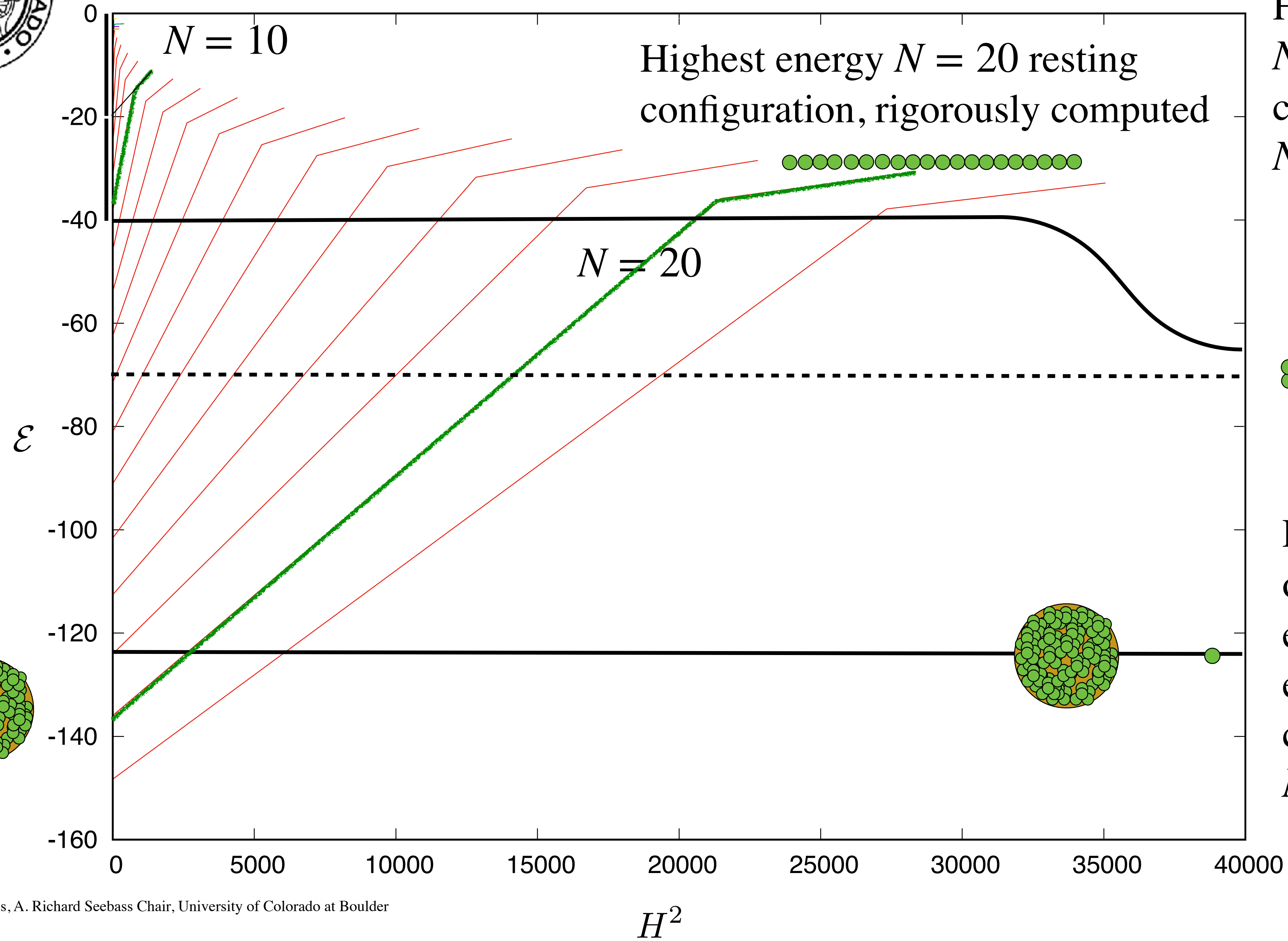


Bounding Curves for End-State Fission

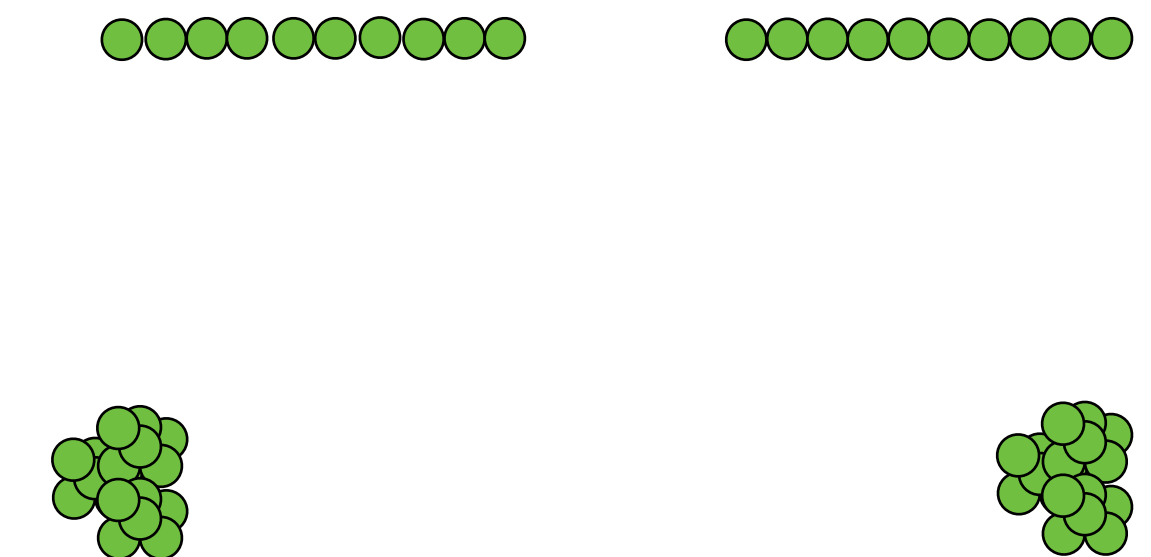




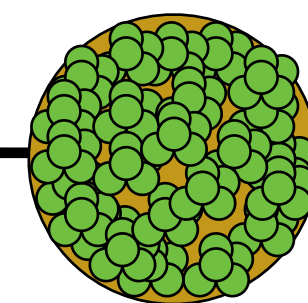
Bounding Energies can be precisely computed up to $N = 21$



Highest energy
 $N = 20$ orbits,
computed from the
 $N = 10$ results



Lowest energy $N = 20$
orbit, and threshold
energy beyond which
ejection can occur,
computed from the
 $N = 19$ results





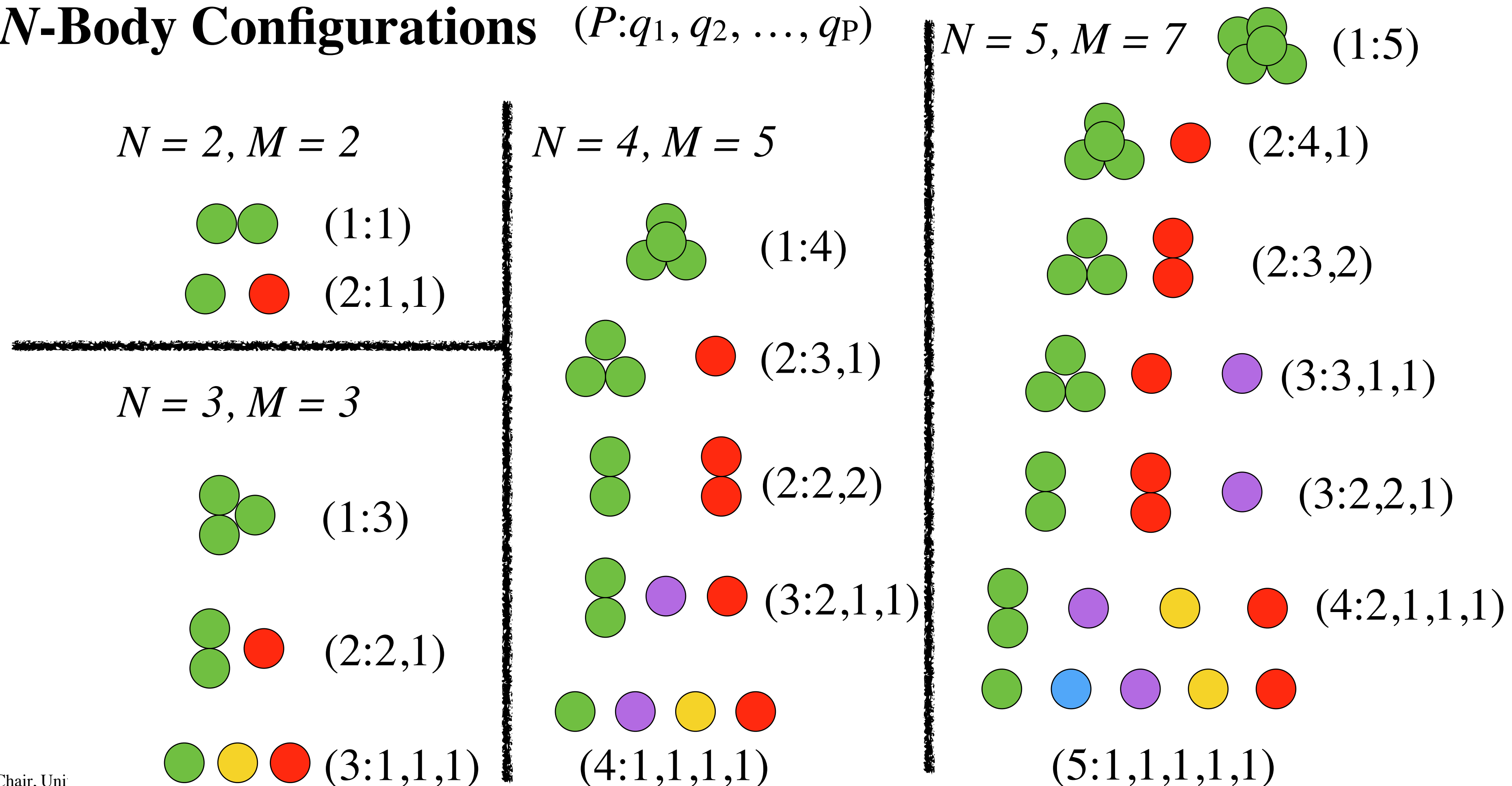
Ejection Limits



Fission and Escape for N Bodies

- Given N bodies, a *Configuration* is a collection of these bodies into P groups, with each group having a set number of bodies q_i . Does not consider the state of the configurations

N -Body Configurations $(P:q_1, q_2, \dots, q_P)$



Finite Density Hill Stability: Definitions & Theorems



System Energy of an
 N -Body Problem

$$E = T + \mathcal{U}$$

Gravitational Potential Energy
for Finite Density Bodies

$$\mathcal{U} = - \sum_{i=1}^{N-1} \sum_{j=i+1}^N \frac{1}{r_{ij}}$$

Minimum Potential

$$\mathcal{U}_m(N) = \min_{r_{ij} \geq 1} \mathcal{U}$$

- A *Configuration* of P components with q_i bodies in each component ($i = 1, 2, \dots, P$) is *Configuration Hill Stable* if none of these components can mutually escape to infinity.
- Theorem: *A given configuration of P components with q_i bodies in each component is Configuration Hill Stable if*

$$E < \sum_i^P \mathcal{U}_m(q_i)$$

$$\mathcal{U}_m(2) = -1 \quad \mathcal{U}_m(4) = -6$$

$$\mathcal{U}_m(3) = -3 \quad \mathcal{U}(5) \sim -10$$

- Proof: *Since the distance between bodies is bounded, there is a minimum gravitational potential energy, which restricts escape.*

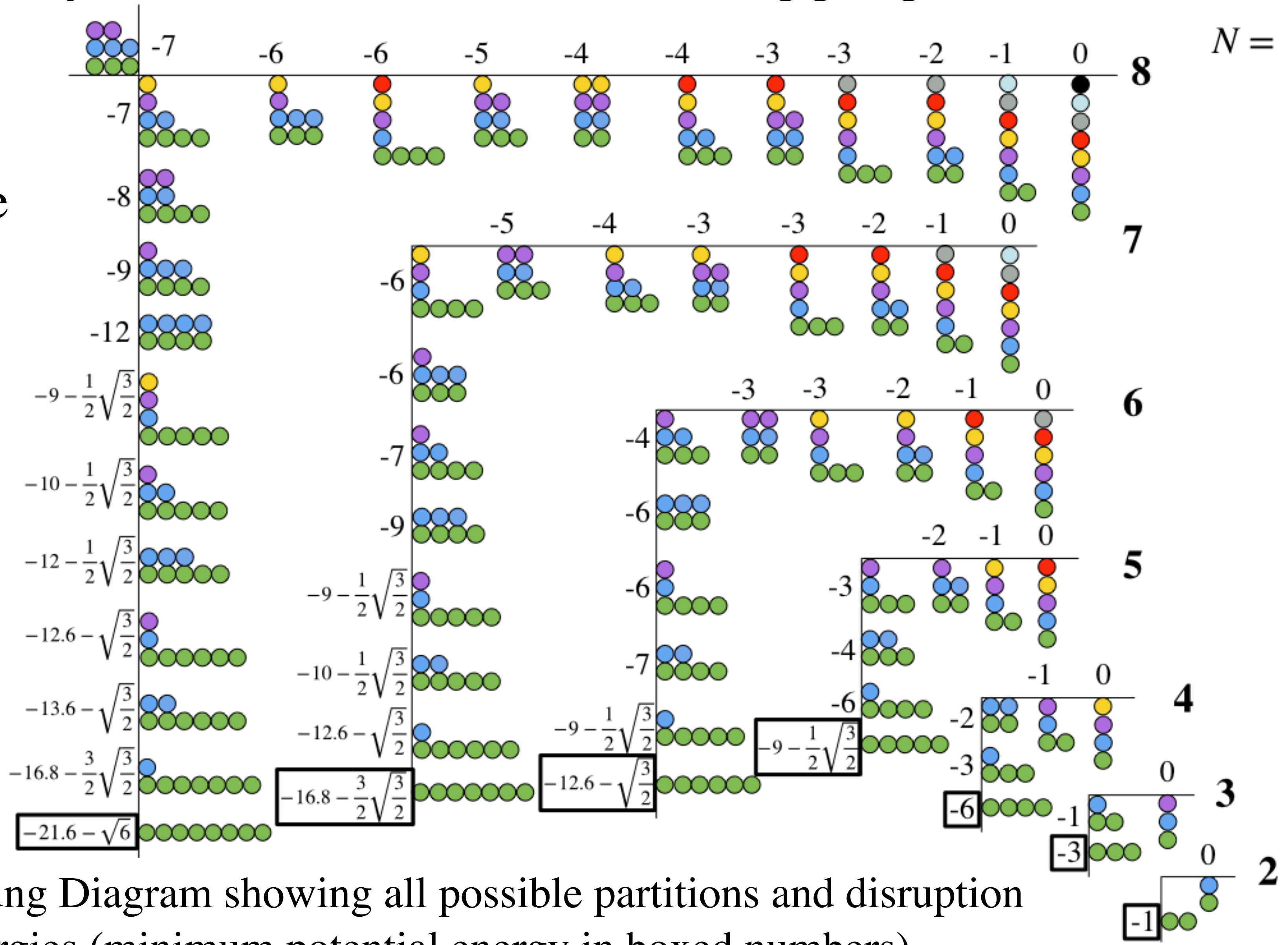


How many ways can a Rubble Pile Disaggregate?

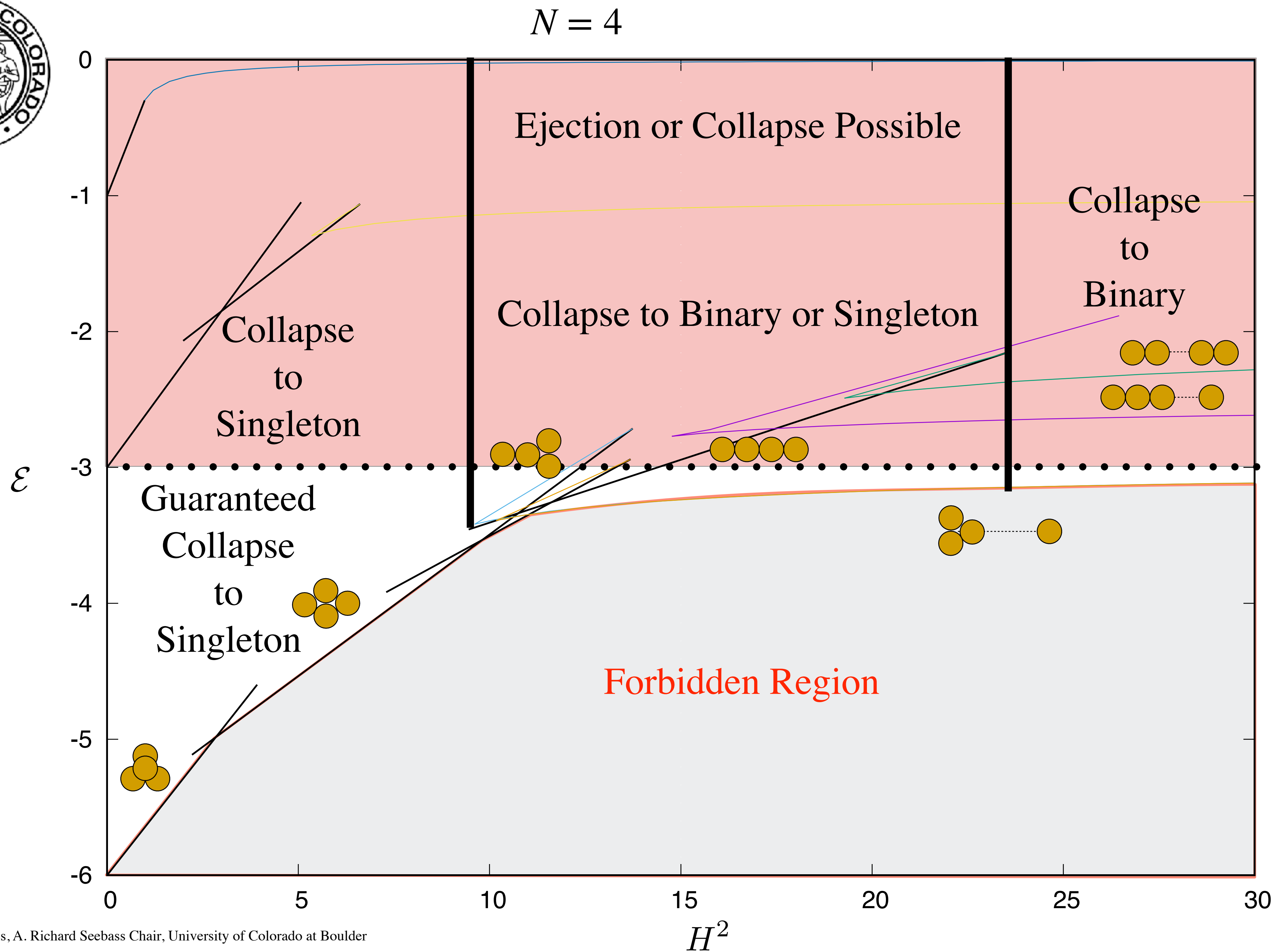


- For a rubble pile with N components... the number of ways it can disaggregate equals the number of “integer partitions” of N

N	Number of Partitions
2	2
3	3
4	5
5	7
6	11
7	15
8	22



Young Diagram showing all possible partitions and disruption energies (minimum potential energy in boxed numbers)



“Collapse” involves loss of energy through internal dissipation, while conserving angular momentum

“Ejection” involves loss of energy and angular momentum, due to the mutual E and AM being lost

$N = 5$

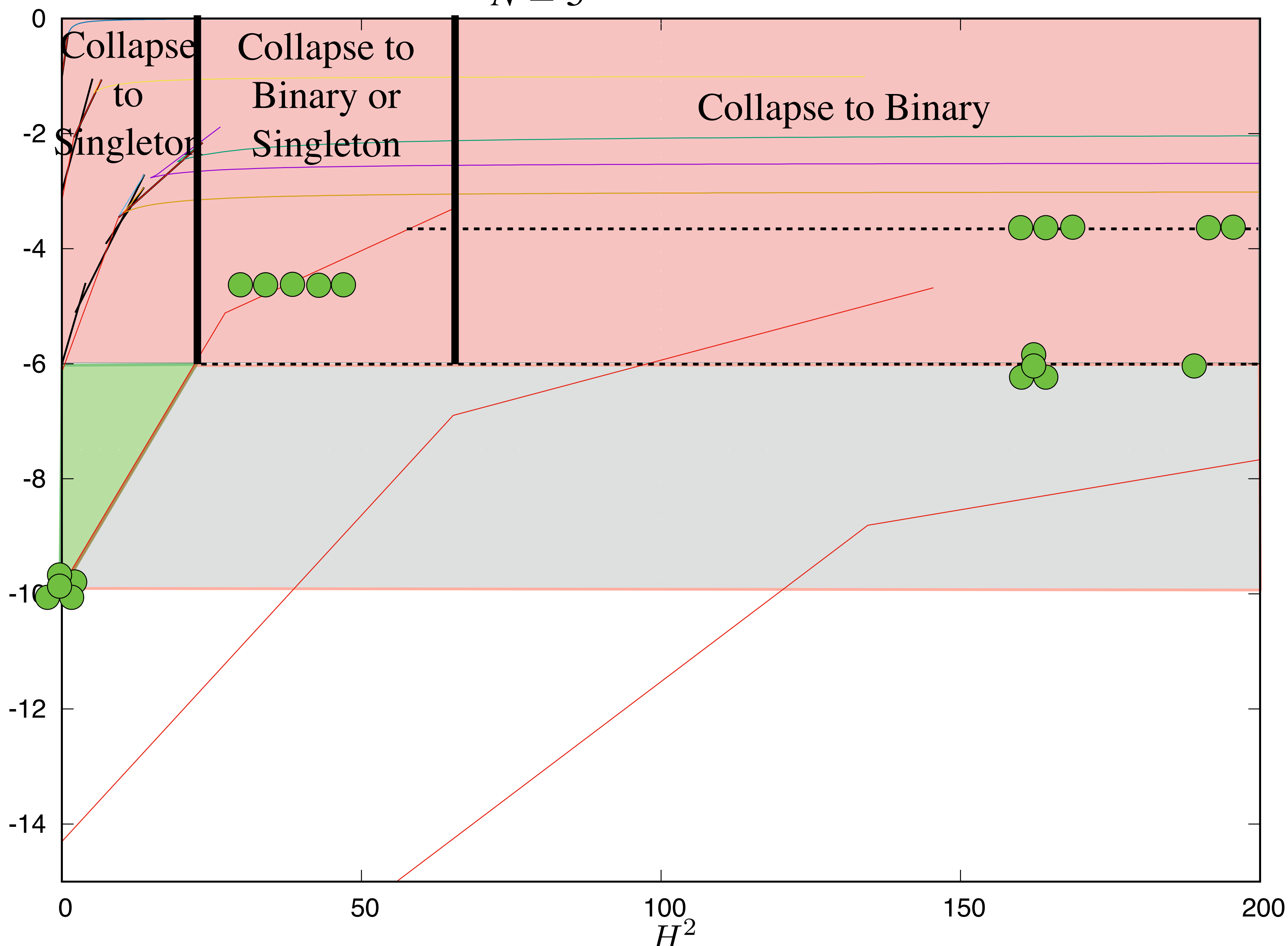
Ejection or Collapse

Guaranteed Collapse

Forbidden Region

For $N > 4$ we
can define
approximate
limits for these
different
regimes...

$\mathcal{E}$



$N = 6$

Ejection or Collapse

Guaranteed Collapse

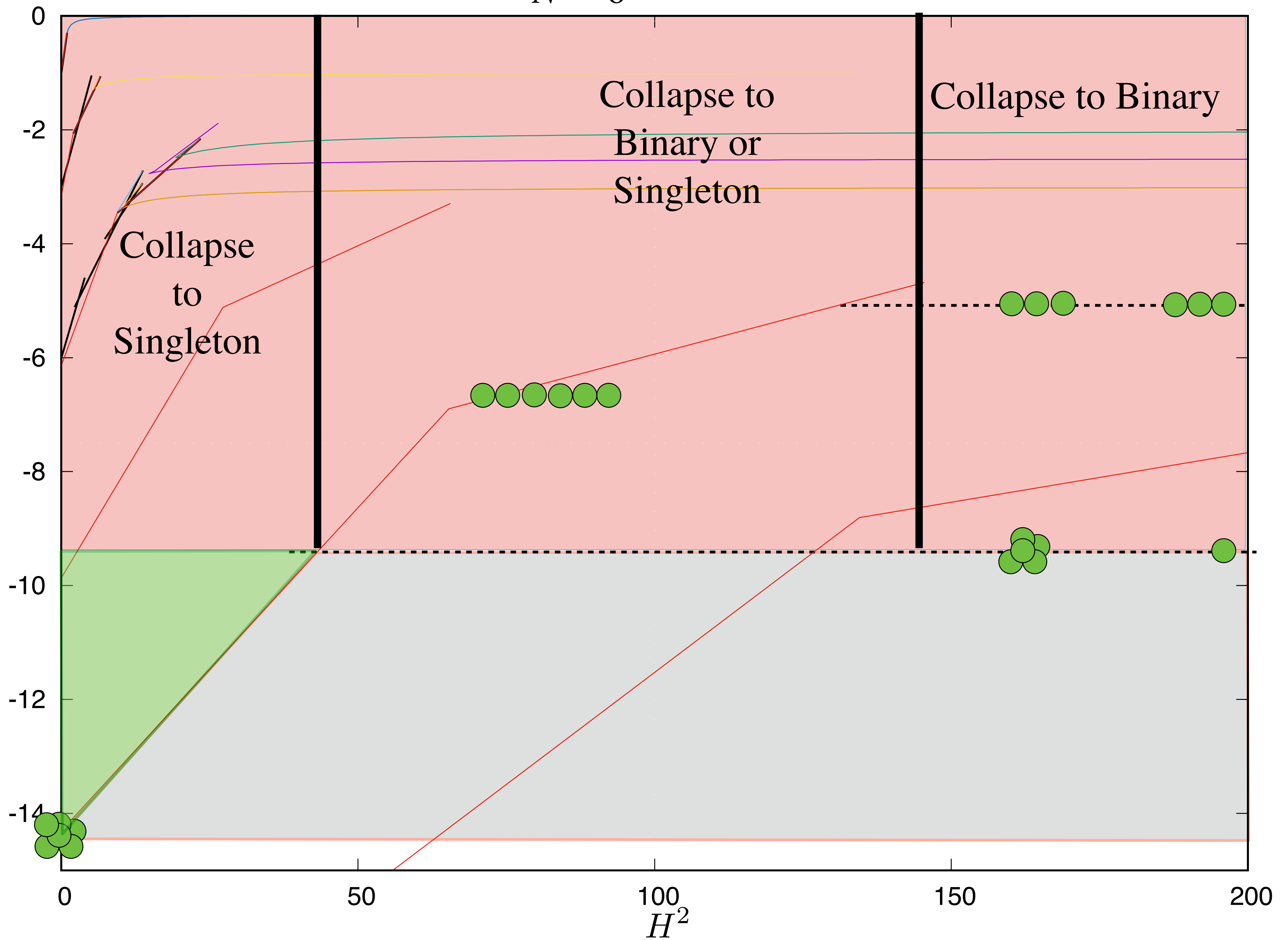
Forbidden Region

$\mathcal{E}$

Collapse
to
Singleton

Collapse to
Binary or
Singleton

Collapse to Binary

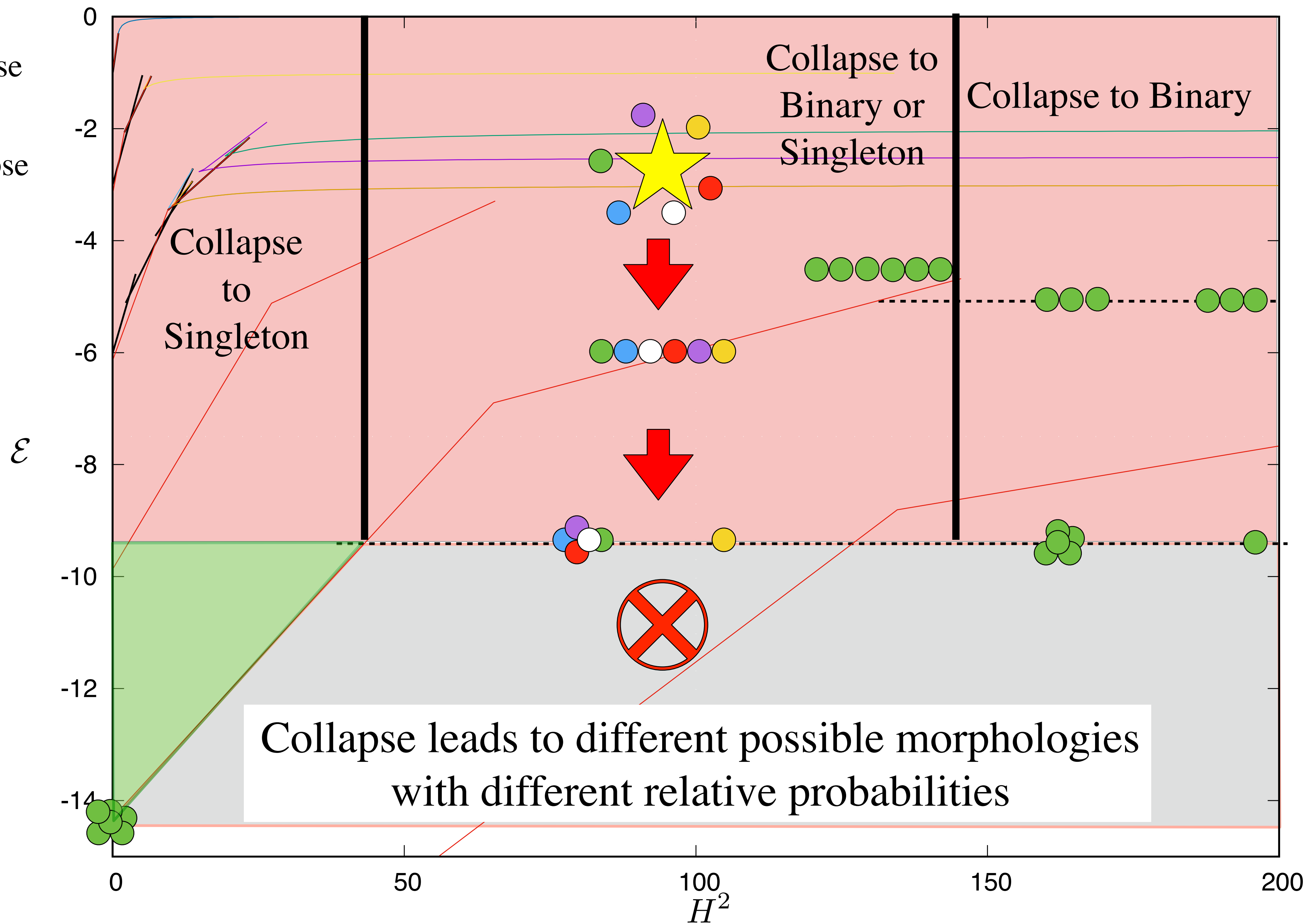


$N = 6$

Ejection or Collapse

Guaranteed Collapse

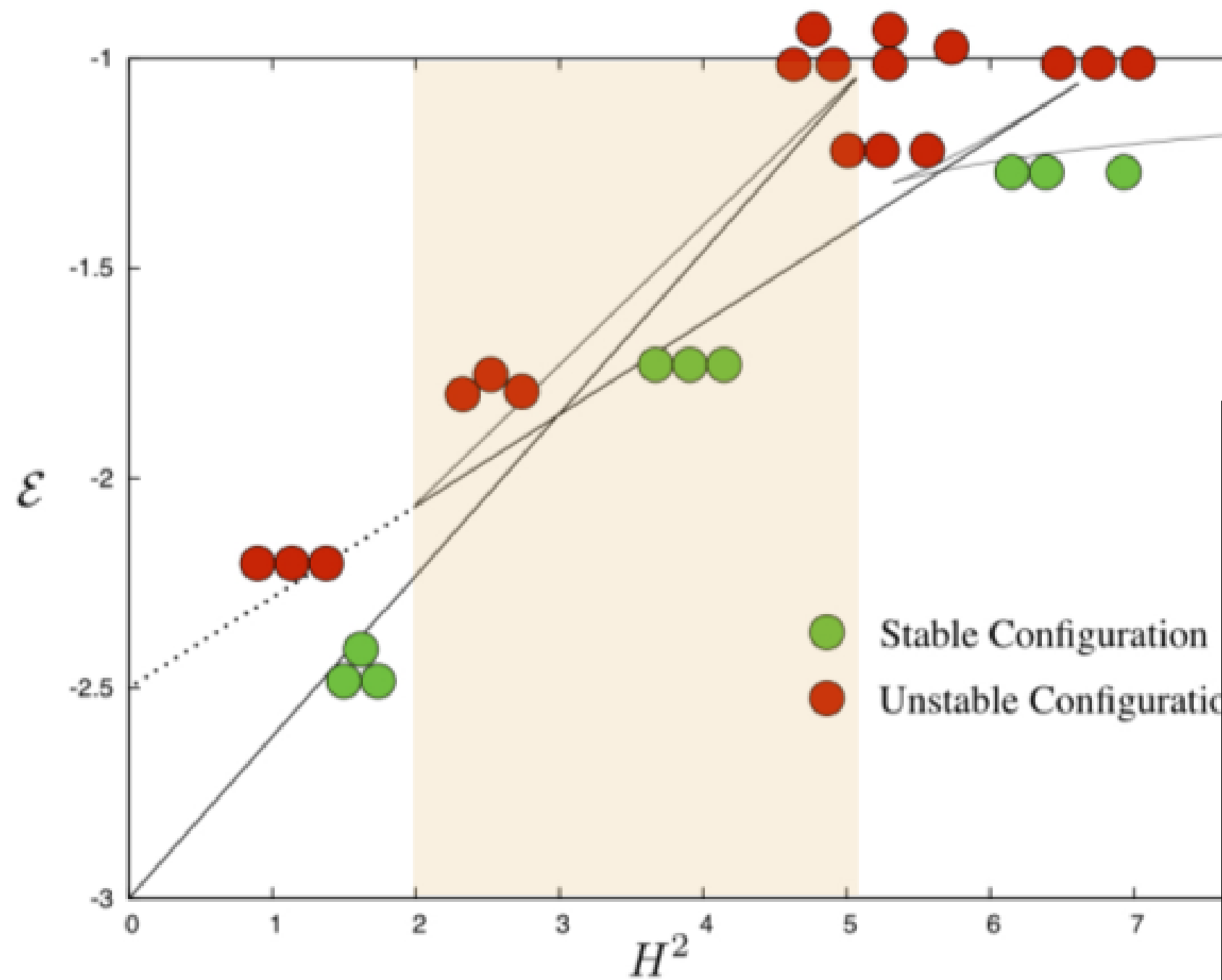
Forbidden Region



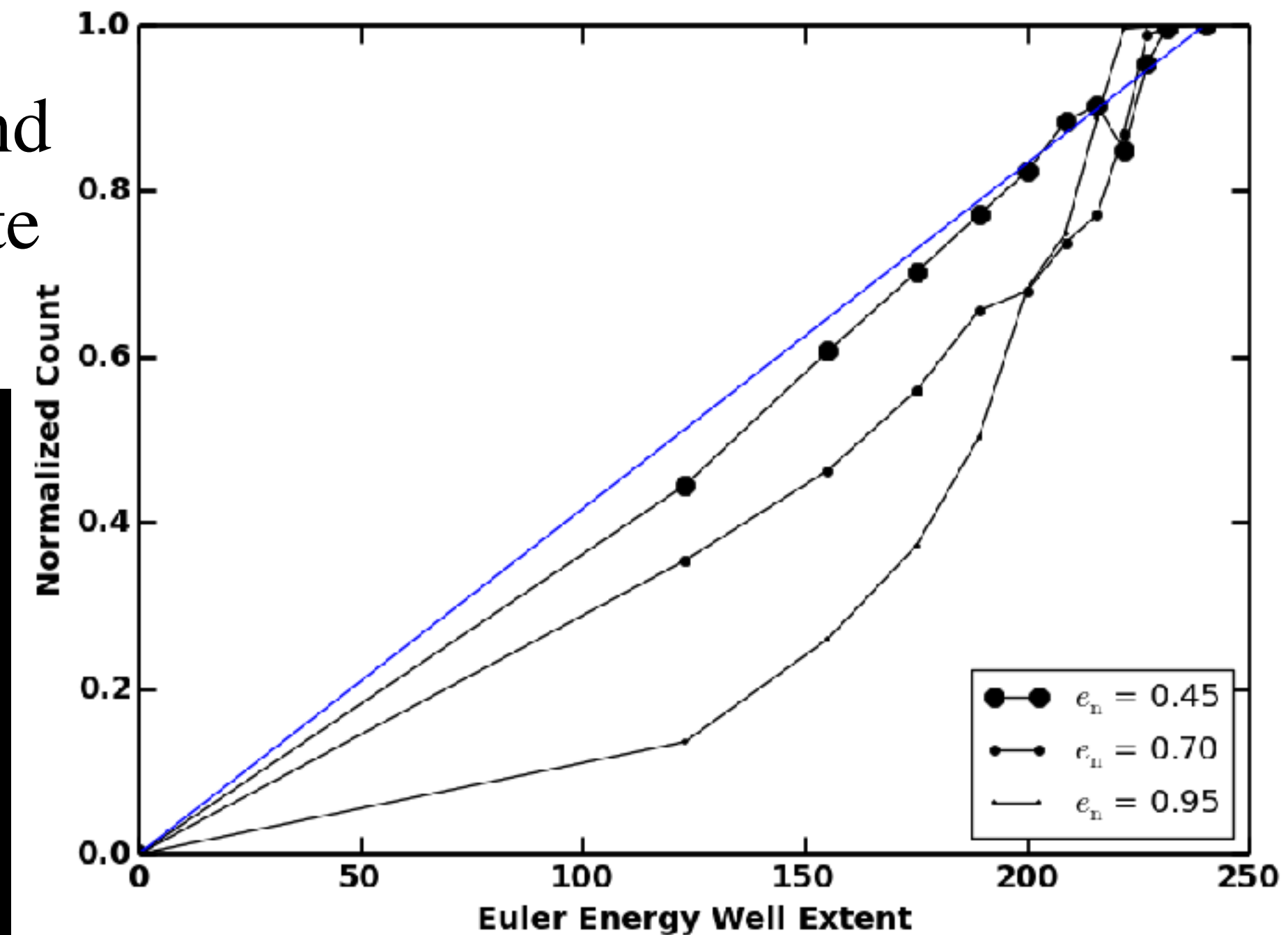
Outcome of Energy Collapse



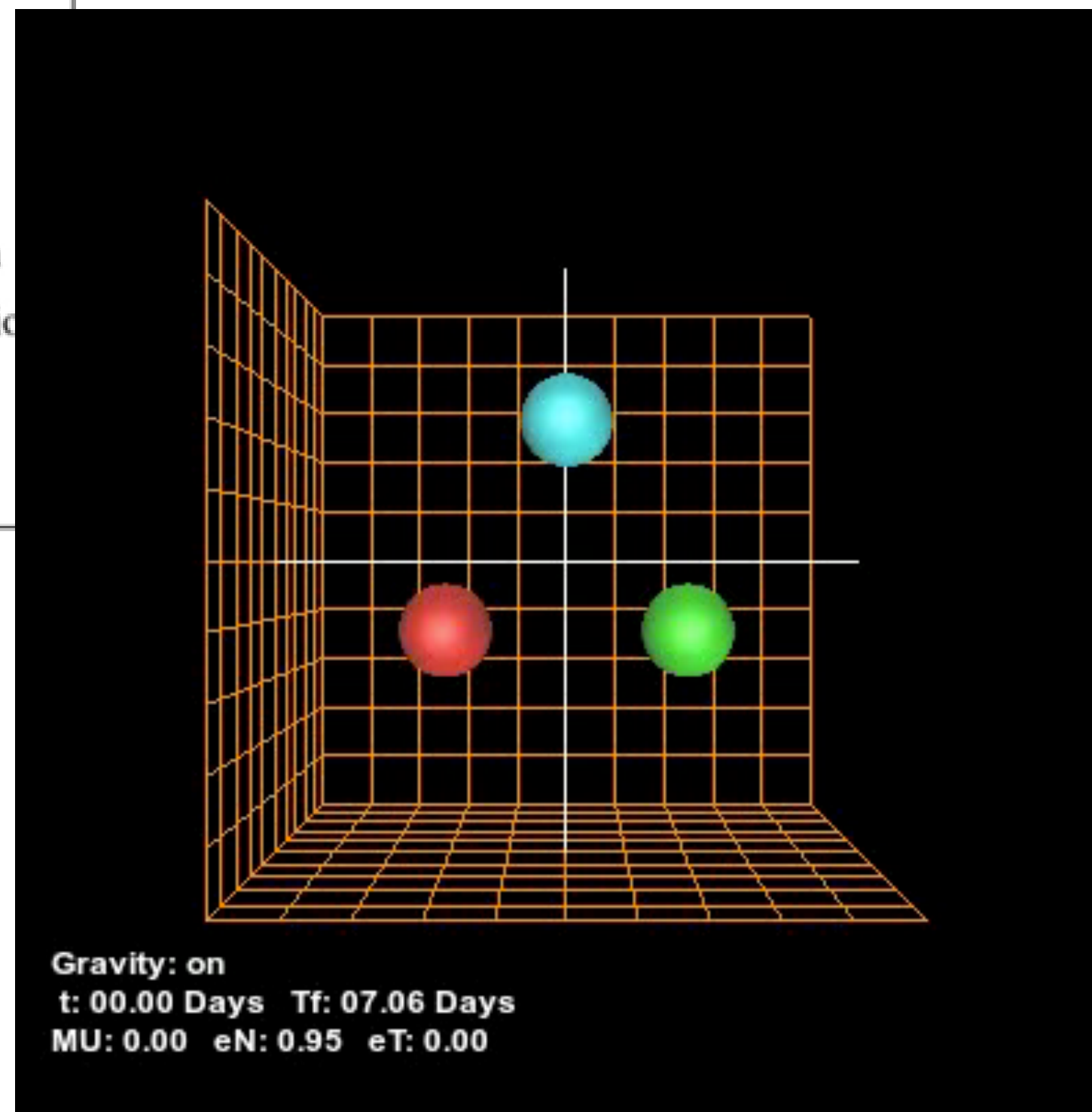
- Results are stochastic but predictable, from a thermodynamics perspective
 - Probability of final state depends on rate of energy dissipation



Fraction that end
in an Euler State



Outcome of a collapse where
either Lagrange or Euler resting
configurations can exist



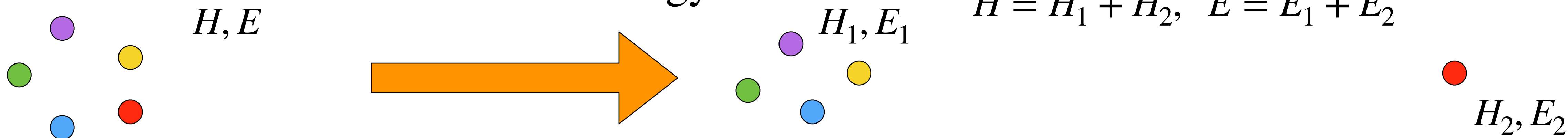
Total Angular Momentum Measure



Dealing with Escape

- The amended potential function can be used to determine the minimum energy needed for a component of the system to escape, however...
- ... once a component escapes the function has limited value and “loses track” of the remaining system energy and angular momentum:
 - $I_H \rightarrow \infty$ when *any* mass of the system escapes
 - This causes $\mathcal{E} = \frac{H^2}{2I_H} + \mathcal{U} \rightarrow \mathcal{U}' =$ the PE of the non-mutually escaped components
 - The minimum energy function loses all information about the total angular momentum, and makes it essentially useless for further computation
- Here we present a simple theorem that allows us to rigorously and recursively continue to use the minimum energy function

$$H = H_1 + H_2, \quad E = E_1 + E_2$$





Background: Jacobi Formulation

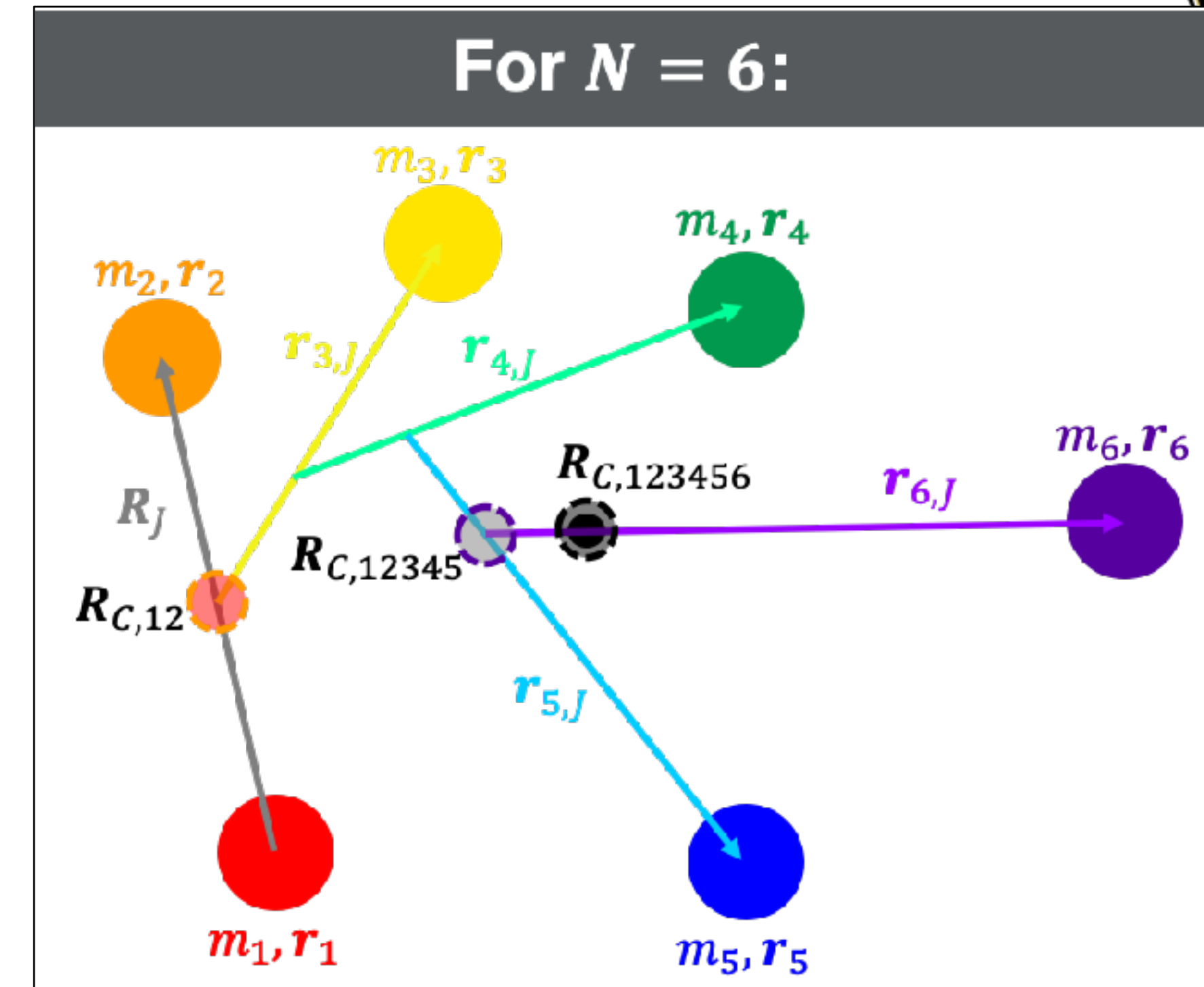


- Given: $i = 1, 2, \dots, N + 1$ bodies with mass, position and velocity: $m_i, \mathbf{r}_i, \mathbf{v}_i$
- The Jacobi formulation defines:
 - Cumulative mass M_i :

$$M_0 = m_1, M_1 = M_0 + m_2, \dots, M_i = M_{i-1} + m_{i+1}, \dots, M_N = M_{N-1} + m_{N+1}$$
 - Cumulative system centers of mass $\mathbf{R}_i^C$:

$$\mathbf{R}_0^C = \mathbf{r}_1, \mathbf{R}_1^C = \frac{1}{M_1} [M_0 \mathbf{R}_0^C + m_2 \mathbf{r}_2], \dots, \mathbf{R}_i^C = \frac{1}{M_i} [M_{i-1} \mathbf{R}_{i-1}^C + m_{i+1} \mathbf{r}_{i+1}], \dots, \mathbf{R}_N^C = \frac{1}{M_N} [M_{N-1} \mathbf{R}_{N-1}^C + m_{N+1} \mathbf{r}_{N+1}]$$
 - Cumulative new relative states R_i, V_i :

$$R_1 = \mathbf{r}_2 - \mathbf{R}_0^C, R_2 = \mathbf{r}_3 - \mathbf{R}_1^C, \dots, R_i = \mathbf{r}_{i+1} - \mathbf{R}_{i-1}^C, \dots, R_N = \mathbf{r}_{N+1} - \mathbf{R}_{N-1}^C$$
, same for velocities
- Leads to the cumulative Angular Momentum and Energy, where each component i only involves the states of bodies $j \leq i + 1$:



$$\mathbf{H} = \sum_{i=1}^N H_i \quad : \quad H_i = \frac{M_{i-1} m_{i+1}}{M_i} R_i \times V_i$$

$$E = \sum_{i=1}^N E_i \quad : \quad E_i = \frac{1}{2} \frac{M_{i-1} m_{i+1}}{M_i} (V_i \cdot V_i) + \sum_{j=1}^i \mathcal{U}_{j,i+1}$$



Theorem: Rigorous Limits on the Disaggregated Minimum Energy Function



- Theorem:

$$\mathcal{E} = \frac{H^2}{2I_H} + \mathcal{U} \leq \sum_{i=1}^N \left[\frac{H_i^2}{2I_i} + \sum_{j=1}^i \mathcal{U}_{j,i+1} \right] \leq \sum_{i=1}^N E_i = E = T + \mathcal{U}$$

- Proof:

– Note that $\mathcal{U} = \sum_{i=1}^N \sum_{j=1}^i \mathcal{U}_{j,i+1}$, $H = \sum_{i=1}^N H_i$, $E = \sum_{i=1}^N E_i$, $I_H = \sum_{i=1}^N I_i$

– The basic item to prove is just a matter of algebra:

$$\frac{\left(\sum_{i=1}^N H_i \right)^2}{\sum_{i=1}^N I_i} \leq \sum_{i=1}^N \frac{H_i^2}{I_i} \Rightarrow \left(\sum_{i=1}^N H_i \right)^2 \leq \left(\sum_{j=1}^N I_j \right) \sum_{i=1}^N \frac{H_i^2}{I_i}$$



Escape using the Jacobi Formulation

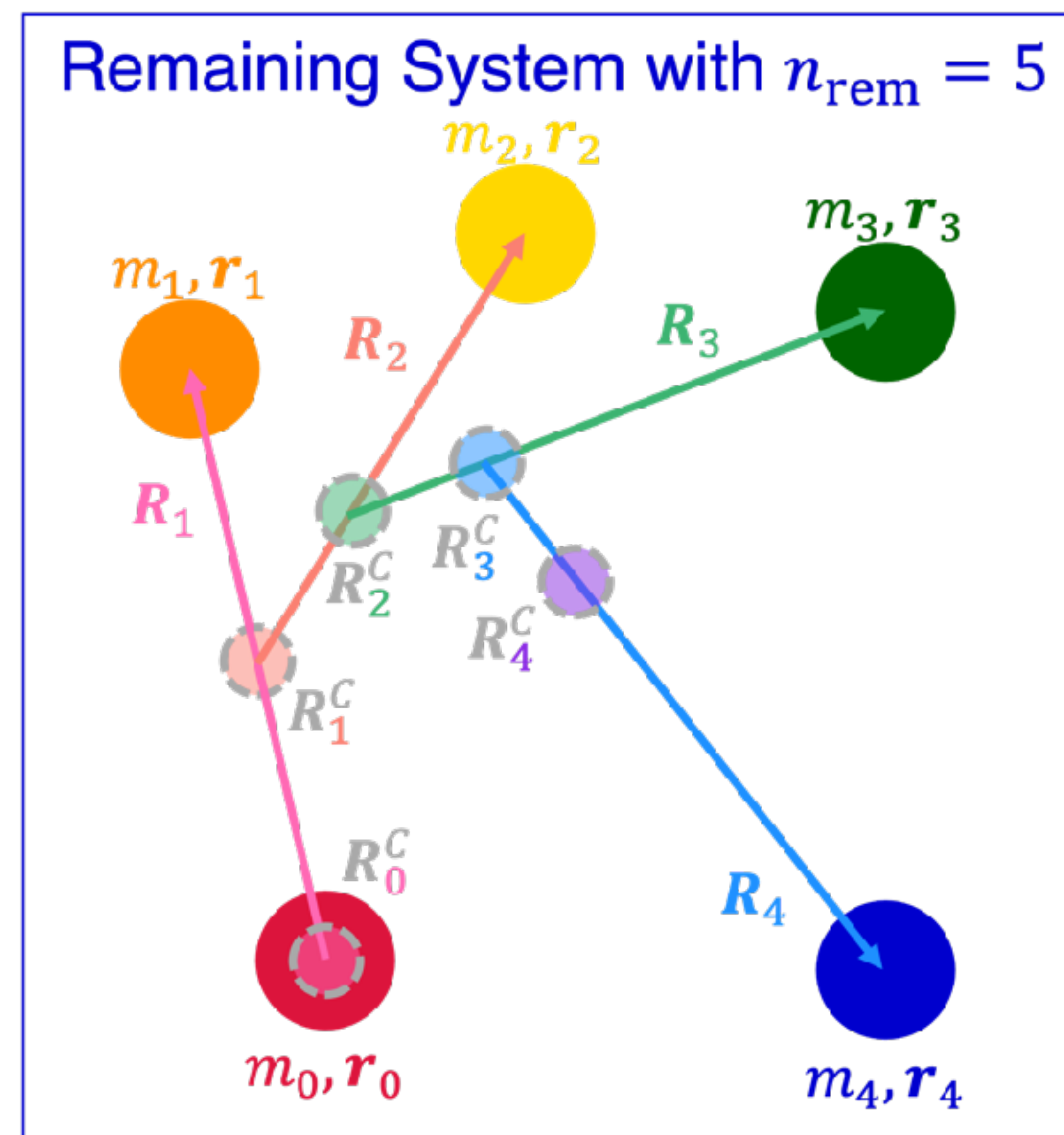
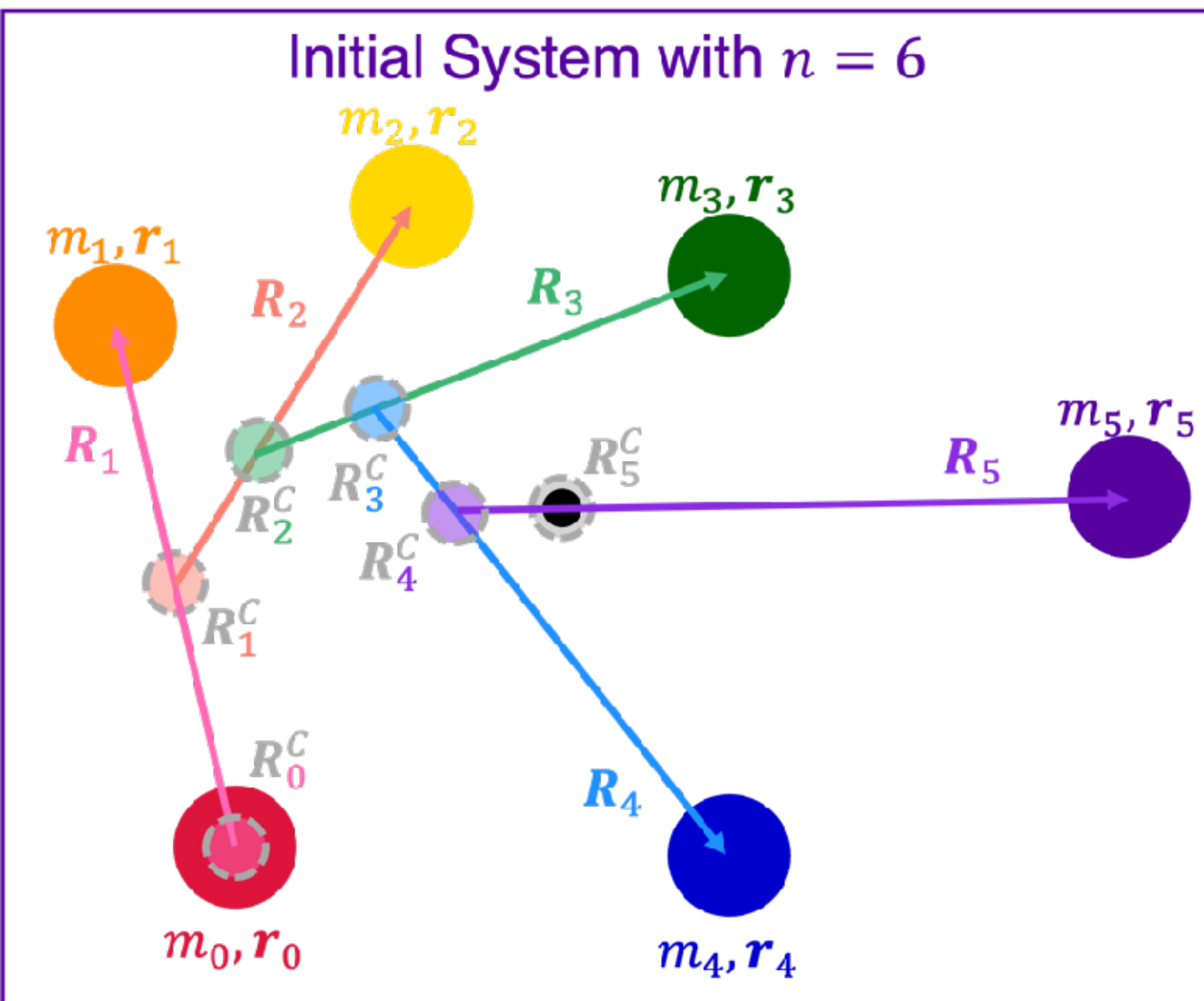
- If a component escapes, the system can always be expressed, using the Jacobi formulation, such that the N th body is the escaping body, and it does not interfere with any of the $i < N$ components.
- If the N th component is the escaping one, we find:

$$H_N \rightarrow \frac{M_{N-1}m_{N+1}}{M_N} V_\infty b_\infty \qquad E_N \rightarrow \frac{M_{N-1}m_{N+1}}{M_N} \frac{1}{2} V_\infty^2$$

- We can rewrite the minimum energy function in terms of these cumulative systems.

$$\mathcal{E} \leq \sum_{i=1}^N \mathcal{E}_i \leq \sum_{i=1}^N E_i = E \text{ where } \mathcal{E}_i = \frac{H_i^2}{2I_i} + \sum_{j=1}^i \mathcal{U}_{j,i+1} \leq E_i$$

- This allows the escaping bodies to be naturally decoupled from the rest of the system



Ejected: P_5
 E_C and H_C
 $h_C \rightarrow v_{\infty, C}$ and $b_{\infty, C}$
 $E_{C(2)} \rightarrow a_C$ and e_C





Application...

- Given an $N + 1$ component system, with its amended potential: $\mathcal{E} = \frac{H^2}{2I_H} + \mathcal{U}$
 - If a component escapes, re-order the Jacobi formulation so that $N + 1$ escapes and $R_N \rightarrow \infty$:

$$\mathcal{U} - \sum_{j=1}^N \mathcal{U}_{j,N+1} \quad \mathcal{E} \leq \mathcal{E}_{1 \rightarrow N-1} + \frac{H_N^2}{2I_N} + \sum_{j=1}^N \mathcal{U}_{j,N+1} \leq E_{1 \rightarrow N-1} + E_N = E$$

0

$$\mathcal{E}_{1 \rightarrow N-1} \leq E_{1 \rightarrow N-1} = E - \frac{1}{2} \frac{M_{N-1} m_{N+1}}{M_N} V_\infty^2$$

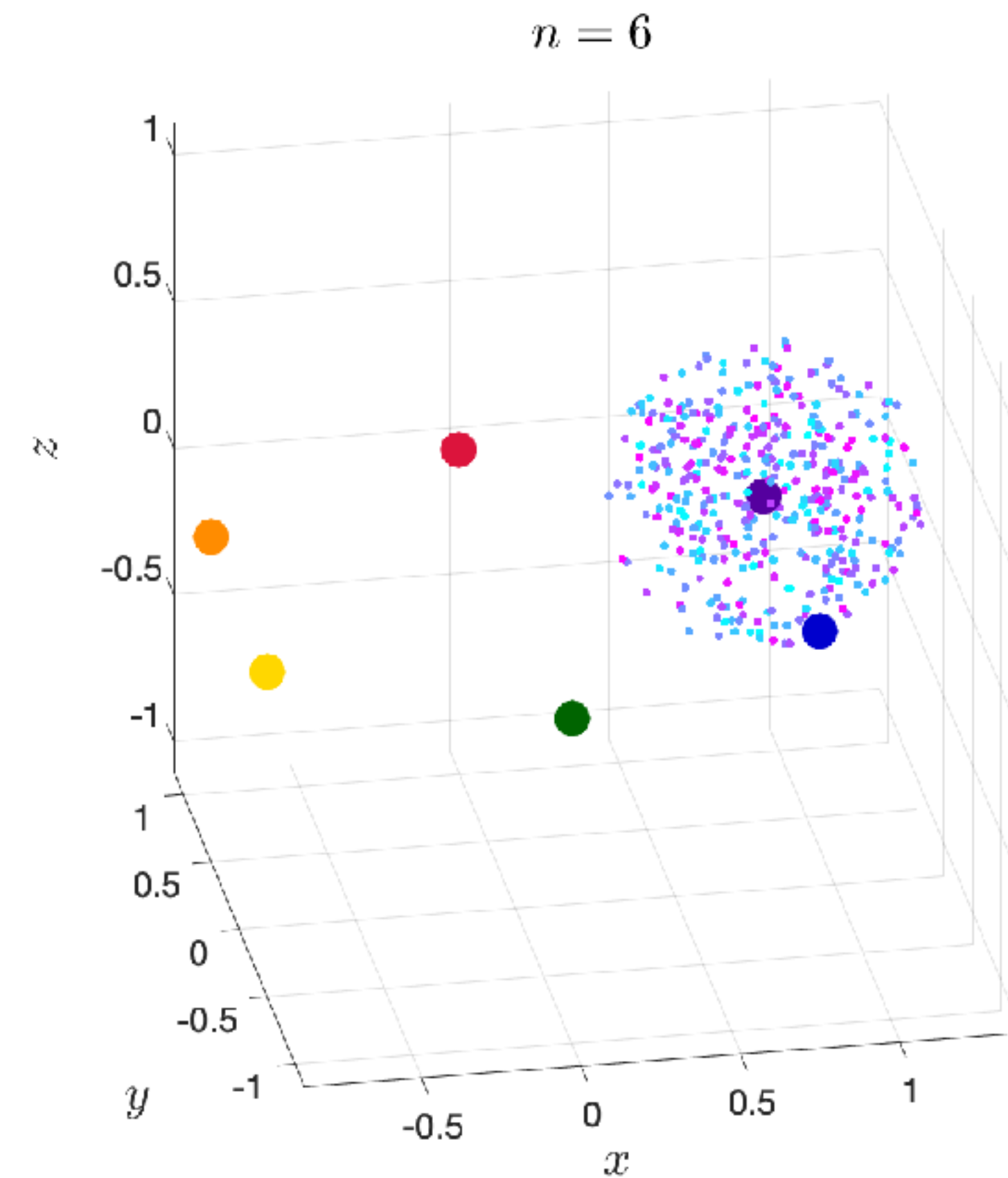
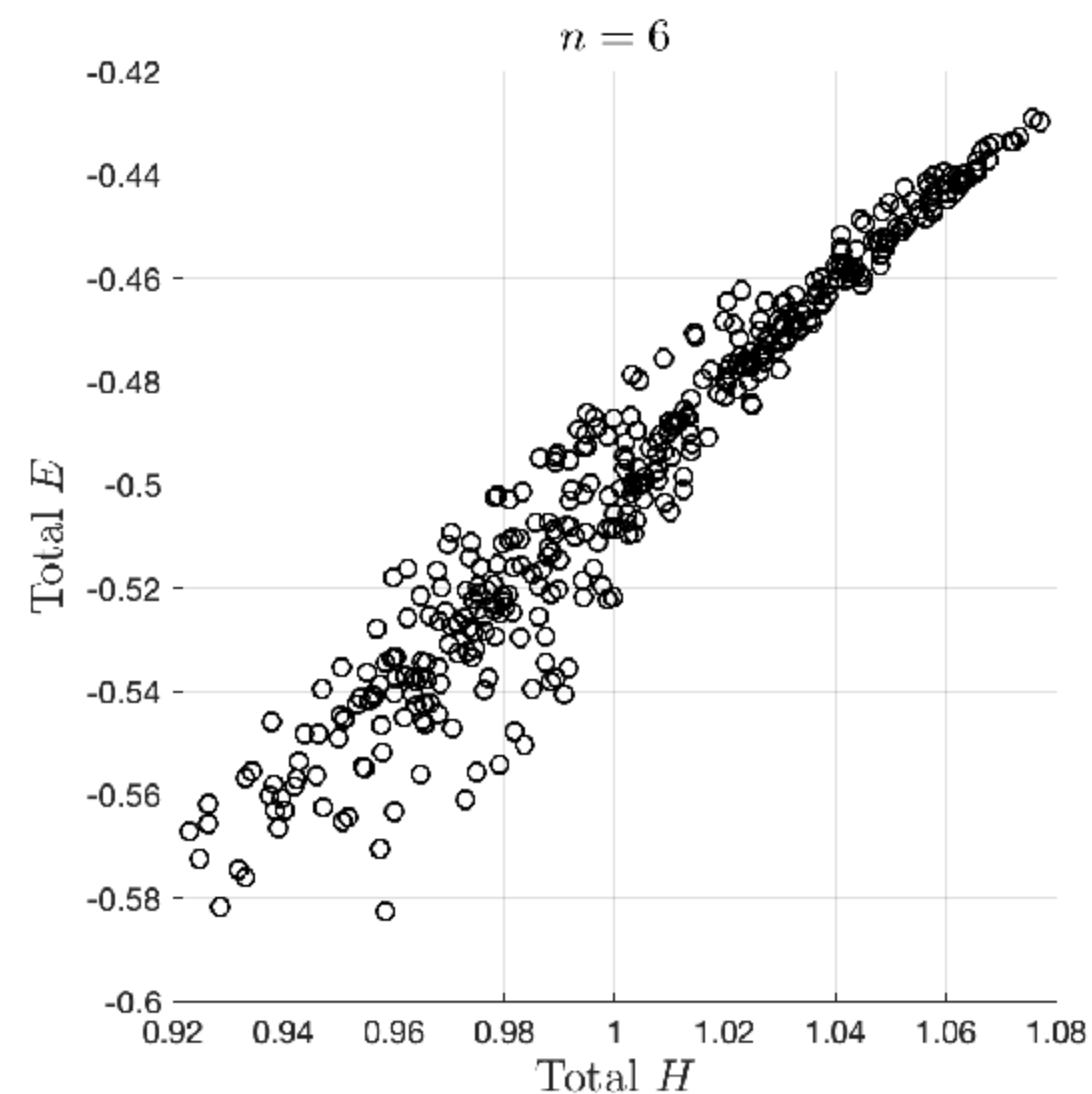
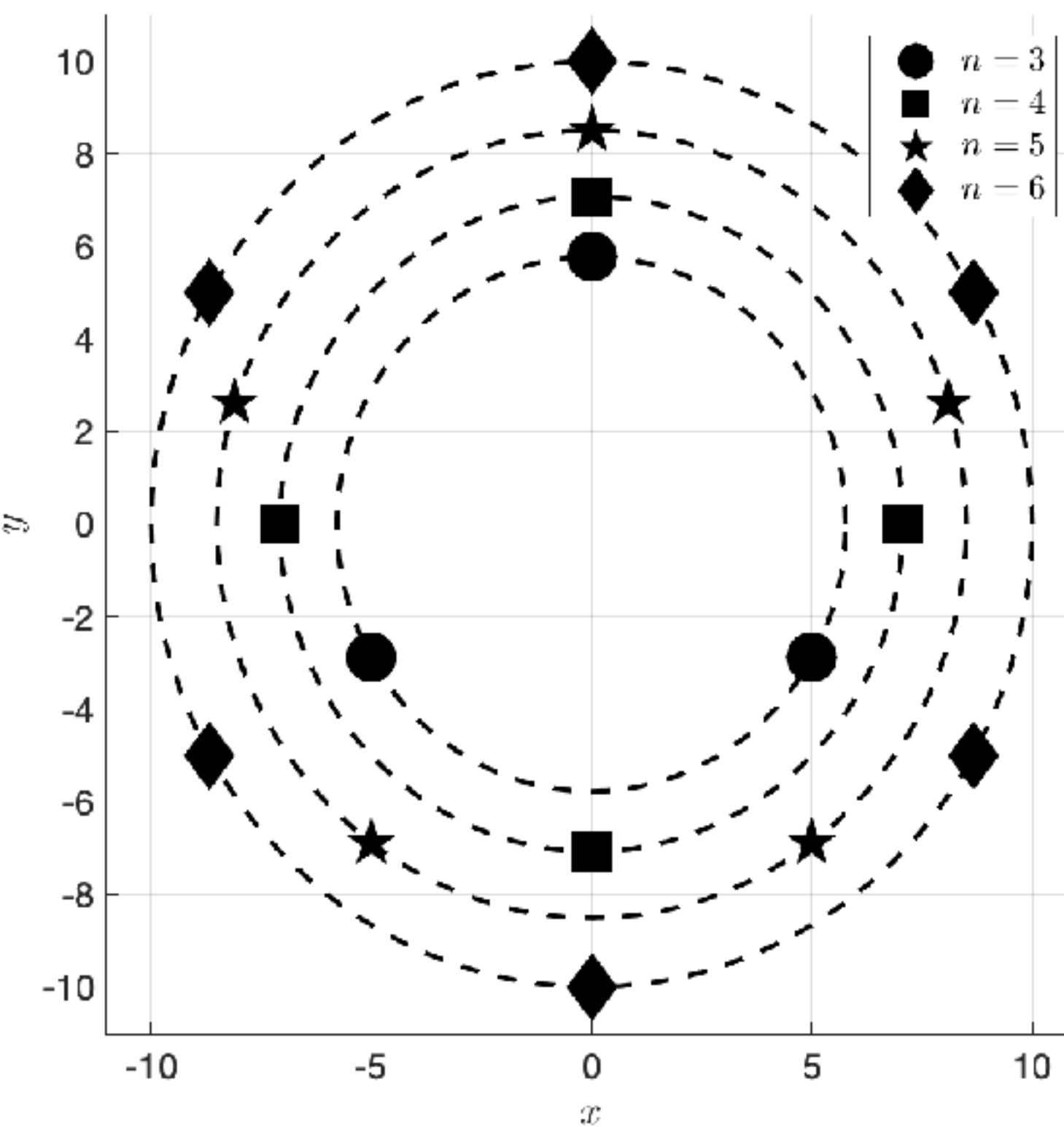
Lost Energy... and
Lost Angular
Momentum...

$$\frac{M_{N-1} m_{N+1}}{M_N} V_\infty b_\infty$$

How Do Systems Shed E & H ?



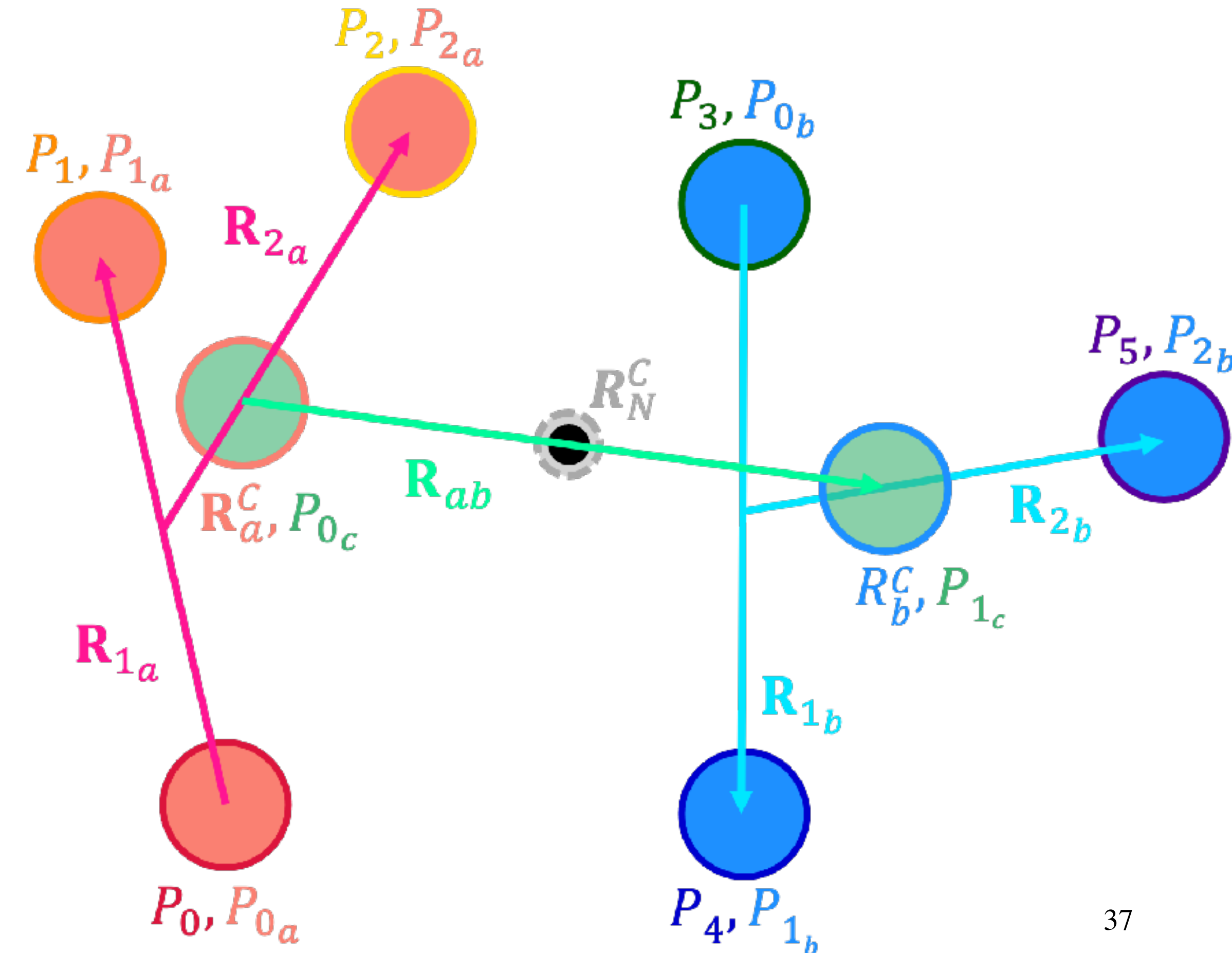
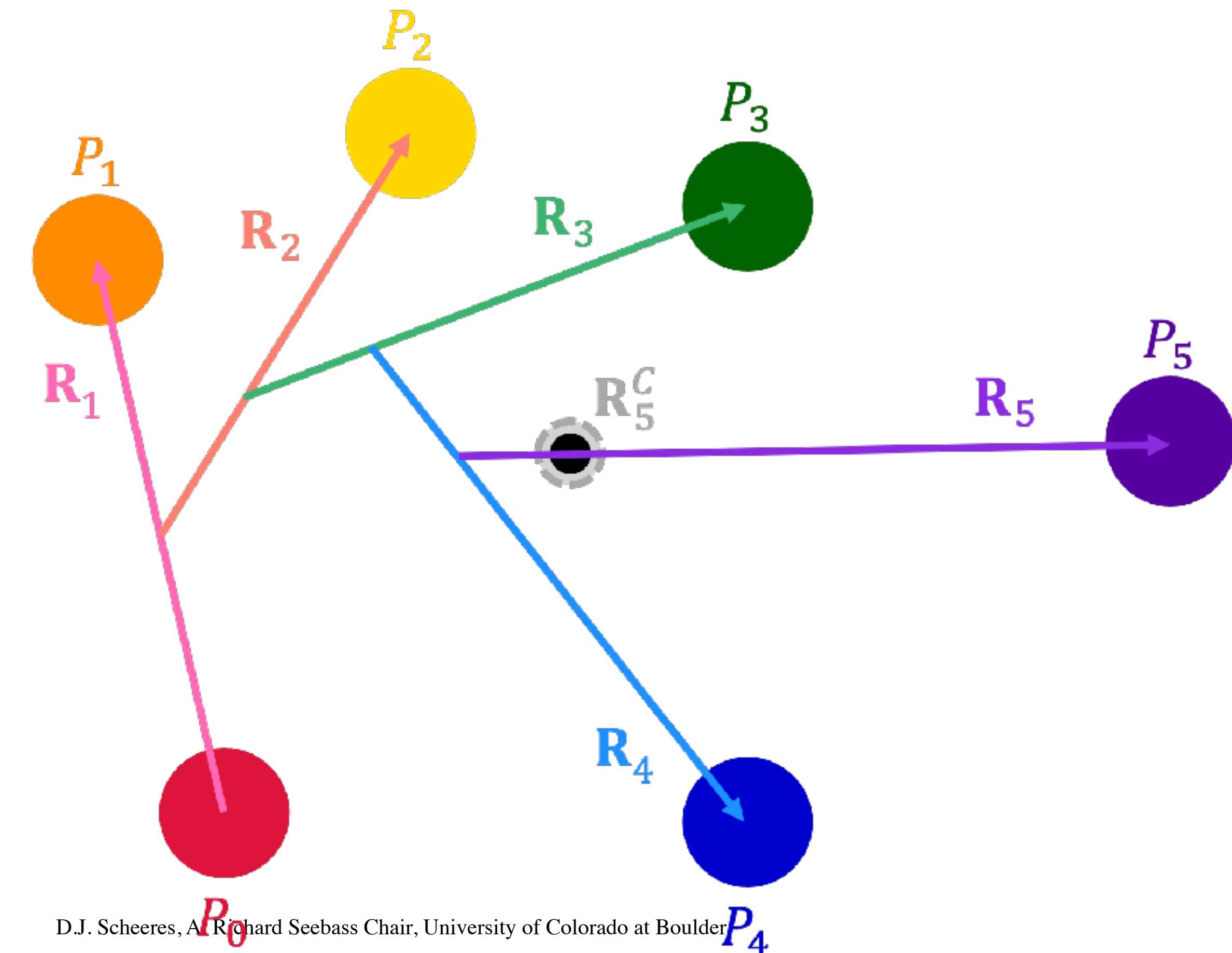
- To understand how escaping bodies remove energy and angular momentum, we carried out some specific studies using the point-mass N -body problem
 - Initial configurations of 3, 4, ..., 6 bodies started in a central, circular configuration with one body displaced randomly



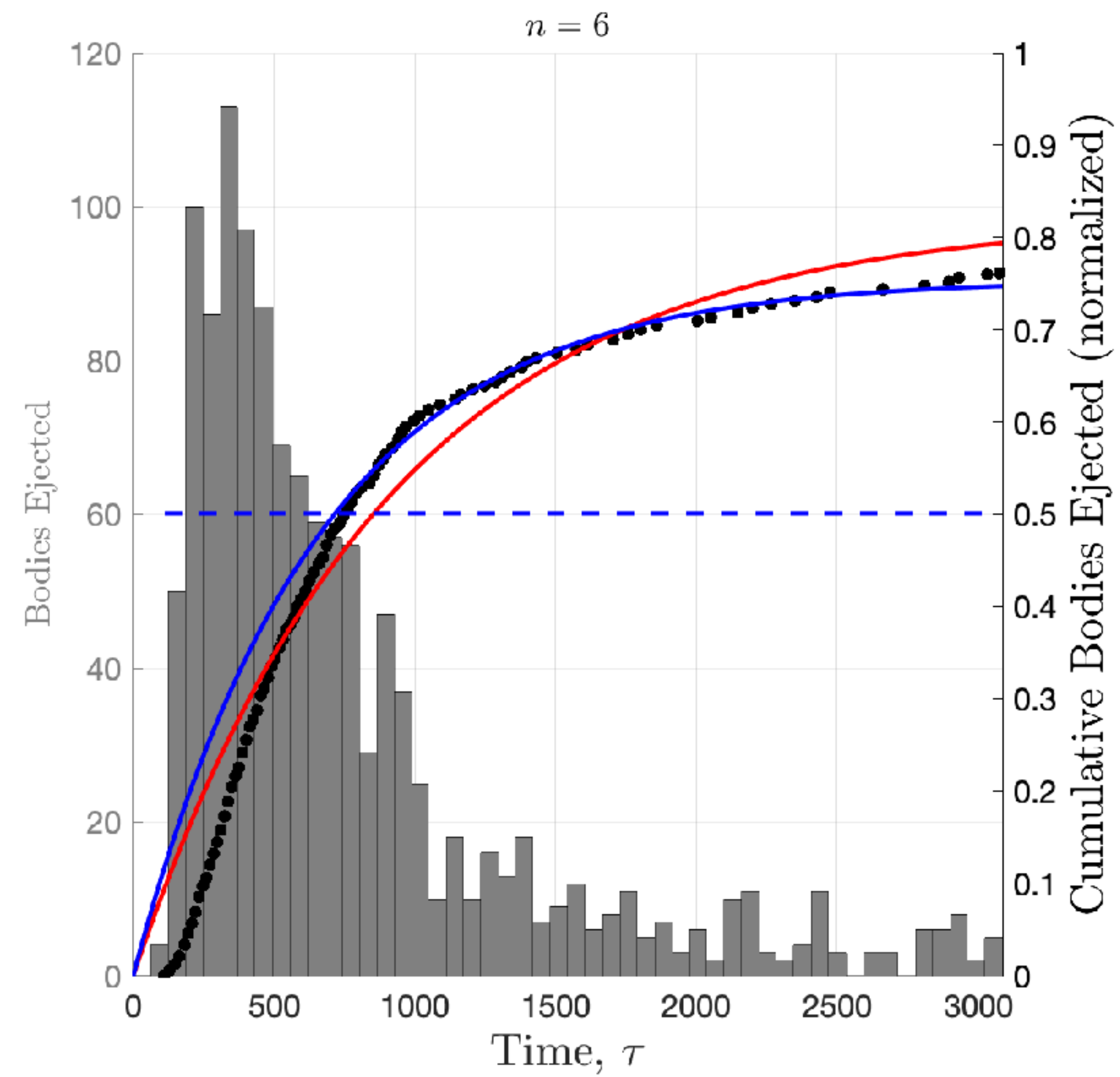
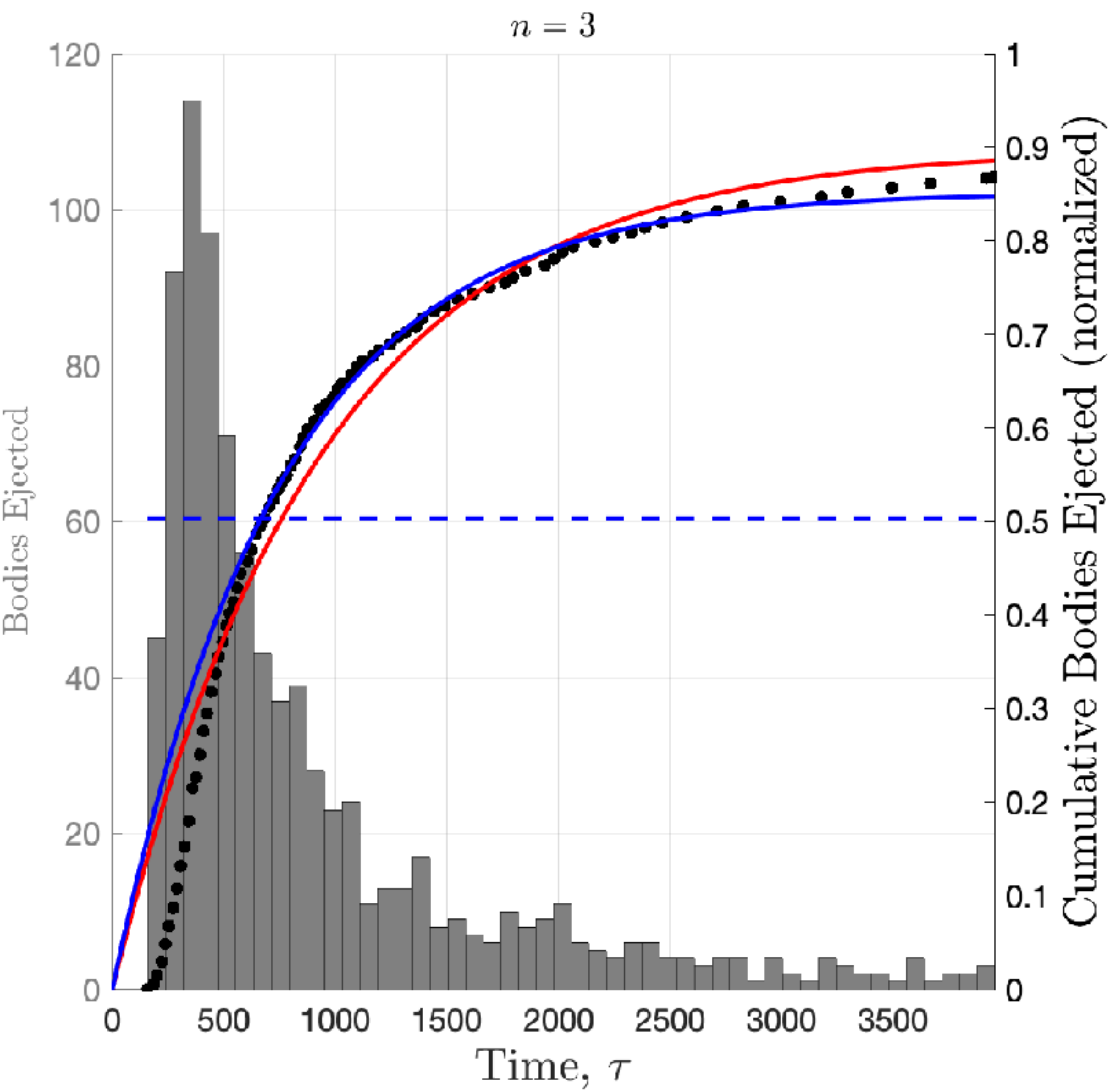
Case Study: $N = 6$



- We track the energy and angular momentum that a system loses due to ejection of single and multiple collections of bodies
 - Angular momentum and energy within an escaping cluster is not considered “lost”
 - For a non-spherical rigid body, individually escaping components will have their spin E and AM



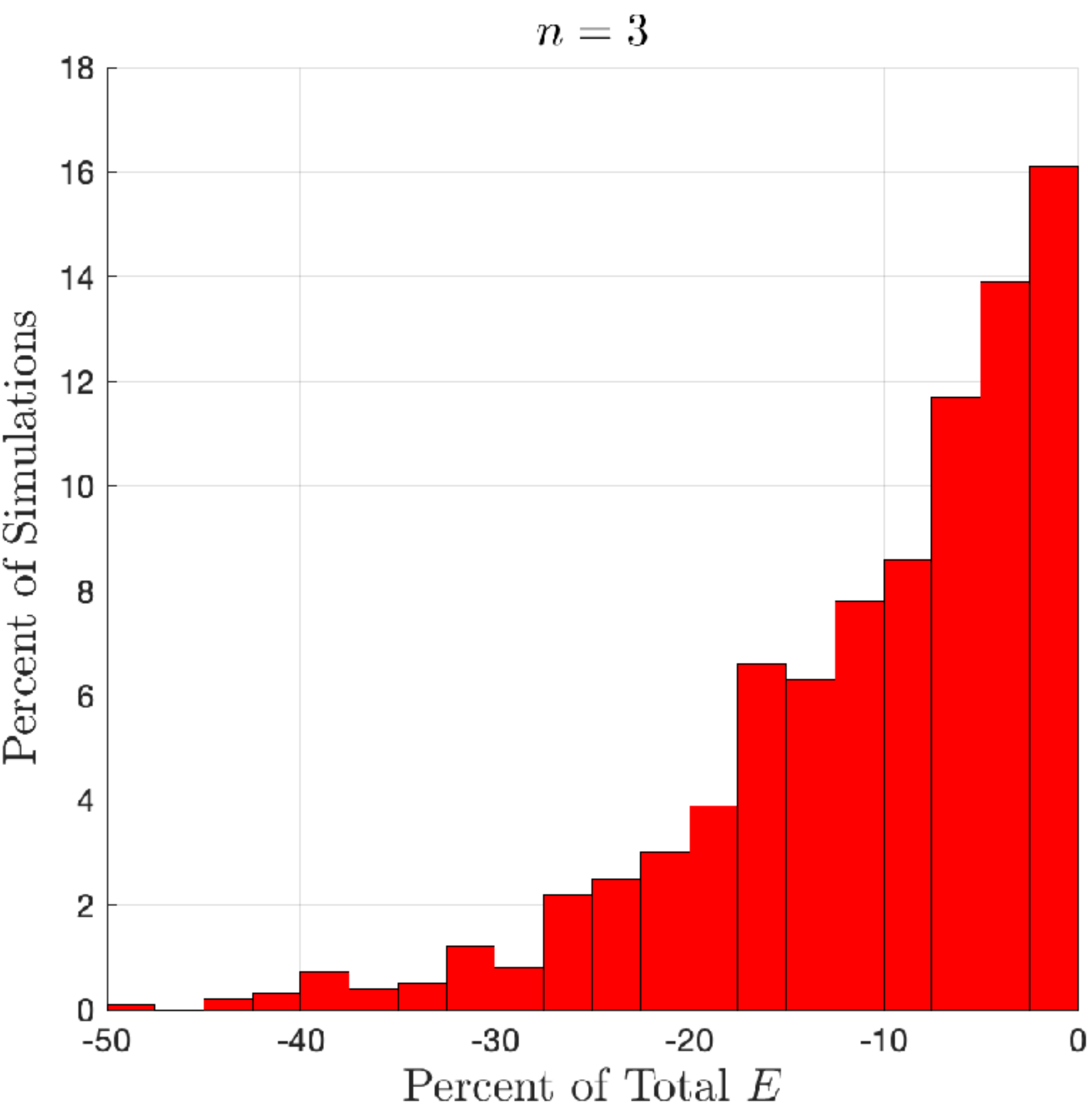
Computational Cost



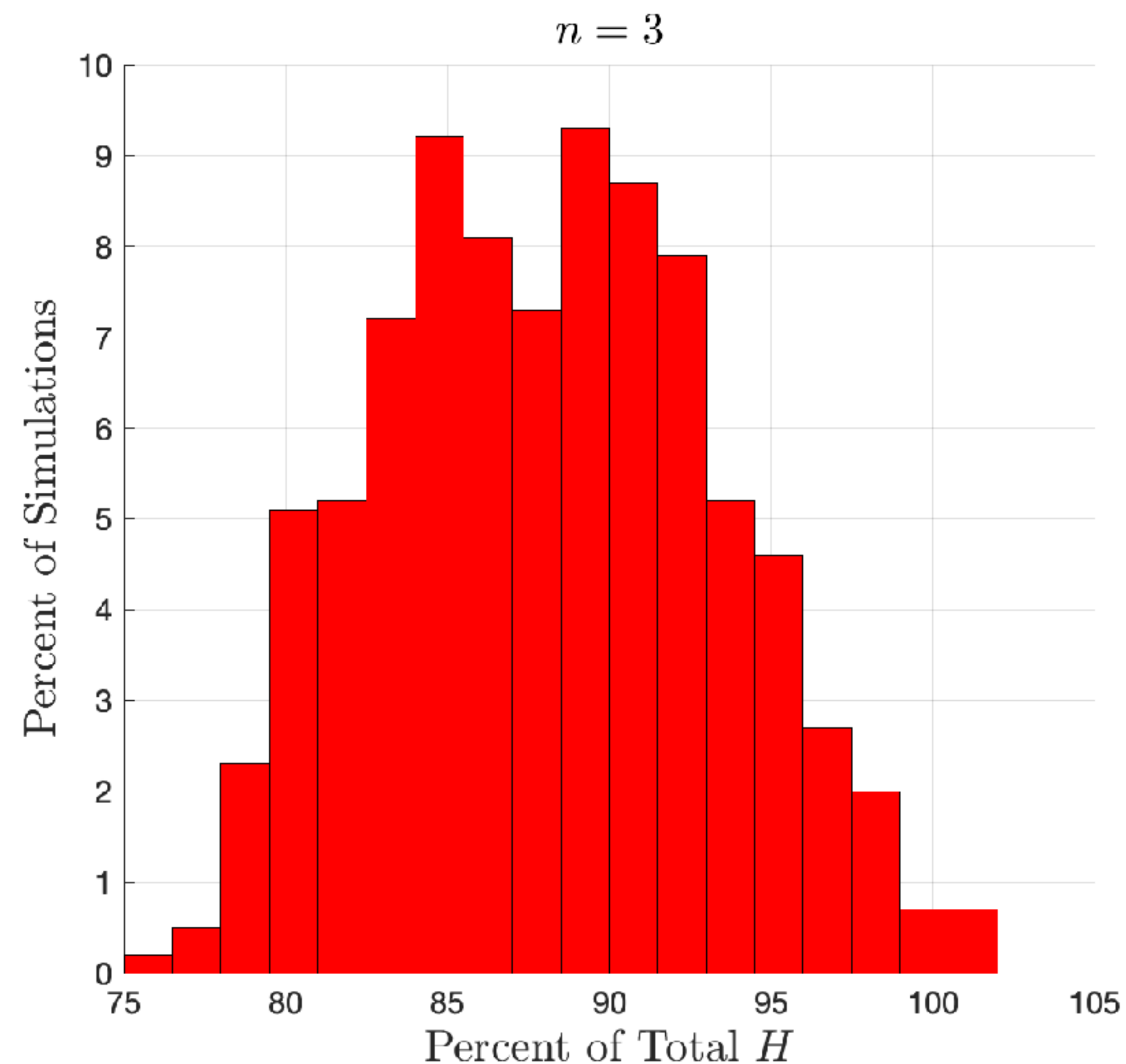
Energy and Angular Momentum Loss: $N = 3$



Energy is always lost due to escape



Lost angular momentum tends to be randomly distributed

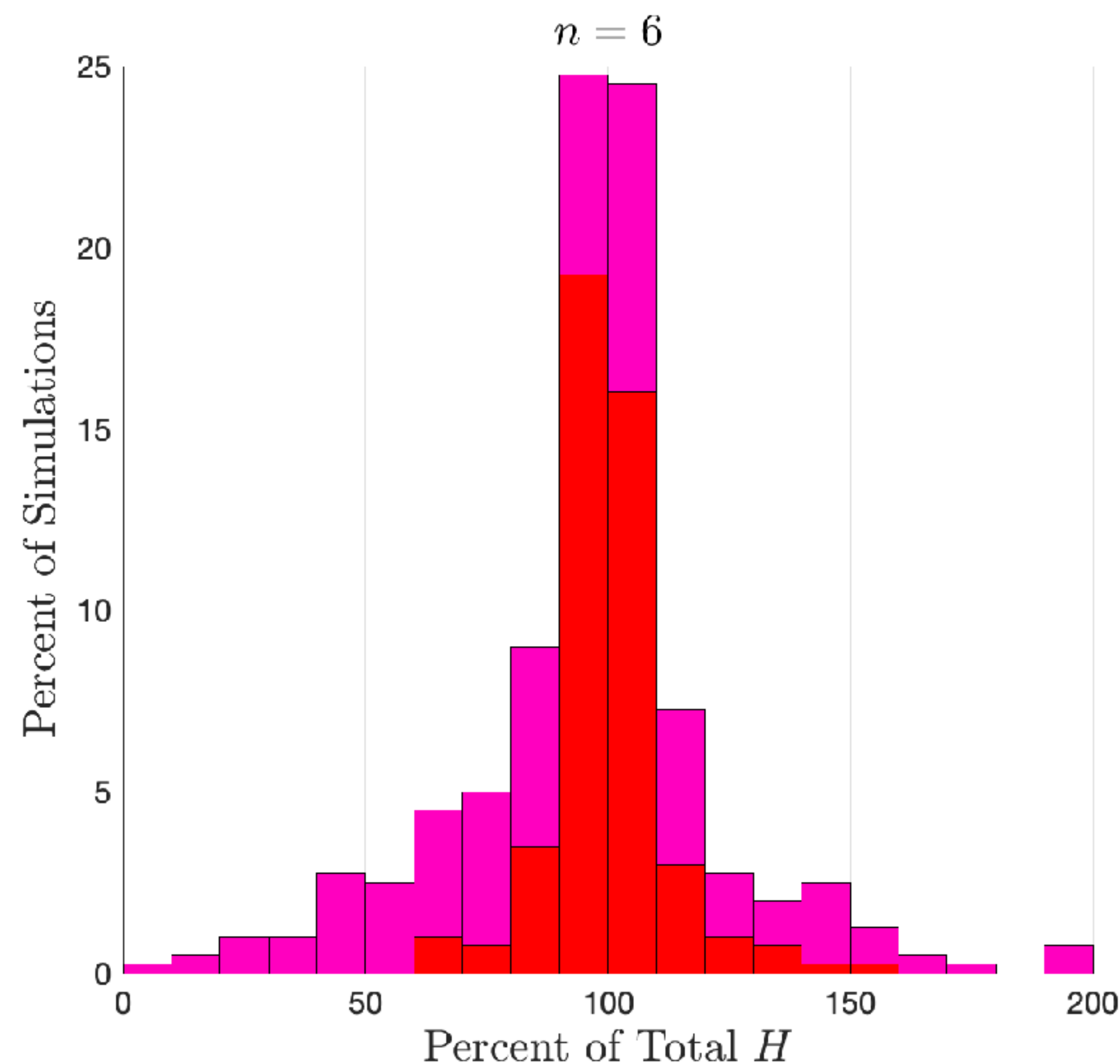
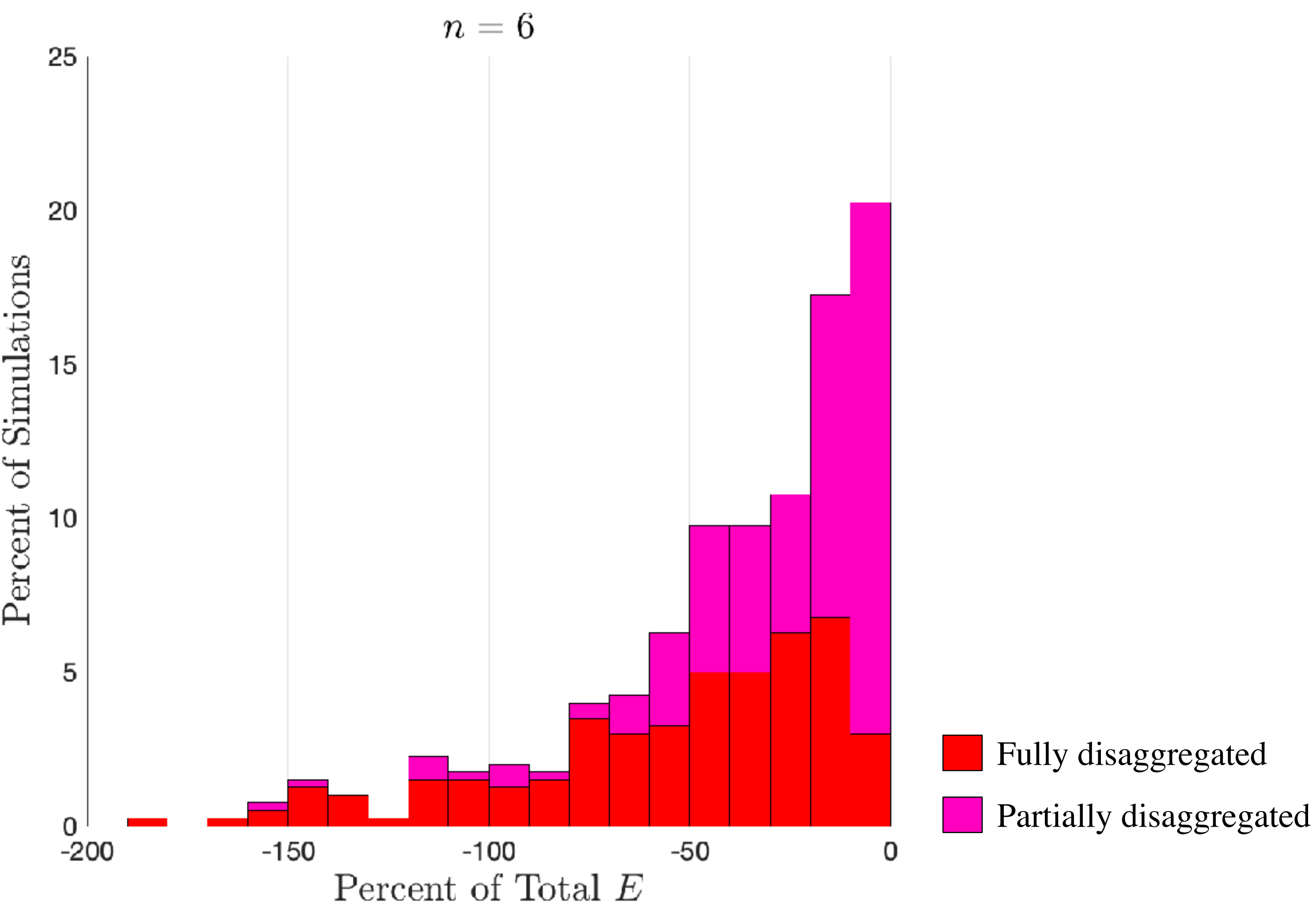


Energy and Angular Momentum Loss: $N = 6$

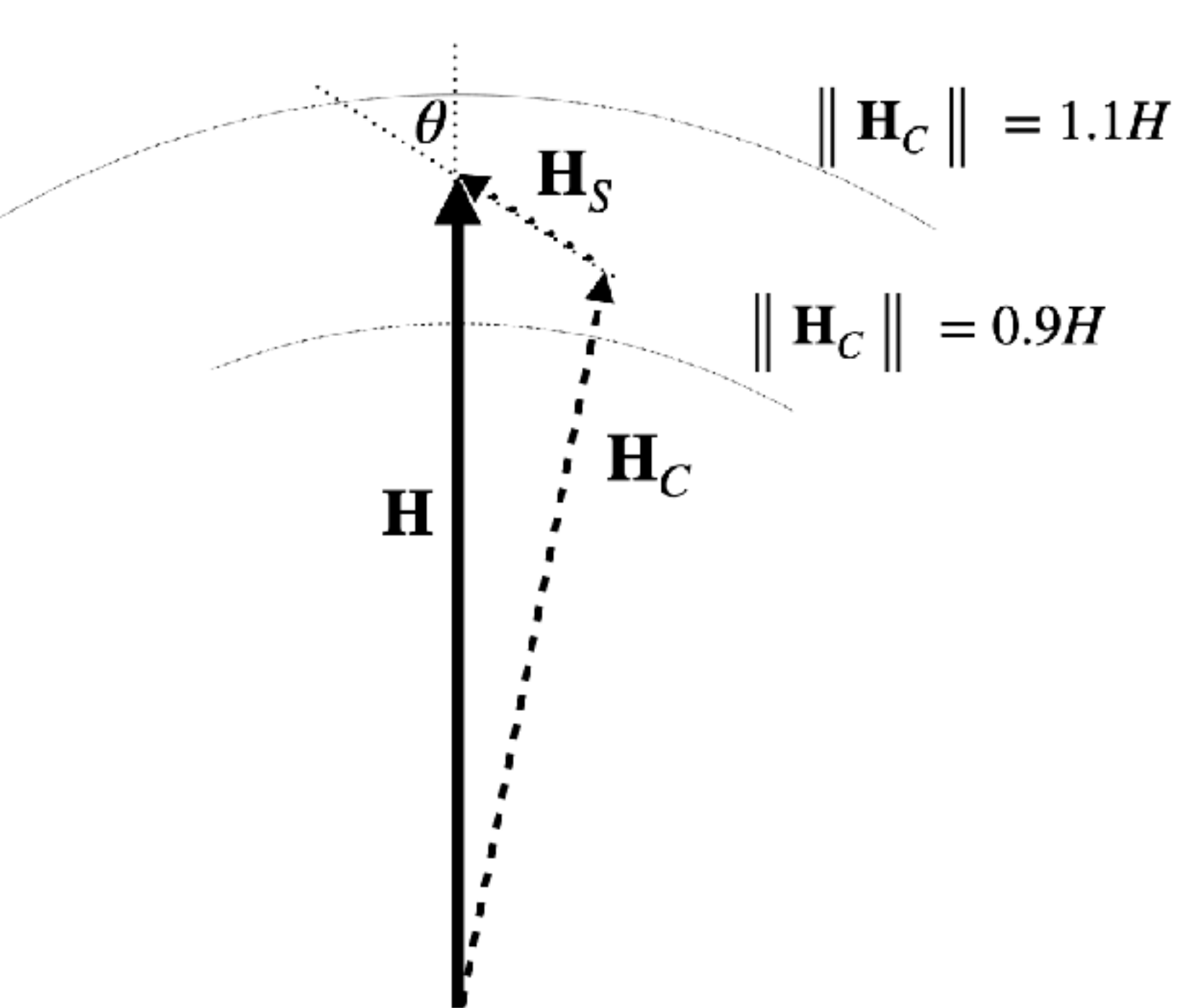


Energy is always lost due to escape

Lost angular momentum tends to be randomly distributed about 100% as N increases



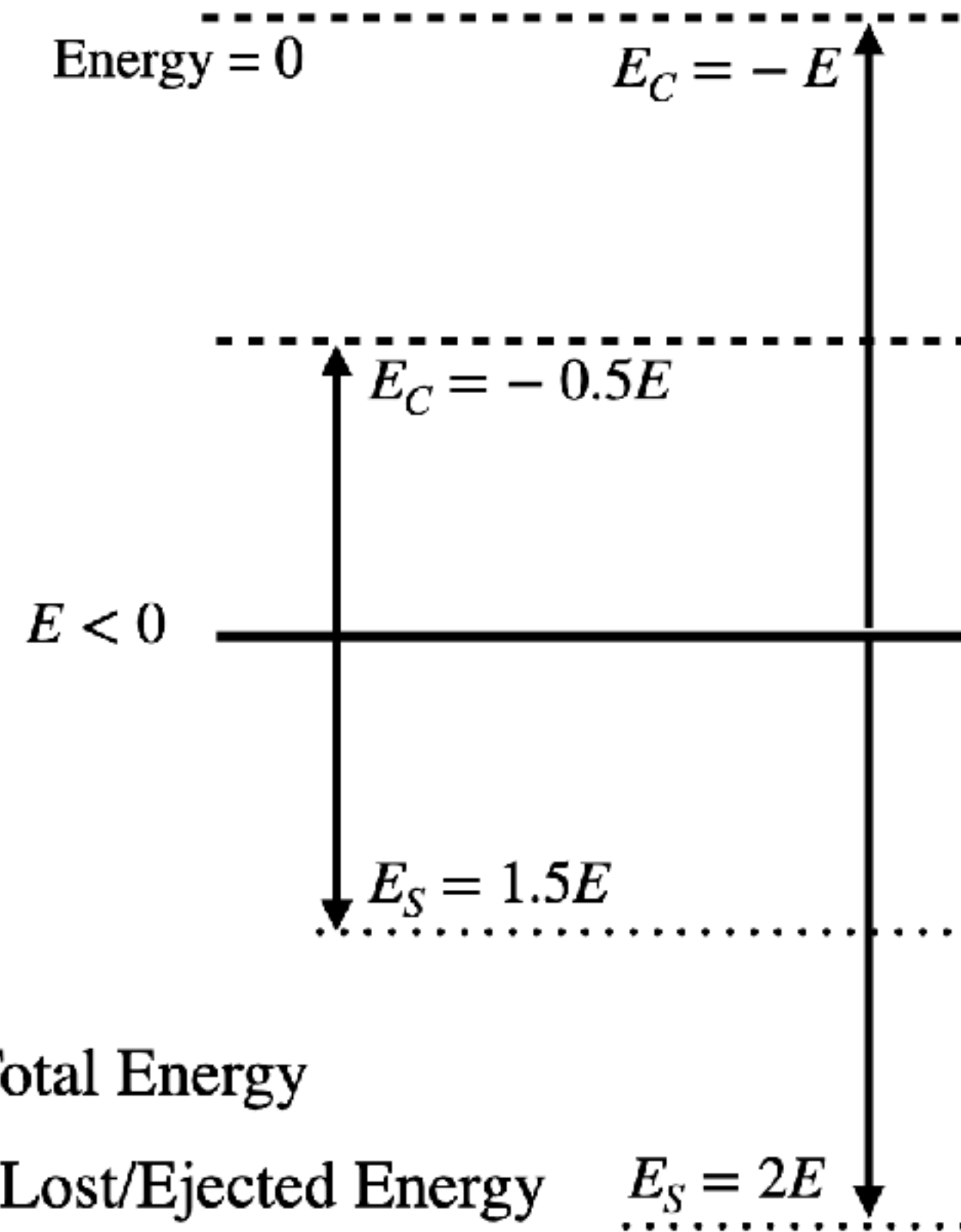
Explaining the Results



$\mathbf{H}$ = Total Angular Momentum

$\mathbf{H}_C$ = Lost/Ejected Angular Momentum

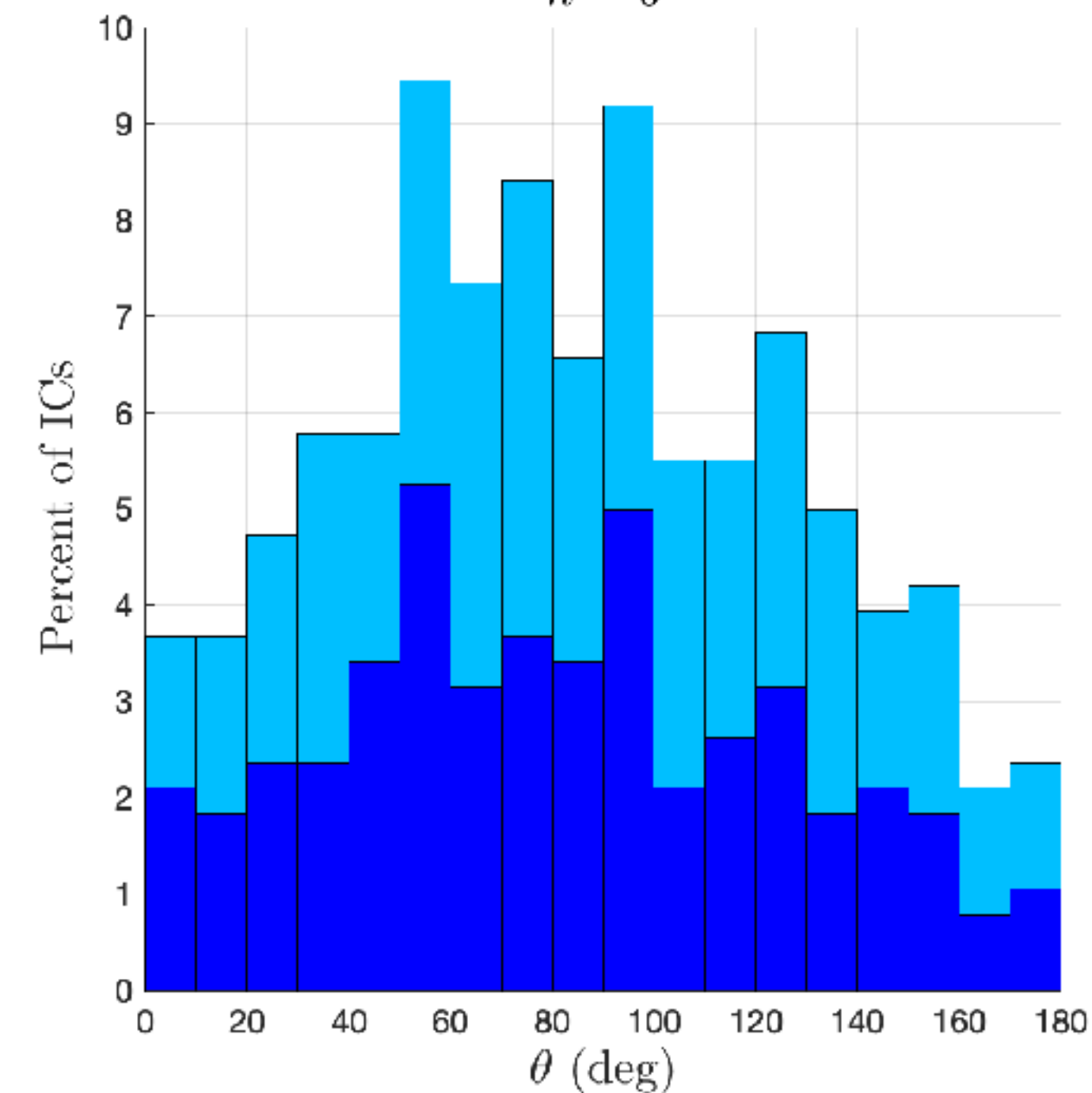
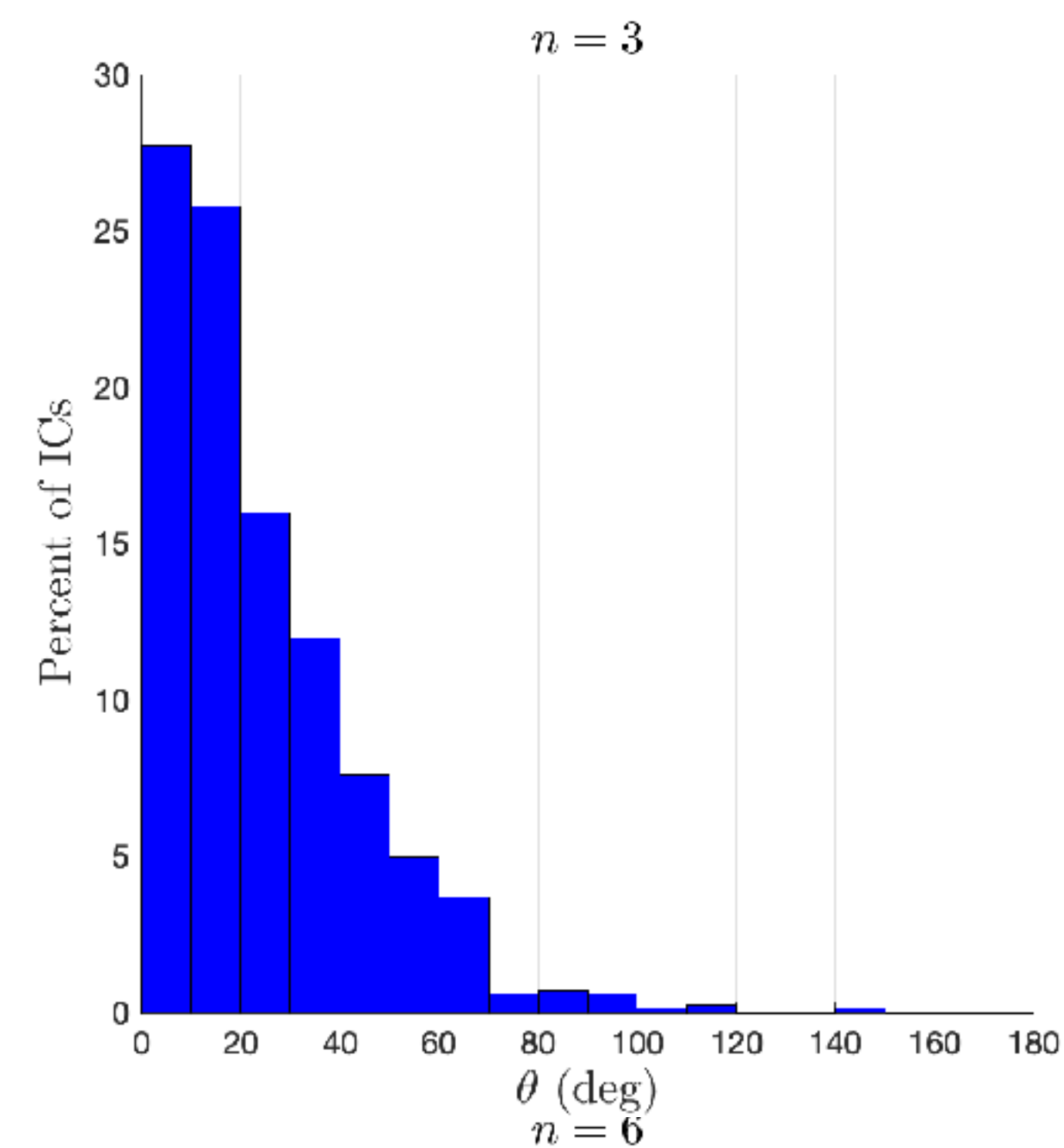
$\mathbf{H}_S$ = Final System Angular Momentum



E = Total Energy

E_C = Lost/Ejected Energy

E_S = Final System Energy



Orientation of the system AM approaches uniform distribution as N increases

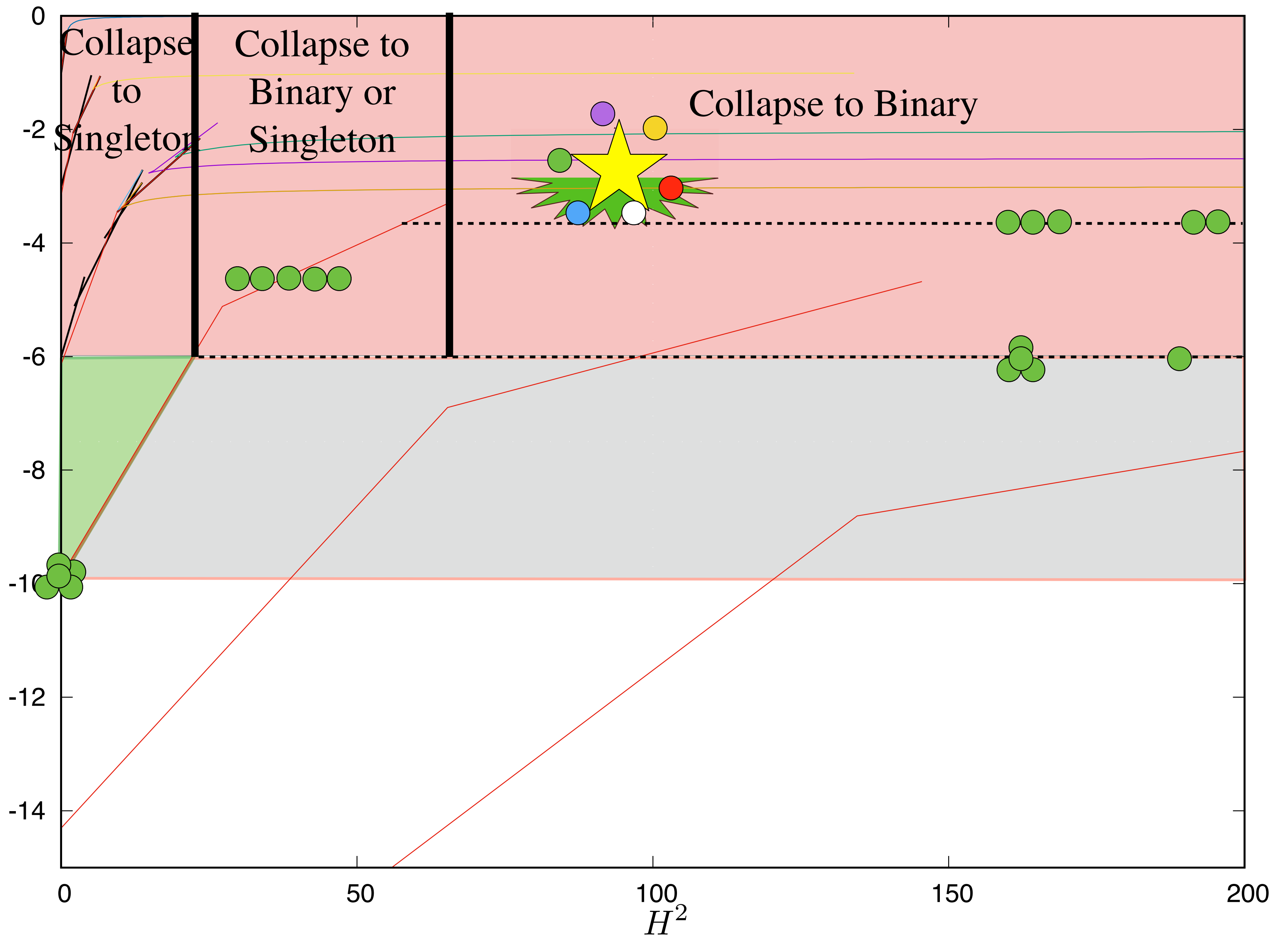
$N = 5$

Ejection or Collapse

Guaranteed Collapse

Forbidden Region

$\mathcal{E}$

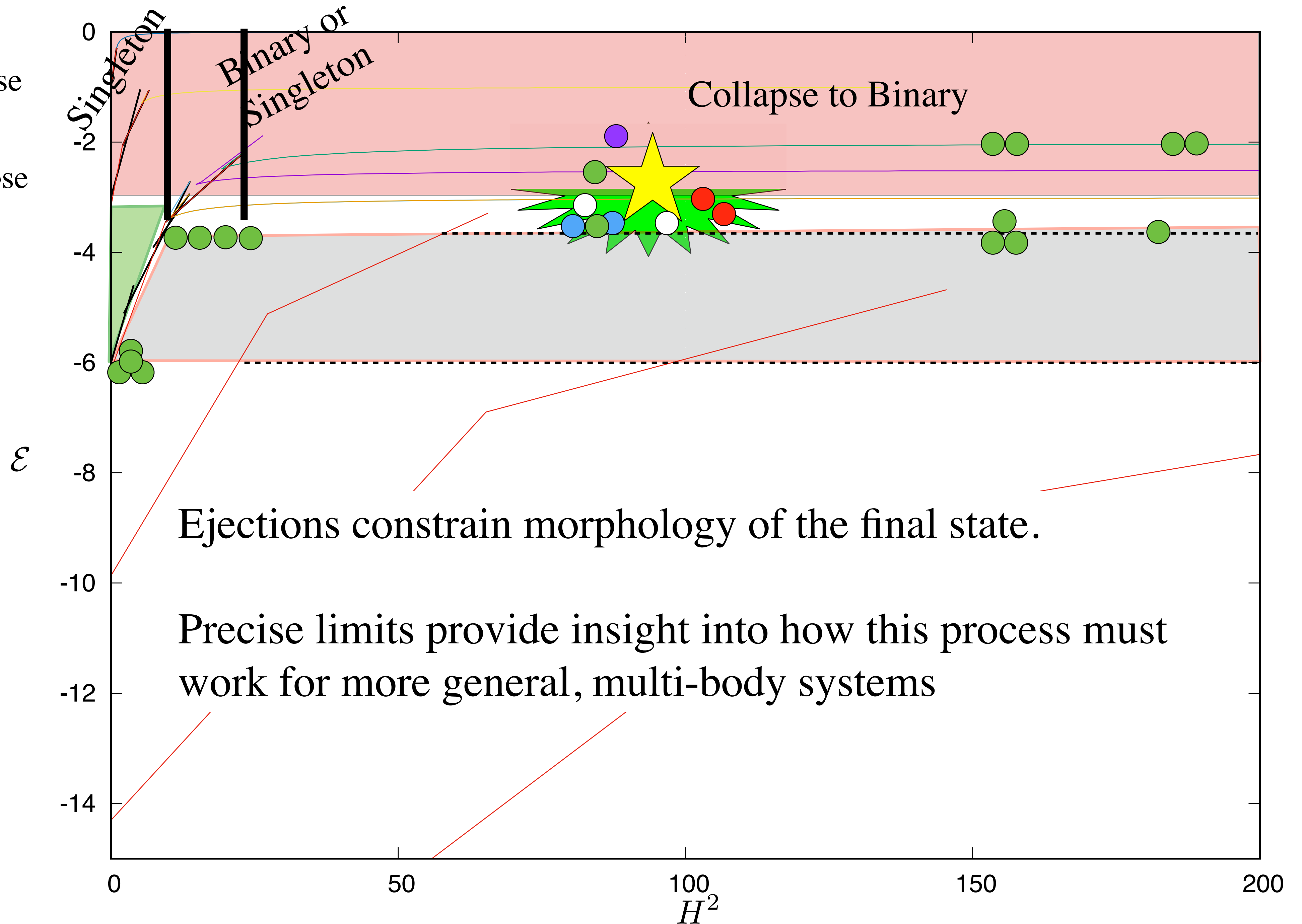


$N = 4$

Ejection or Collapse

Guaranteed Collapse

Forbidden Region



Future Work



- Current focus is on the energetics of the ejection and collapse process
 - Goal is to understand the formation constraints on rubble pile asteroids which can lead to their observed morphologies such as shape, binarity and internal mass distributions
 - A more general approach to using the Amended Potential is developed which allows it to be rigorously modified when escape occurs
- Simple ejection numerical experiments show expected and unexpected results
- Expected:
 - Energy is lost to the “background”, decreasing the energy left in the individual components and serving as a stabilizing dissipation route
 - Most energy loss is relatively small, accumulated at zero
- Unexpected:
 - Angular momentum loss has a wide variance, and can leave the escaping components with random system angular momentum

Supporting Papers



- D.J. Scheeres & G. Brown. “Bounds on Energy and Angular Momentum Loss in the Full n-Body Problem,” submitted to CMDA, 12/22.
- D.J. Scheeres. 2020. “Disassociation energies for the finite density N-body problem,” *Celestial Mechanics and Dynamical Astronomy* 132:4.
- D.J. Scheeres. 2018. “Stability of the Euler Resting N-Body Relative Equilibria,” *Celestial Mechanics and Dynamical Astronomy* 130:26.
- D.J. Scheeres. 2017. “Constraints on Bounded Motion and Mutual Escape for the Full 3-Body Problem,” *Celestial Mechanics and Dynamical Astronomy* 128(2-3): 131-148.
- T. Gabriel and D.J. Scheeres. 2016. “Energy Dissipation End States of the Sphere Restricted Planar Three Body Problem with Collisional Interaction,” *Monthly Notices of the Royal Astronomical Society* 463, 794-801.
- D.J. Scheeres. 2016. “Relative Equilibria in the Spherical, Finite Density 3-Body Problem,” *Journal of Nonlinear Science* 26: 1445-1482.
- D.J. Scheeres. 2012. “Minimum Energy Configurations in the N-Body Problem and the Celestial Mechanics of Granular Systems,” *Celestial Mechanics and Dynamical Astronomy* 113: 291-320.

Current Analysis



- Research with Erick White (CU)
 - Study the sphere-restricted 3-body problem to determine outcomes as a function of initial condition
 - Implemented using the REBOUND numerical computation tool
- Procedure:
 - Specify an initial angular momentum and energy
 - Find a configuration that has this value, including spins, and start integration
 - Evaluate impacts assuming a coefficient of restitution and relative surface friction to damp energy





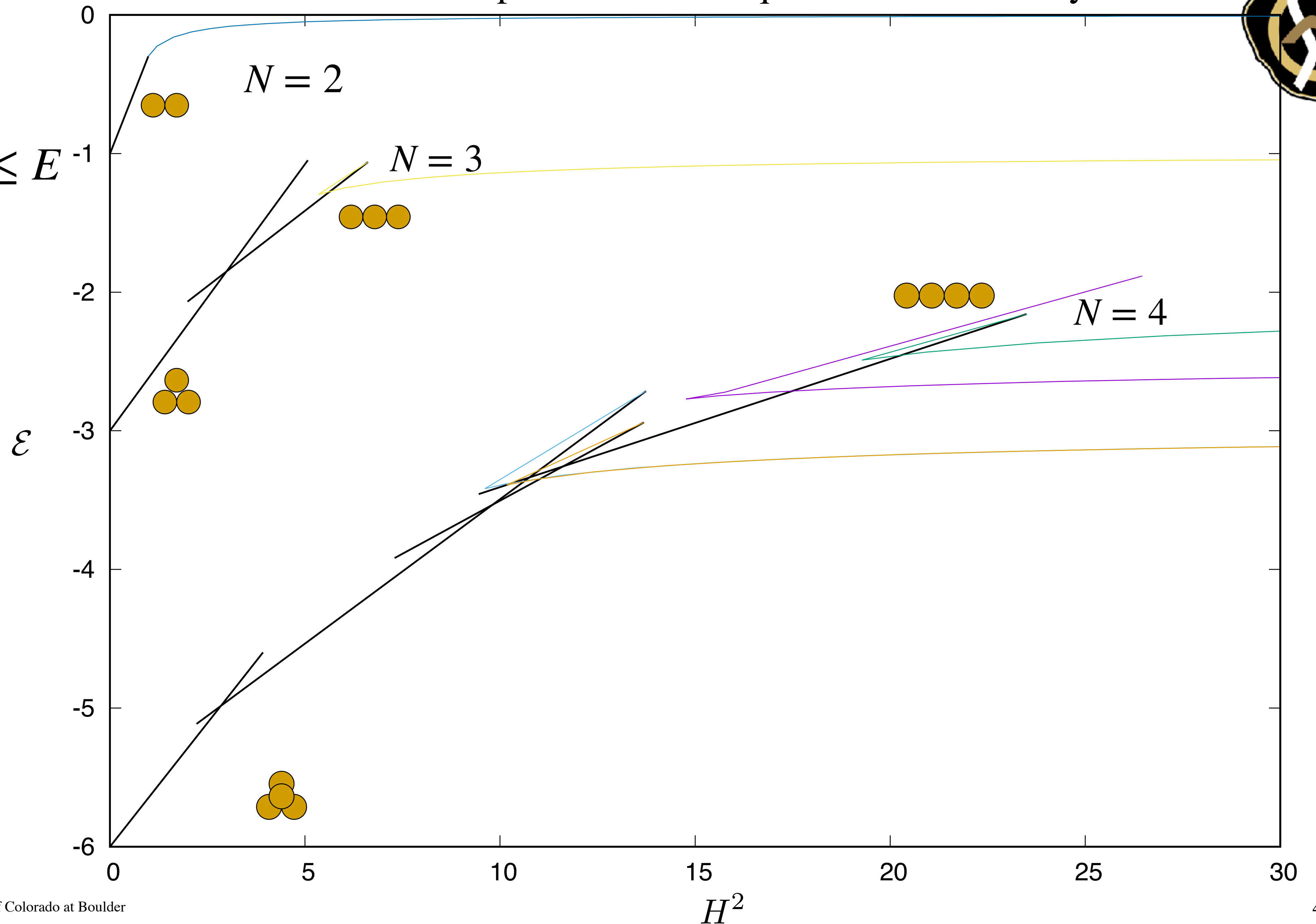
Combined Graph of Relative Equilibria and Stability

$$\mathcal{E} = \frac{H^2}{2I_H} + \mathcal{U} \leq E$$

Can normalize to directly compare energies, opens up the analysis of inclusion and ejection of bodies.

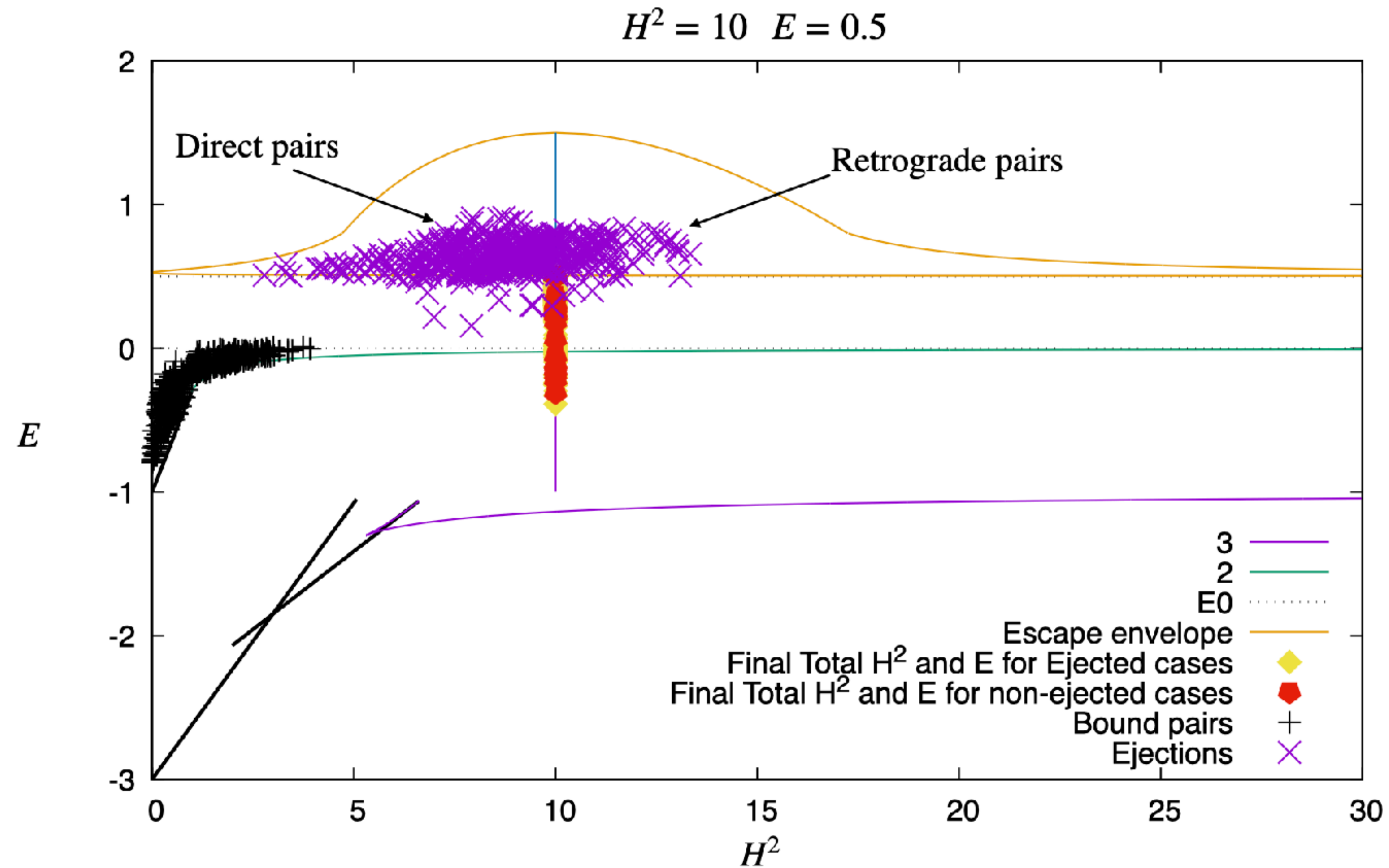
How do we increase N ?

How do we account for ejection?

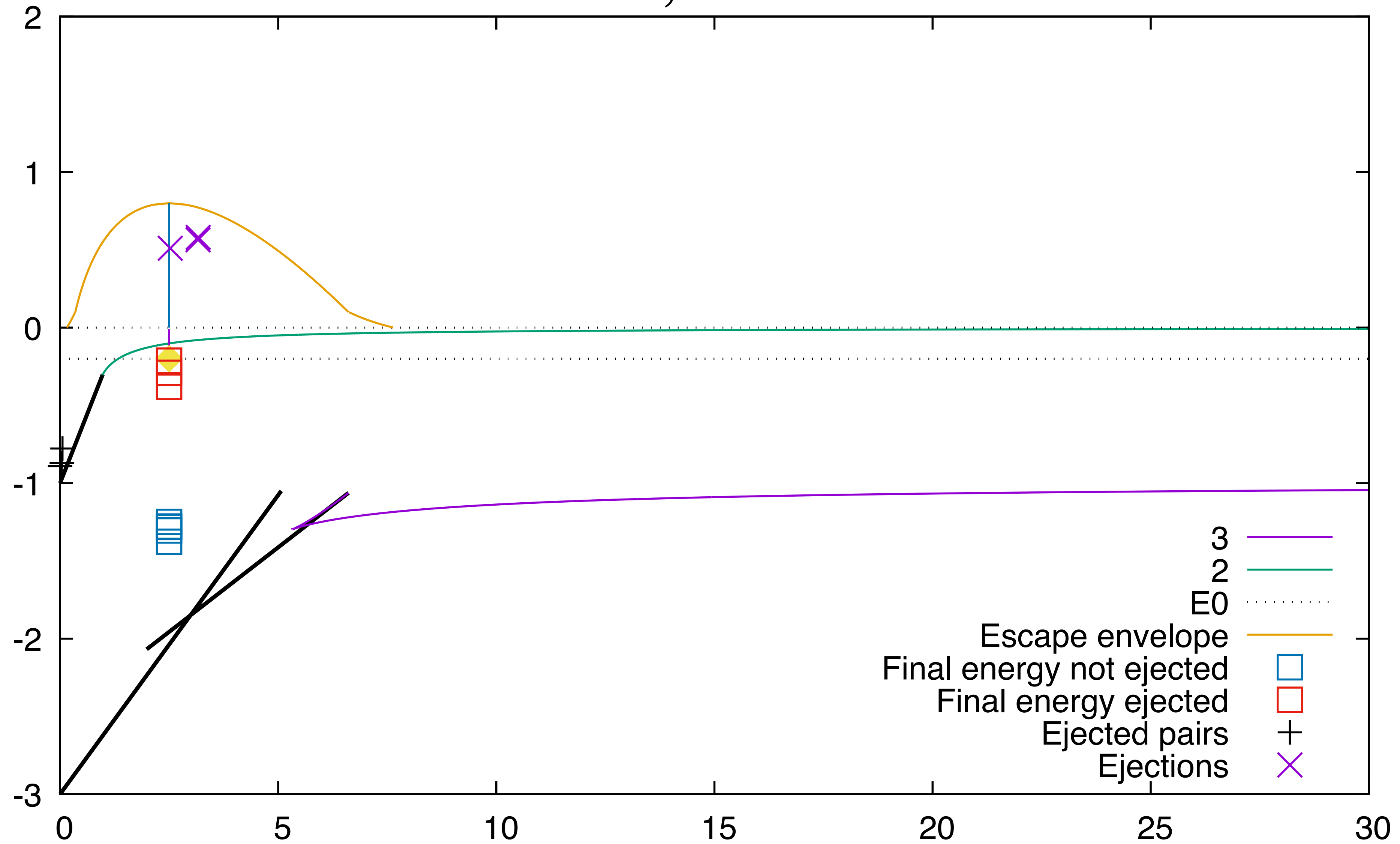




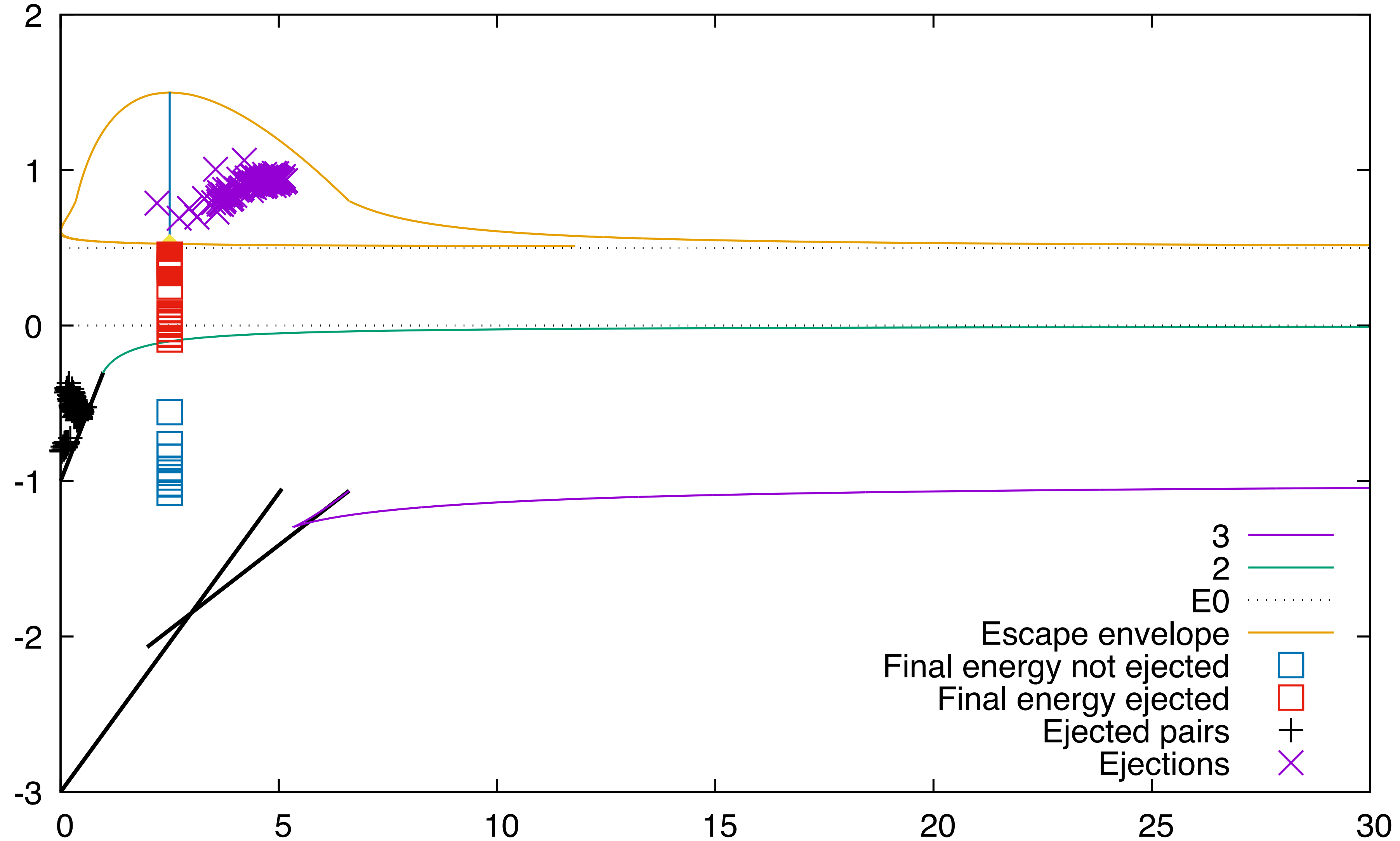
Outcomes



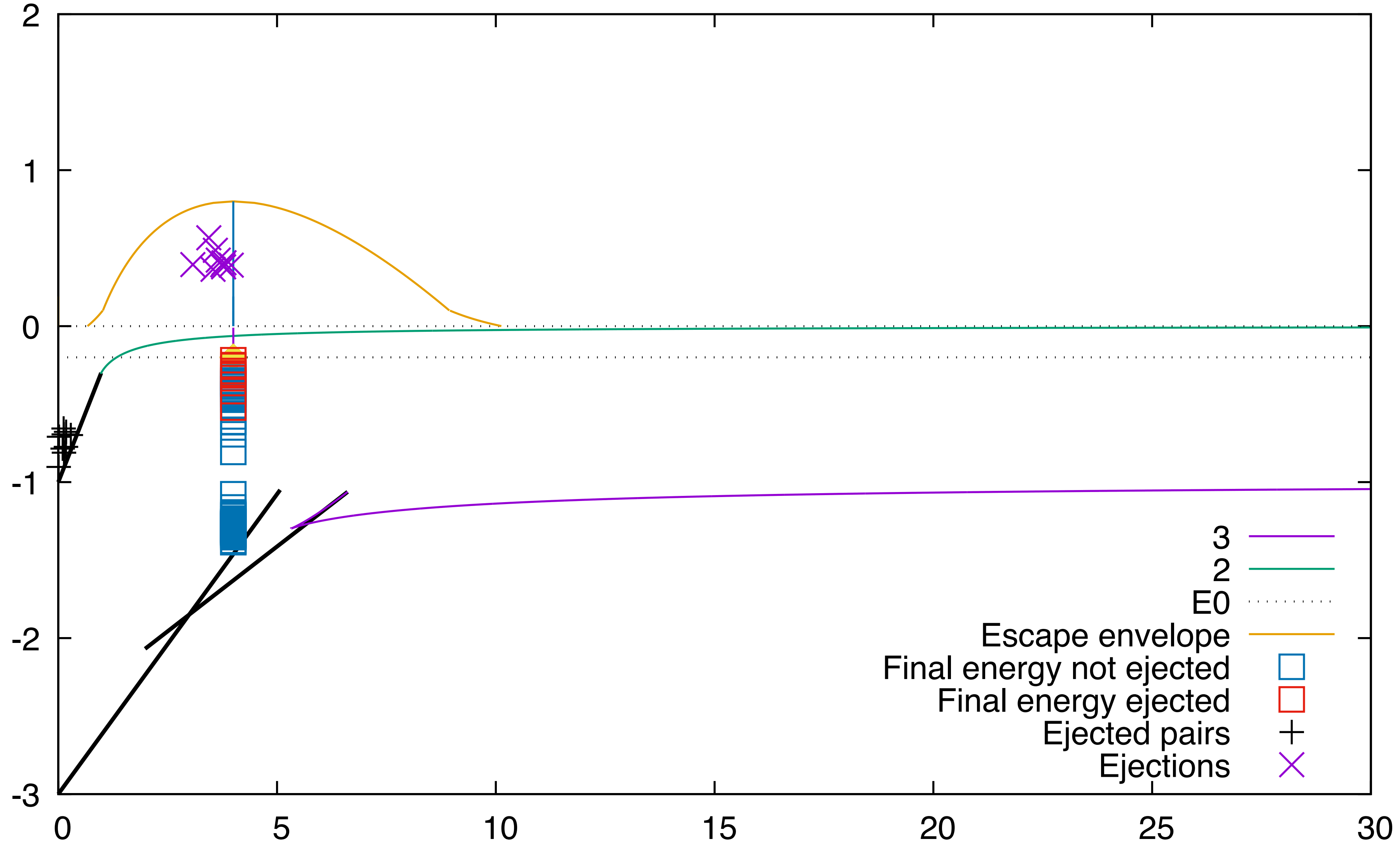
$$H^2 = 2.5 \ ; \ E = -0.2$$



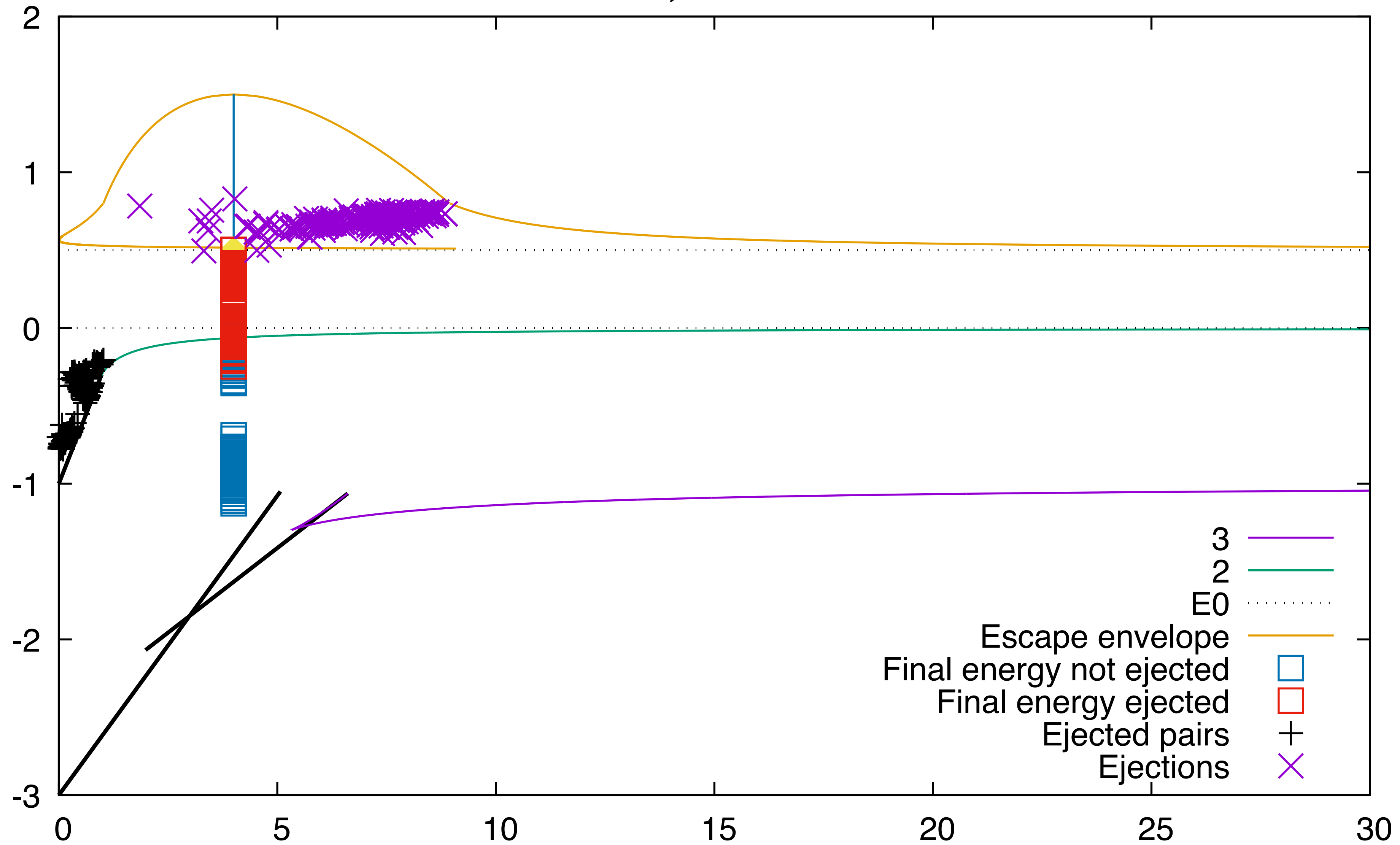
$$H^2 = 2.5 ; E = 0.5$$



$$H^2 = 4 ; E = -0.2$$

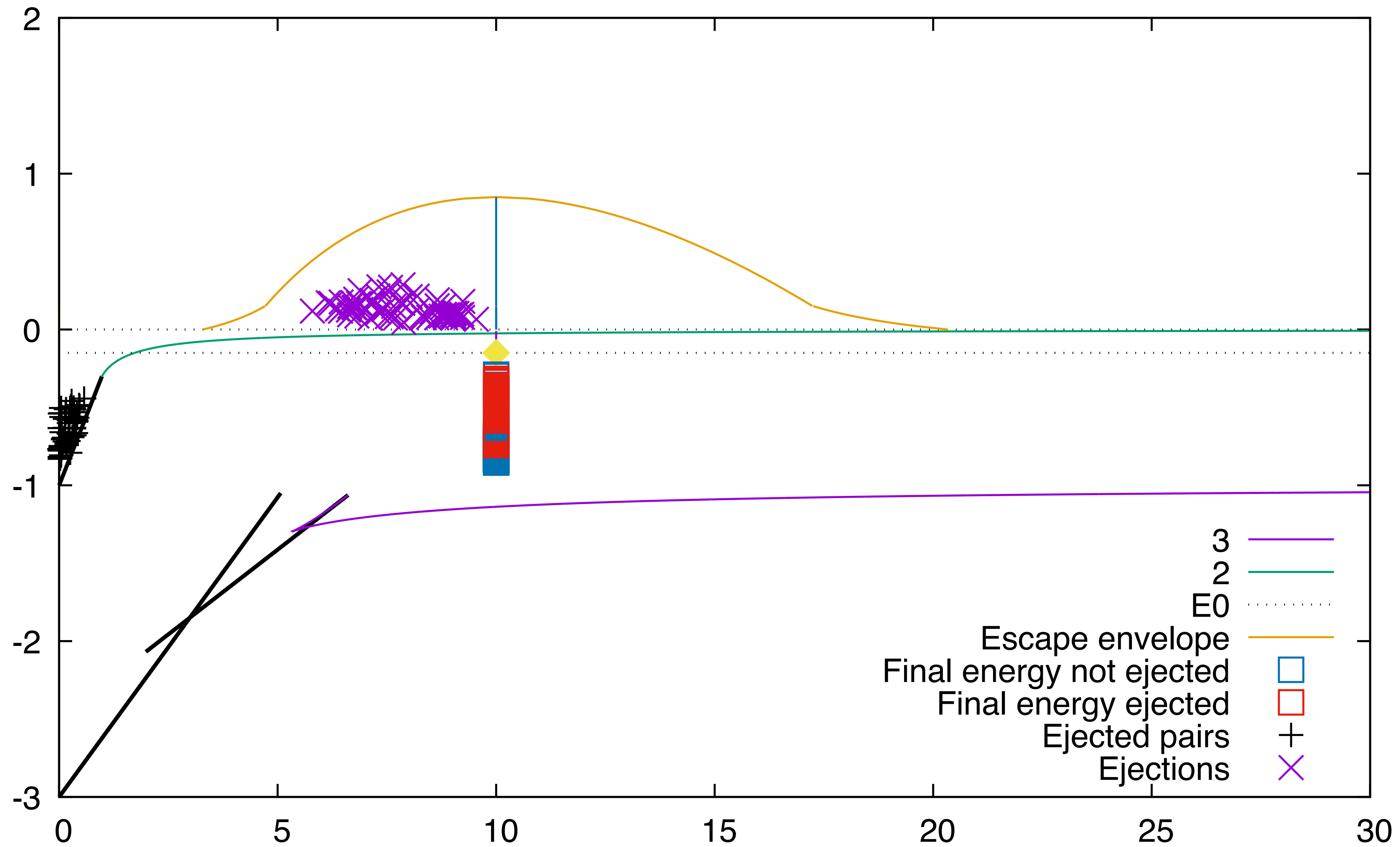


$$H^2 = 4 ; E = 0.5$$

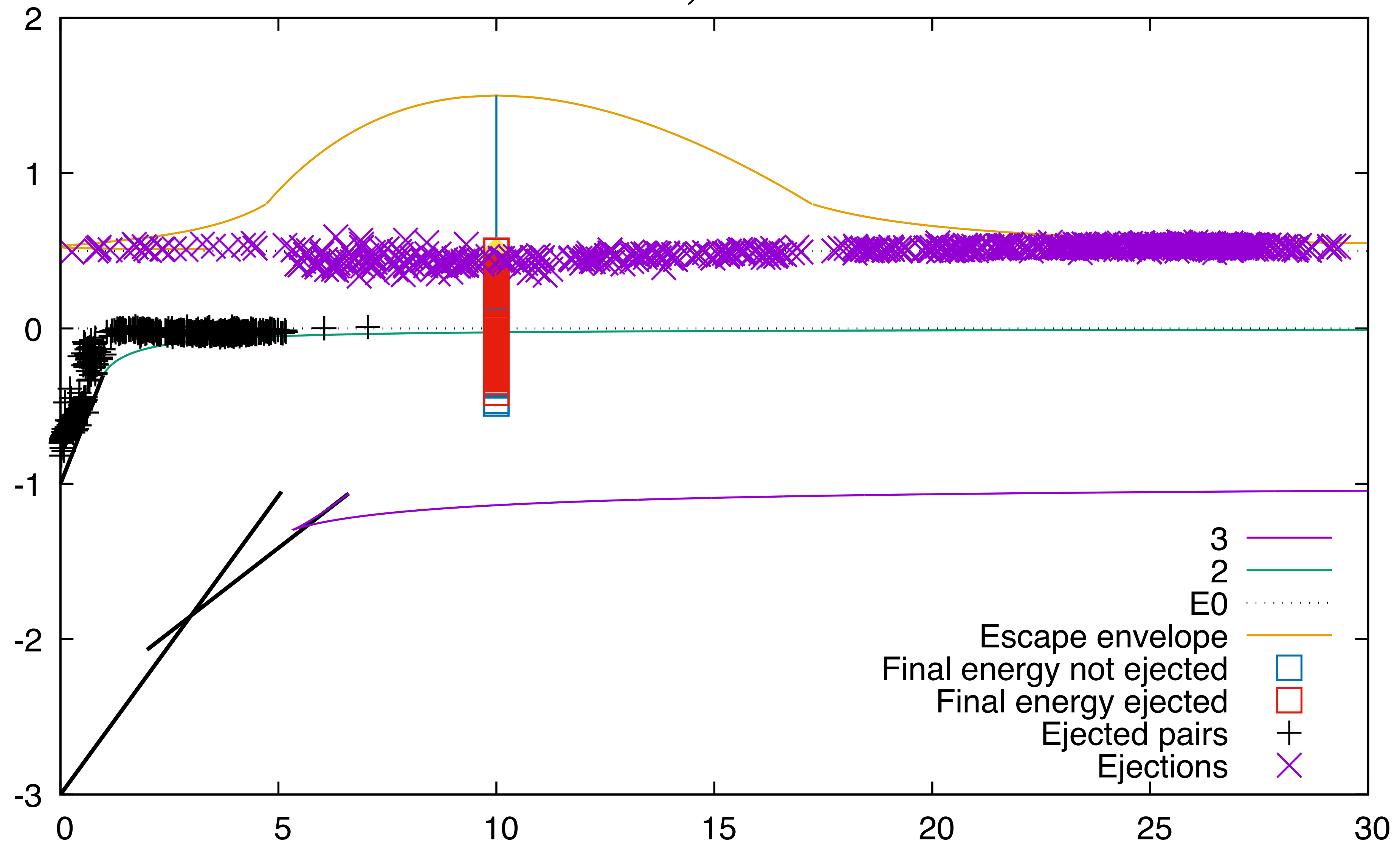


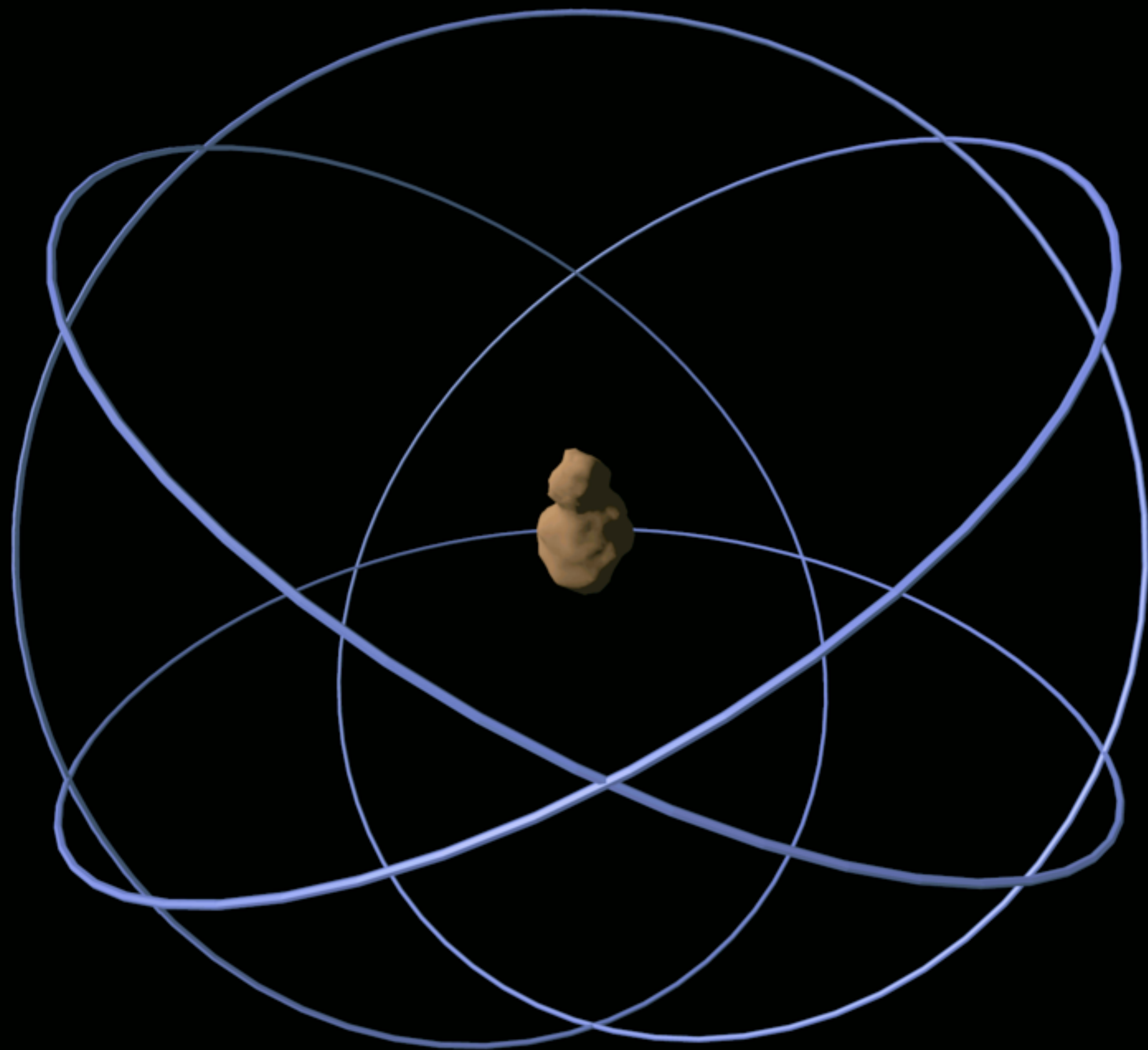


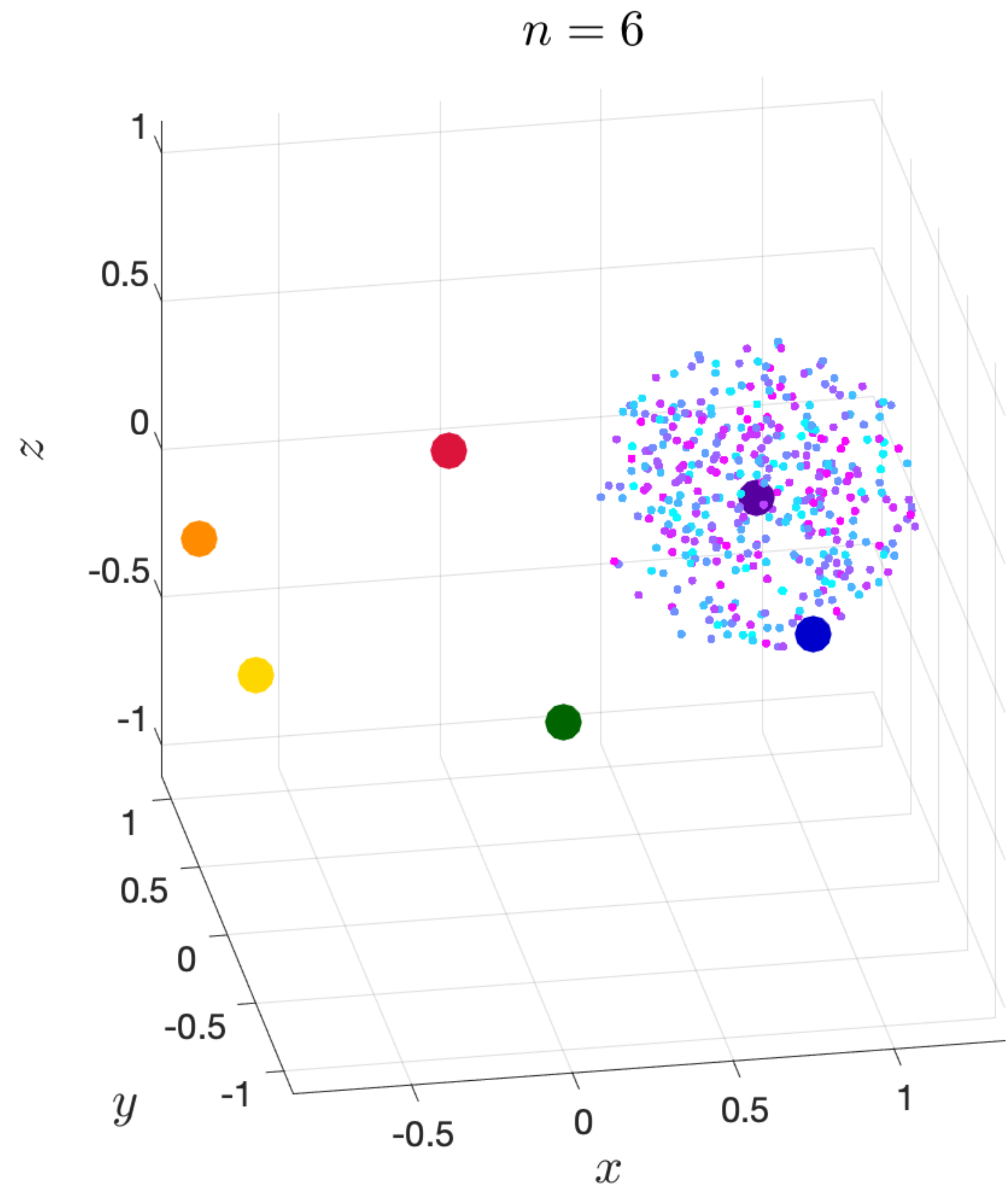
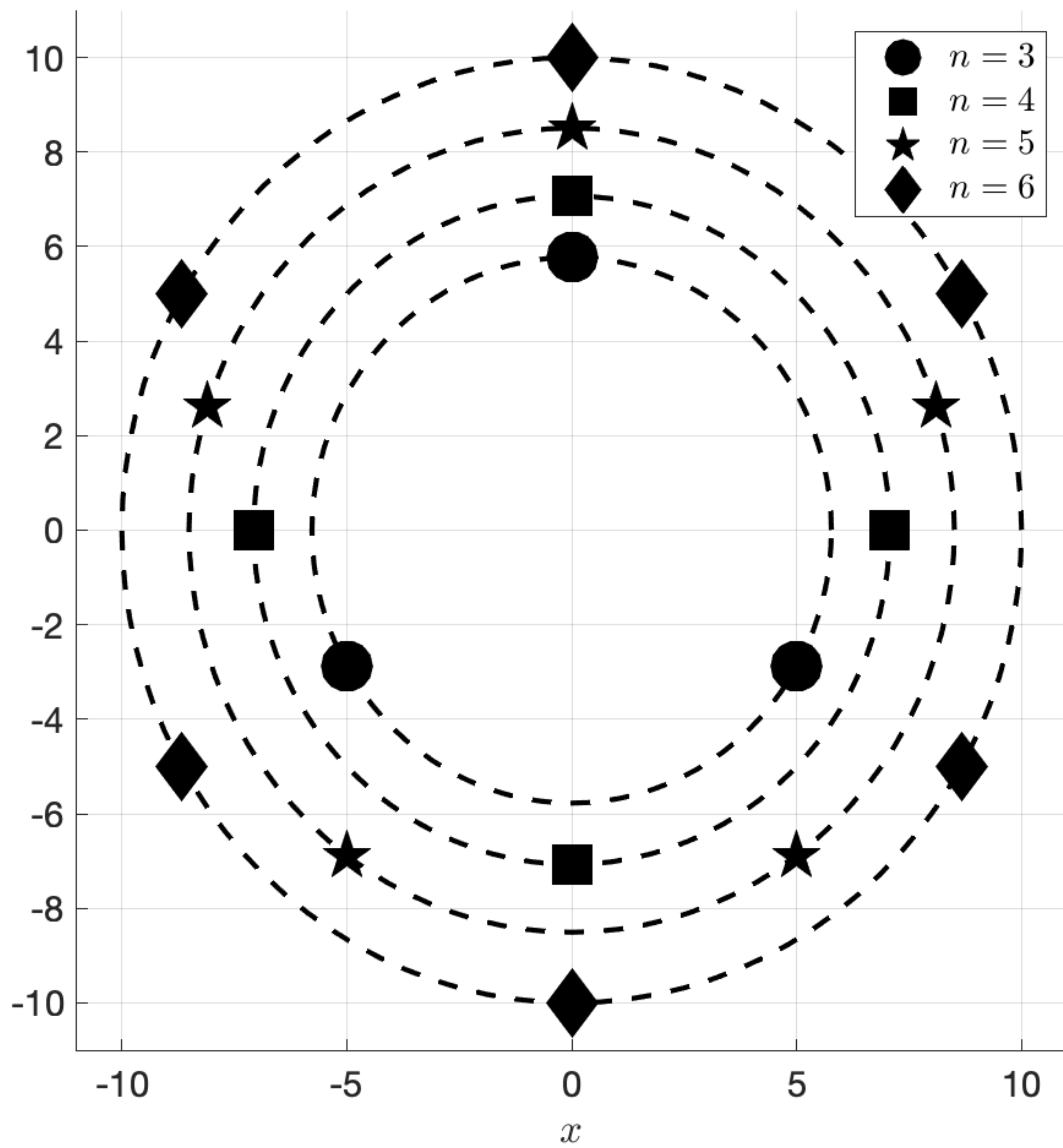
$$H^2 = 10 ; E = -0.2$$



$$H^2 = 10 ; E = 0.5$$

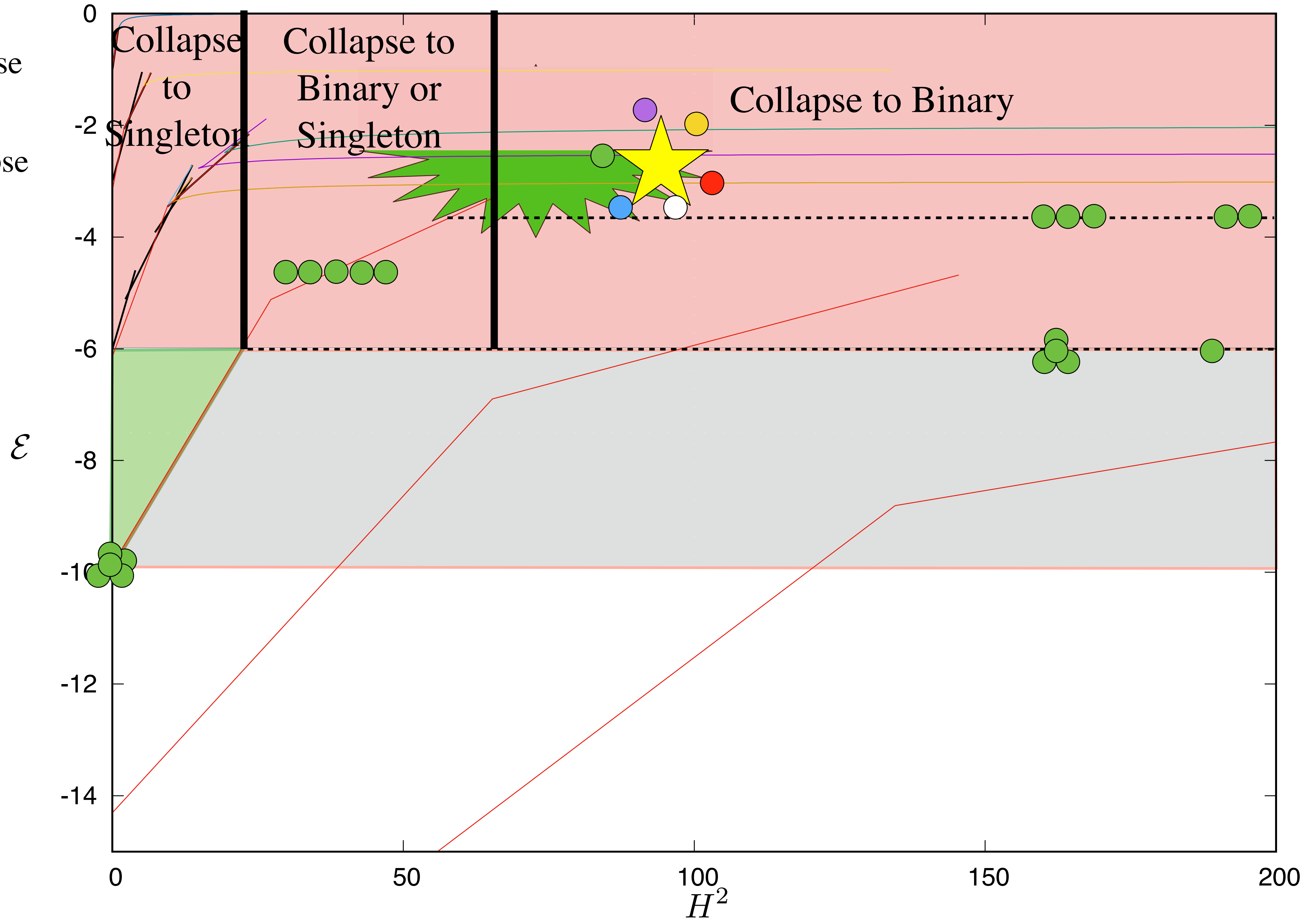






$N = 5$

- Ejection or Collapse
- Guaranteed Collapse
- Forbidden Region



$N = 4$

Ejection or Collapse

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